Knowing When Not to Adapt
A Label-Free Adapt/Freeze/Abstain Certificate for Safe Test-Time Adaptation

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***Abstract*—Test-time adaptation can silently degrade a deployed model on unlabeled target data, and with no target labels the system cannot tell a helpful shift from a harmful one. Deciding *whether* to adapt is possible only relative to a declared bound on calibration drift—a drift budget β—so the question that decides the method is where that budget comes from. This paper answers it, and the answer is that it cannot come from the deployment.**

We prove that for every declared class of target laws the exact minimax label-free budget is the fibre radius Γ_z, attained by a constant audit that reads none of its data; for the deployment class $\mathcal{C}_\beta$ the fibre radius equals β exactly, so the best possible label-free audit returns its own input, and the number it returns is simultaneously the radius of the band on which a strict adapt-or-freeze commitment is unsound. Two consequences follow. Any δ-valid label-free audited rule commits with probability at most δ over the unrestricted class—decision yield is bounded by the error budget, at every batch size and for every evidence map. And a *fully labeled* calibration sample from any other domain leaves the minimax budget unchanged, absent a separately declared coupling. One lemma—that the label kernel on the disagreement region is unconstrained by every label-free observable—carries the matched-evidence impossibility, the abstention band, and the budget impossibility as readings of the same one-parameter family at three radii.

We then measure the prediction. Declaring β as a high quantile of realized drift on source-like development cells, exactly as this paper's own method section prescribes, fails on $6{,}885$ real evaluation cells: on CIFAR-10-C the declared budget is $1.4\times$ to $50\times$ smaller than soundness requires, 24–73% of deployment cells fall outside the declared class, and committed actions are wrong at 0.4–16.6% where the frontier promises 0%; on ImageNet-C a large enough budget is derivable and returns zero commitments on all 405 cells in five of ten configurations. This is not a quantile-estimation failure: across 470 splits the conformal order statistic that replaces the plug-in quantile differs from it by a median 0.34%, with a median coverage difference of exactly zero. We withdraw the operational reading of the frontier; the proofs are untouched.

What escapes the impossibility is a population of deployments rather than a deployment. Under retrospective outcome logging and exchangeability with K logged episodes, β is identified as an episode quantile and estimable by a conformal order statistic, with the episode requirement bracketed to a factor of e: $1/(e\alpha)-1 \le K^\star \le 1/\alpha-1$, between 3 and 9 logged deployments at $\alpha = 0.10$, and first feasible on real data at exactly $K = 9$. The escape is average-over-episodes instead of supremum-over-the-fibre, paid for with historical labels, and neither ingredient suffices alone; the guarantee it buys is marginal across deployments, not conditional on the one in hand. The assumption that buys it is the one a shifted deployment violates: on CIFAR-10-C the episode budget attains its nominal 0.900 coverage under an exchangeable control (0.905) and along nuisance axes, and collapses to 0.626 when the corruption family changes and 0.454 when severity changes, with half the severity splits admitting no budget at any declared level; weighted conformal restores coverage to 1.000 by driving decision yield to 0.000.

What survives is a finite-sample certificate that trades decision yield for a false-adapt guarantee. Knowability-Guided Adaptation wraps a fixed candidate adapter, estimates the benefit $\Delta = R_T(f_0) - R_T(f_a)$ from label-free evidence, and commits only when a calibrated interval $\widehat{\Delta} \pm \varepsilon$ excludes zero. The population and empirical layers share the target functional Δ, but KGA does not compute M, γ, or β, and ε is not an estimate of β. Across the eligible Tent and EATA CIFAR-10-C stress arms, KGA makes $2{,}845$ strict decisions among $4{,}320$ cell–seed evaluations (65.9% decision coverage), including $1{,}113$ and $1{,}244$ adapt decisions with no observed false adaptations. Tent's shipped operating point beats both fixed policies with intervals that exclude zero down to the six corruption-family clusters; EATA's point estimate beats both, but its adapt-side interval includes zero once corruption is the resampling unit. Under leave-one-corruption-out refitting the radius inflates $4.3\times$–$6.9\times$ and the false-adapt count remains zero in all ten runs, while the utility comparison survives only descriptively for Tent and fails for EATA. The completed SAR rebuild is reported separately as a negative arm rather than pooled into this headline.

The empirical panel is scoped, not uniform, and we report the losses. On the one-sided natural tracks the supported conclusion is safety rather than dominance: Camelyon17 OOD reproduces always-adapt, the primary iWildCam and Office-Home panels reproduce always-freeze, and RxRx1 freezes throughout; these observations remain weak evidence—Camelyon17's promoted subset contains no harmful cell; RxRx1 and the primary iWildCam/Office-Home panels make 0 adapt decisions; and the source-replayed Office-Home/iWildCam transfer scores lack a predeclared uniform A7 full-fit-versus-LOO stability bound. A separate Office-Home replication has a small point edge over both fixed poli-

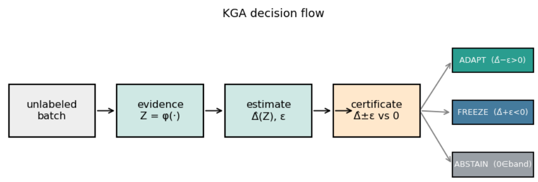

Fig. 1. **K-Bound previews the decision.** From an unlabeled batch under shift, the certificate estimates the adaptation benefit $\widehat{\Delta}$ with a finite-sample radius ε and returns ADAPT (knowably helpful), FREEZE (knowably harmful), or ABSTAIN (sign unidentifiable), with false-adapt/false-freeze $\leq \alpha$.

cies, but its seed interval includes zero. The controlled CIFAR-10-C result is candidate-specific: Tent has a corruption-cluster-robust beats-both result; EATA has a point-estimate result whose adapt-side clustered interval includes zero; and the completed SAR rebuild loses to always-adapt. ImageNet-C SAR is likewise a point estimate, not a CI-supported win. PACS, ImageNet-R, and CIFAR-10.1 are negative or transfer-failure results and remain in the panel. KGA is a safety, validity, and abstention layer for test-time adaptation, not a universal accuracy booster; the population drift budget is an external specification, while the deployable guarantee is conditional on coverage transfer at the declared inference unit.

***Index Terms*—test-time adaptation, safe adaptation, abstention, selective prediction, distribution shift, identifiability, label-free risk estimation**

Scope. We report positive and negative results; see §XII and §XI for honest limitations and excluded alternate wins.

I. Introduction

Should a model adapt to unlabeled target data, or would adapting make it worse? Test-time adaptation (TTA) updates a model on the test stream—through entropy minimization, sample filtering, or stability-aware updates—and in favorable shifts recovers lost accuracy. In unfavorable shifts the same unlabeled objective silently degrades it: the model sharpens wrong predictions, adapts to label imbalance, or collapses under non-stationary streams. Because target labels are unavailable at deployment, the system usually cannot tell whether adaptation is helping or hurting.

Most work on TTA asks how to design a better adaptation mechanism. This paper asks the prior question: should the system adapt at all? We study label-free adaptation as a decision problem. Given a frozen baseline, an adapted candidate, and an unlabeled target batch, the system must choose one of three actions—adapt when adaptation is knowably helpful, freeze when knowably harmful, and abstain when the available evidence cannot identify the sign of the benefit. Figure 1 previews this decision certificate.

This decision is constrained by identifiability, and the constraint has a name and a price. That label-free deployment quantities cannot in general pin down a target risk—and hence cannot in general decide adaptation without some assumption on the shift—is established. Garg et al. [13] show identifying out-of-distribution accuracy is as hard as identifying the optimal predictor, and Kalai and Kanade [14] characterize the optimal predict-versus-abstain trade-off under shift against a shift budget. We do not rederive this general impossibility; we *specialize* it to the adaptation-benefit sign and make its boundary exact. The sign of the benefit is identifiable from label-free observables precisely when an observable margin on the disagreement region exceeds a declared bound on calibration drift—a *drift budget* β—so that $|M| > \beta$. The margin is how confidently the label-free evidence points toward "adapt" or "freeze"; the budget is how far that signal can be thrown off by calibration drift; the decision is trustworthy exactly when confidence outweighs drift. This makes precise why common label-free heuristics work in benign regimes and fail under drift: they live on the β=0 face of the frontier.

The question that decides the method is therefore where β comes from, and this paper answers it: it cannot come from the deployment. For every declared class of target laws the exact minimax label-free budget is the fibre radius Γ_z, and it is attained by a constant audit that reads none of its data (Theorem 6). For the paper's own deployment class that radius equals β exactly (Corollary 4): the best possible label-free audit of the drift budget returns the budget it was given, and the number it returns is simultaneously the radius of the band on which a strict adapt-or-freeze commitment is unsound. Two riders make this operational rather than philosophical. Over the unrestricted class any δ-valid audited rule commits with probability at most δ—decision yield is bounded by the error budget, at every batch size, every calibration sample size and every evidence map. And a *fully labeled* calibration sample from any other domain leaves the minimax budget unchanged absent a separately declared coupling (Theorem 7). One lemma carries all of it—that on the disagreement region the label kernel is a free parameter no label-free observable constrains (Lemma 3)—and the same lemma, read at three different radii, is the matched-evidence impossibility, the abstention band, and the budget impossibility.

We then measure the prediction on our own method. Declaring β as a high quantile of realized drift on source-like development cells, exactly as this paper's method section prescribes, fails on 6,885 real evaluation cells (§VII), and **we report that negative result about our own parameter** rather than withdraw it. It is not a quantile-estimation failure: across 470 splits the conformal order statistic that replaces the plug-in quantile differs from it by a median 0.34%, with a median coverage difference of exactly zero. What escapes the impossibility is a *population* of deployments rather than a deployment (§VIII): under retrospective outcome logging and episode exchangeability β is identified as an episode quantile and estimable by a conformal order statistic, with the episode requirement bracketed to a factor of e. The assumption that buys it is the one a shifted deployment violates, and we measure that too. What remains deployable is a finite-sample certificate that trades decision yield for a false-adapt guarantee, and we measure that frontier exactly (§X-A), including the track where it buys nothing.

In the test-time-adaptation setting studied here we further quantify *how much* untestable information the decision needs: the minimal supplement that no label-free-checkable assumption can remove is exactly *one bit* of orientation, which can only be supplied through structural assumptions that are falsifiable but not verifiable. (In full generality the irreducible supplement is problem-dependent—an interval under affine ambiguity—so we state the one-bit count for the TTA/panel setting under our stated hypothesis, not for arbitrary functionals.)

We instantiate the framework as KGA, a mechanism-agnostic wrapper around existing adaptation candidates such as Tent, EATA, and SAR. KGA does not introduce a new adaptation objective. Instead, it estimates the label-free benefit of a candidate, attaches a calibrated uncertainty radius, and returns adapt, freeze, or abstain. This makes KGA a safety layer for TTA rather than a replacement for TTA methods.

Empirically, KGA behaves according to the theory. **Headline (natural shifts):** under a valid out-of-fold conformal radius KGA is *no-harm on the four one-sided natural tracks with locked held-out artifacts*—Camelyon17, iWildCam, Office-Home, and RxRx1—where it matches the better fixed policy and beats the worse one at zero observed false adaptation, with *no* natural-shift beats-both. It is **not** uniformly no-harm across the panel: on the domain-generalization and rendition tracks it is a conservative null that *loses* to always-adapt, by 2.45× on PACS (0.0431 versus 0.0176) and on 7 of 10 backbones on ImageNet-R (mean 0.0112 under this manuscript's interpolated radius, 0.0151 under the exact-rank rule, against always-adapt's 0.0064 either way), and CIFAR-10.1 fails the declared transfer bar outright ($\mathrm{FA_u} = 0.167$). Those three are reported here, not withdrawn. Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt (CIs exclude zero) but *tie* always-freeze (Tables XXI, XXII); their earlier in-sample-radius "beats-both" was a calibration bug, corrected here to no-harm. On the one-sided WILDS shifts Camelyon17 (Protocol G) and RxRx1, KGA likewise matches the better fixed policy at false-adapt $\leq \alpha$ without beating both; the earlier Camelyon17 Protocol-G beats-both was a separate pooling artifact (in-distribution `id_val` cells mixed into the held-out set, §X-D). The completed three-seed PACS leave-one-domain-out study is a null action-safety diagnostic rather than a promoted no-harm or beats-both result (Table XXVIII). **Confirmatory (identifiable synthetic/mixed):** on the pre-registered CIFAR-10-C stress grid, where harmful adaptation is frequent and detectable, KGA beats both always-adapt and always-freeze for Tent and EATA while making zero false-adapt decisions; the unreconciled CIFAR-10-C SAR artifact is withheld. On the same mixed stream it **beats POEM and AETTA** (Table XV) with Holm-corrected paired-bootstrap CIs. In online non-stationary CIFAR-10 streams, always-adapt collapses in the harm-dominated regime and always-freeze forgoes gains in the helpful regime, while KGA remains near-best across both. At ImageNet scale, KGA also improves over both trivial policies on a harmful SAR stream. ImageNet-R remains a completed weak-evidence diagnostic, not a win.

a) Contributions.: **Submission hierarchy.** The primary claim stack is deliberately narrow: the impossibility spine below, the finite-sample certificate, and the registered empirical panel. Multicandidate, anytime, multiclass-capacity and regression results are companion extensions with their own stated assumptions (Appendix A); none is needed to validate any empirical claim.

- **One lemma, three radii.** We isolate the construction the paper previously carried out three times: on the disagreement region the label kernel is a free parameter that no label-free observable constrains (Lemma 3). Read at radius δ it is the matched-evidence impossibility (Theorem 1); at radius β it is the abstention band of the strict-commitment frontier (Theorem 3); at its extreme points it is the vacuity of label-free budget audits (Theorem 18). Naming it once is what makes the rest of the paper a single statement rather than three. The configuration it constructs is also *observed*: §VI-F exhibits deployment pairs in the logged CIFAR-10-C and ImageNet-C artifacts whose label-free evidence cannot be told apart while their benefit has resolvably opposite sign, and measures that such pairs are 39× *rarer* than chance — so the evidence is informative and merely insufficient, which is exactly the gap the lemma describes.
- **The exact minimax label-free drift budget, and the identity that closes the loop.** For every declared class $\mathcal{C}$ the least label-free-certifiable budget is the fibre radius $\Gamma_z(\mathcal{C})$, attained by a constant audit that reads none of its data (Theorem 6(a),(b)). For this paper's own deployment class, $\Gamma_z(\mathcal{C}_\beta) = \beta$ exactly (Corollary 4): *the best possible label-free audit of the drift budget returns the budget you declared, and that number is the radius of the band on which the frontier must abstain.* No label-free audit ever converts an abstention into a commitment (Theorem 6(c)), and over the unrestricted class the audited rule's decision yield is at most the false-commitment budget δ, at every batch size, every calibration sample size and every evidence map (Theorem 6(d)).
- **Labels at the wrong domain buy nothing, and the three currencies a budget costs.** A fully labeled calibration sample of any size leaves the minimax budget at $\frac{1}{2} + |M|$ unless a coupling to the deployment domain is separately declared (Theorem 7). The exact decomposition (Proposition 3) shows a budget is the sum of three terms bought in three different

currencies—labels at an anchor, a label-free transport term, and a declared invariance defect—and that the declared term is itself a drift budget of the same type, so declaring ρ instead of β is a renaming and not a reduction. With deployment-domain labels the rate is closed: an $n^{-1/2}$ floor (Theorem 8) matching the paper's own upper bound, and a three-way dichotomy (Theorem 9) separating no-rate, root-n and degenerate regimes.

- **Exact benefit-sign frontier and the β=0 face.** The sign is identifiable over the declared drift-budget class *iff* $|M| > \beta$, with the three-way adapt/freeze/abstain rule as the unique maximal sound certificate; ATC, DoC, GDE, COT, AETTA and agreement-on-the-line are its β=0 face, sound precisely when calibration does not drift.
- **A negative result about our own parameter, and it is the predicted one.** Declaring β as this paper's method section prescribes fails on 6,885 real evaluation cells (§VII): on CIFAR-10-C the declared budget is 1.4× to 50× smaller than soundness requires, 24–73% of deployment cells fall outside the declared class, and committed actions are wrong at 0.4–16.6% where the frontier promises 0%; on ImageNet-C a large enough budget is derivable and returns zero commitments on all 405 cells in five of ten configurations. Theorem 7 says why: the procedure supplies the first term of Proposition 3 and sets the other two to zero by an undeclared exchangeability assumption.
- **Where a budget can come from, priced.** Under retrospective outcome logging and episode exchangeability, β is identified as an episode quantile and estimable by a conformal order statistic (§VIII), with the episode requirement bracketed to a factor of e $(1/(e\alpha) - 1 \le K^\star \le 1/\alpha - 1$; between 3 and 9 logged deployments at $\alpha = 0.10$, first feasible on real data at exactly $K = 9$). The escape is one logical move—average over the episode law instead of taking a supremum over the fibre, paid for with historical labels—and neither ingredient suffices alone (Proposition 4). The assumption that buys it is measured and it fails on the axis that defines a new deployment: against a nominal 0.900, coverage is 0.905 on an exchangeable control, 0.626 when the corruption family changes and 0.454 when severity changes; weighted conformal restores coverage to 1.000 by driving decision yield to 0.000.
- **Split-observability (honest generality).** We isolate the structural property our proofs actually use—a benefit functional is *split-observable* if $\mathrm{sign}(\Delta) = \mathrm{sign}(M + \gamma)$ with M observable and $|\gamma| \le \beta$—and show the impossibility, finite-n floor, certificate, and minimax/necessity results hold for *any* split-observable functional, with TTA (via the disagreement-region reduction) as the demonstrated instance. The $|M| > \beta$ criterion and the one-bit count are stated for split-observable functionals, not for arbitrary functionals.
- **A finite-sample certificate, with its frontier measured rather than asserted.** The certificate controls false adaptation and false freezing under risk-aligned evidence. On the CIFAR-10-C stress grid it commits on 66.8% of 6,480 cells at zero false adaptations over 1,113 and 1,244 ADAPT decisions, cuts regret 5.00× below the better fixed policy and 3.1× below a hindsight-tuned label-free drift heuristic held to $\mathrm{FA_u} \le 0.10$, and places its commitments on cells whose true effect is 24.5× larger than the ones it declines (§X-A); under leave-one-corruption-out calibration the radius inflates 4.3×–6.9× and regret worsens 3.4×–7.6× while the false-adapt count stays at zero in all ten runs. Where the shift is one-sided it buys nothing and we report that: on ImageNet-R it is 1.76× worse than always-adapt.
- We validate on a multi-regime panel with bootstrap confidence intervals. **Headline:** no-harm on the locked held-out natural shifts Office-Home, iWildCam, Camelyon17, and RxRx1 under their declared protocols; completed PACS and ImageNet-R runs remain null diagnostics. **Confirmatory:** beats-both on synthetic stress grids (CIFAR-10-C, ImageNet-C SAR) and a pre-registered mixed head-to-head win over POEM and AETTA (Table XV). Controlled synthetic validations and an additional breadth dataset (PACS) are reported in Appendix 0a, with CIFAR-10.1 retained in-body as a supporting low-margin check (§X-C). A separately constructed cross-protocol aggregate uses one pooled benefit model and out-of-fold radius and is reported only as a routing construction (§X-E); it is not a universal deployment result and does not change any per-dataset verdict.

Two bars, stated once. Throughout, the **pre-registered headline bar** is a point-estimate beats-both at false-adapt $\le \alpha$; the **CI-robust** criterion (both bootstrap regret-gap CIs exclude zero) is a strictly stronger layer we report as a separate column. Under a valid out-of-fold radius, no natural shift clears either bar: Office-Home and iWildCam beat always-adapt with CIs excluding zero but *tie* always-freeze, so both are no-harm, not wins. The only beats-both rows are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR).

K-Bound does not claim universal improvement, and it does not claim no-harm across the whole panel. **The headline deployment claim is no-harm on the four one-sided natural tracks with locked held-out artifacts** — Camelyon17, iWildCam, Office-Home and RxRx1 — where KGA matches the better fixed policy and beats the worse one at zero observed false adaptation. On the domain-generalization and rendition tracks it is a conservative null that *loses* to always-adapt (PACS by 2.45×; ImageNet-R on 7 of 10 backbones), and CIFAR-10.1 fails the declared transfer bar outright; see the qualification at §I and Table I. Where the claim does hold, the mechanism is the obvious

TABLE I

Retained-outcome evaluation panel: nine datasets under declared protocols. Regret-to-oracle (lower is better) for always-freeze, always-adapt, and KGA; oracle regret is 0 by construction. **Headline rows** are held-out natural shifts. No-harm holds on the four one-sided tracks with locked held-out artifacts (Office-Home, iWildCam, Camelyon17, RxRx1); it does *not* hold uniformly — PACS, in this very block, loses to always-adapt by 2.45× (0.0431 versus 0.0176). **Confirmatory rows** are identifiable synthetic/mixed regimes. **Pre-reg WIN** applies the single headline bar uniformly: held-out test scored once, false-adapt $\leq \alpha$=0.10, *and* $\text{regret}_{\text{KGA}} < \text{regret}_{\text{ADAPT}}$ *and* $< \text{regret}_{\text{FREEZE}}$ (point estimate). **CI-robust** is the strictly stronger layer (`statistical_policy_v1.md`): both regret-gap 95% bootstrap CIs exclude zero. Under the declared protocols the only promoted beats-both rows are synthetic stress grids (CIFAR-10-C, ImageNet-C SAR); locked natural-shift results are no-harm or diagnostic. KGA regret values are reported under the stated out-of-fold radius; earlier in-sample-radius Office-Home/iWildCam "wins" were a calibration bug and are corrected to no-harm (§X-D). The seven core datasets plus a supporting null (CIFAR-10.1) and a domain-generalization suite (PACS) are all shown; heuristic / AETTA / ATC gate comparisons appear in App. 0a (Table XXVI).

Dataset (protocol)	regime	always-freeze	always-adapt	**KGA**	false-adapt	pre-reg WIN
Headline: natural / held-out shifts						
Office-Home (Prot. M v2)	covariate, freeze-favorable	0.01582	0.04681	0.01570	$0 \leq \alpha$	no[b]
iWildCam (Prot. H v2)	natural, freeze-favorable	0.00410	0.10283	0.00367	$0 \leq \alpha$	no[b]
Camelyon17 (Prot. G)	natural, adapt-favorable	0.1381	0.0000	0.0000	$0 \leq \alpha$	no[c]
RxRx1 (Prot. J)	natural, harmful-detectable	0.0000	0.2531	0.0000	$0 \leq \alpha$	no[d]
PACS (3-seed diagnostic)	domain generalization	0.0446	0.0176	0.0431	0.009	no[g]
Confirmatory: identifiable synthetic/mixed						
CIFAR-10-C stress (Tent/EATA)	synthetic, harmful-detectable	0.1232/0.1311	0.0086/0.0037	**0.0016/0.0015**	$0 \leq \alpha$	**yes**
ImageNet-C (SAR)	synthetic, harmful-detectable	0.0319	0.0625	**0.0108**	$0 \leq \alpha$	**yes**
Diagnostic (not headline)						
ImageNet-R (Prot. D)	natural, weak evidence	0.0325	0.0064	0.0112[e]	1/480	no
CIFAR-10.1 (Prot. K)	natural, low-margin	0.00171	0.01902	0.00206	$0.444 > \alpha$	no[f]

[a]Synthetic corruption. The full-scale re-run (2026-07-09; 27 cells = 3 noise corruptions × severities $\{1, 3, 5\}$ × regimes) is now *CI-confirmed* beats-both: gap to always-freeze $[-0.032, -0.011]$, Holm-corrected across the wave (p=0.0003); $\text{FA}_{\text{u}} = 0$; SAR harmful on 14.8% of cells. This supersedes the earlier 36-cell point-only mechanism-faithful configuration (Table XXVII), retained for provenance. [b]No-harm under a valid out-of-fold radius: beats always-adapt with CI excluding zero (Office-Home +0.031 $[0.004, 0.062]$; iWildCam +0.099 $[0.080, 0.118]$) but *ties* always-freeze (Office-Home +0.0001 $[0, 0.0003]$; iWildCam exactly +0.0000 $[0, 0]$). The earlier in-sample-radius beats-both was a calibration bug; corrected here (§X-D). [c]On held-out OOD test (hospital domain, n=18) online EATA is helpful-dominated (0% harmful), so KGA adapts and *ties* always-adapt. The previously reported pooled Protocol-G beats-both (regret 3.6×10^{-5}, FA 2.6%, n=54) mixed in-distribution `id_val` cells into the held-out set; on genuine OOD it vanishes (§X-D). [d]KGA freezes all 360 conditions, *tying* the freeze oracle; adapting is catastrophic (0.253). No-harm, not beats-both. [e]Mean across ten backbones and four seeds; 0/10 candidates pass both Holm-corrected comparisons (App. 0a). [f]$\text{regret}_{\text{KGA}} >$ always-freeze and FA$>\alpha$ cross-seed. [g]PACS row is the descriptive mean over four leave-one-domain-out targets and three seeds (216 decisions). The reported mean false-adapt rate is 0.0093; a raw pooled numerator was not retained, so no binomial interval is reconstructed. No target earns a multiseed beats-both claim. Per-domain means are in Table XXVIII.

one: when adapting is already near-oracle, a certificate can only tie or slightly trail always-adapt; when freezing is already optimal, it should match freeze rather than force an unnecessary update. Its distinctive value—and the confirmatory wins on synthetic grids—appear when harmful adaptation is frequent, detectable, and costly, the regime where blind adaptation is most dangerous.

Scope of the guarantee. K-Bound does not guarantee that a newly encountered deployment is exchangeable with the calibration data, that label-free evidence is risk-aligned, or that a fitted benefit estimator transfers. Its guarantee is conditional: *if* the declared benefit interval attains marginal target coverage, *then* strict adaptation on $\widehat{\Delta} - \varepsilon > 0$ controls the unconditional false-adapt event at level α, and strict freezing on $\widehat{\Delta} + \varepsilon < 0$ the false-freeze event. Deployment diagnostics can identify support shift, instability, leakage and insufficient effective sample size, but passing them does not prove coverage. When required checks fail or cannot be evaluated, the prescribed output is ABSTAIN or FREEZE — not a certified adaptation decision.

II. Related work and positioning

We position K-Bound as a precise delta over five adjacent lines of work. In each case the distinction is the one the theory formalizes: between *improving* an unlabeled objective and *certifying the sign* of its benefit before acting.

A. Test-time adaptation methods

Entropy-based TTA updates a frozen model on unlabeled target data. **Tent** [1] minimizes prediction entropy over the batch-norm affine parameters; **SHOT** [2] combines information maximization with pseudo-labeling for source-free adaptation; **TTT** [3] trains a self-supervised auxiliary task and updates it at test time; **EATA** [4] adds entropy filtering of unreliable samples plus a Fisher penalty against forgetting; **SAR** [5] adds sharpness-aware updates with a reset that aborts incipient collapse; **CoTTA** [6] stabilizes continual adaptation with teacher-student ensembling; and **MEMO** [7] adapts via test-time augmentation and entropy minimization (see [66] for a recent survey). These methods make adaptation *more robust*, and several detect or suppress instability, but all commit to adapting: none emits a *pre-adaptation* certificate separating helpful, harmful, and unknowable evidence, and none can *abstain* on the adapt/freeze decision itself. K-Bound is mechanism-agnostic—it *wraps* any such candidate (we evaluate Tent/EATA/SAR). Tempora [12] further shows that unconstrained offline rankings can invert under latency and compute pressure, with no fixed policy robust across temporal scenarios. K-Bound is complementary: rather than selecting one always-on method, it certifies per-condition ADAPT/FREEZE/ABSTAIN decisions.

B. Protected and guarded adaptation

A second line guards adaptation *during* the update. **POEM** [11] protects TTA by online entropy matching—a betting-style monitor that halts or dampens a single failure mode—and **Schirmer et al.** [9] run anytime-valid risk monitors over the TTA stream, raising a sequential alarm when adaptation degrades a proxy risk; both descend from sequential harmful-shift tracking for deployed models (**Podkopaev & Ramdas** [10]). **PeTTA** [73] senses drift toward collapse and adjusts the adaptation process on the fly, and sample-reliability gates such as **DeYO** [74] decide *which samples* are safe to adapt on. These are the nearest neighbours of our certificate, and the delta is precise: they act *reactively, during or after* the update and control a false-*alarm* rate in one failure direction, whereas K-Bound decides *before* committing, poses the *three-way* decision with an explicit *unknowable* region (where abstention is mandatory, Cor. 1), controls the false-*adapt* rate on the benefit sign itself, and proves the matching impossibility floor (Prop. 1). The two are complementary: a pre-adaptation certificate can gate what a runtime monitor then tracks.

C. Label-free accuracy and error estimation

A third line predicts target *performance* without labels: **ATC** [13] thresholds softmax confidence calibrated on source; **Agreement-on-the-Line** [16], [15] exploits the linear correlation between in- and out-of-distribution model *agreement* to extrapolate accuracy; **AETTA** [18] estimates TTA accuracy from dropout-prediction disagreement. Crucially, Garg et al. [13] also establish the governing limit for this entire line: identifying OOD accuracy is *as hard as* identifying the optimal predictor, so any label-free accuracy estimate is sound only under an assumption on the shift. Our frontier *organizes* these estimators rather than competing with them: ATC, DoC, GDE, COT, AETTA, and agreement-on-the-line are all the zero-budget (β=0) face of the benefit-sign frontier (Thm. 3(iv))—each is sound precisely when calibration does not drift (γ=0), and each silently fails exactly where our budget is positive. Closest in spirit, **Kim et al.** [17] use agreement-on-the-line to decide *when* TTA helps and to select TTA hyperparameters without labels—a genuine decision use of an accuracy estimate, but a plug-in one on that same β=0 face, sound only when calibration transfers, with no finite-sample false-adapt control and no abstain region. Estimating $R_T(f)$—or detecting that TTA has failed *post hoc*—is strictly different from certifying $\operatorname{sign}\big(R_T(f_0) - R_T(f_a)\big)$ *before* adapting with a controlled false-adapt rate, the object K-Bound bounds.

D. Impossibility, abstention under shift, and identifiability

The general impossibility our work specializes is *owned by prior work*, and we cite it as such. **Garg et al.** [13] prove that identifying out-of-distribution accuracy from unlabeled target data is as hard as identifying the optimal predictor—so point identification provably requires an assumption on the shift—and **Kalai & Kanade** [14] give the matching decision theory, characterizing the optimal abstain-versus-predict rule under shift via an observable-quantity-versus-shift-budget threshold with a minimax certificate. These are the sources of both the impossibility and the abstention-under-budget decision we build on; K-Bound does not re-derive them but *specializes* them to the sign of the adaptation benefit and makes the boundary exact. This hardness is also visible across unsupervised risk estimation. **Steinhardt & Liang** [21] estimate risk without labels only under a conditional-independence structure; **Ben-David et al.** [22], [23] bound target risk by source risk plus an $\mathcal{H}$-divergence; **Lipton et al.** [24] (BBSE) correct label shift under an invertible confusion matrix; **Rosenfeld & Garg** [25] bound target error under an explicit, unverifiable assumption (their Remark 3.9 states informally the same obstruction our Theorem 4 makes exact for the benefit sign); the rank-one agreement algebra underlying our Theorem 4 is classical [26], [27], [29], [30], its dependent-classifier diagnostics appear in [28], and label-free algebraic evaluation with consistency alarms is developed in the NTQR program [31]. Each shows risk is identifiable only under structural assumptions and impossible in general—the landscape our Theorem 1(ii) specializes and makes quantitative for the benefit sign. We weaken the target from a *risk* to the *sign of a difference* of risks, which is identifiable on a strictly larger set (the observable disagreement region, Lemma 1).

E. Selective prediction, conformal, and anytime-valid inference

Selective prediction [37] abstains on individual *predictions*; we instead abstain on the *adaptation decision*. Our finite-sample certificate reuses split-conformal prediction (**Vovk et al.** [38]; **Angelopoulos & Bates** [39]) to build the radius ε, and its streaming form (Appendix A) is a testing-by-betting e-process in the safe-anytime-valid-inference tradition (**Shafer** [40]; **Ramdas et al.** [41]); confidence-sequence and concentration tools (**Howard et al.** [43], **Maurer & Pontil** [46], **Hoeffding** [47]); and Vovk's conformal test martingales [42] embody, for exchangeability, the falsify-not-verify stance our budget audit (Thm. 4(iv)) takes for the drift budget. Because deployment breaks exchangeability, we connect to conformal-under-shift and risk control: weighted conformal corrects coverage under known covariate shift [67], adaptive conformal tracks coverage online [69], and beyond exchangeability the coverage gap is bounded by the total-variation drift of the calibration pairs [68]—the population counterpart of our declared budget β. Our marginal false-adapt guarantee sits in the risk-control lineage (RCPS [71]; Learn-then-Test [72]), specialized to the adapt/freeze/abstain loss, and PAC prediction sets give set-valued analogues under covariate shift [70].

The surviving gap. No prior method provides a single adapt/freeze/abstain certificate with an *explicit, proven* unknowable region and a false-adapt guarantee. Table II makes the gap precise. There is a second gap, and it is the one this paper is organized around. Every line of work above that gives a guarantee under shift—weighted and adaptive conformal, the beyond-exchangeability coverage gap, the abstention-under-budget rule—prices its guarantee against a declared shift or drift quantity, and none asks where that declaration comes from at a deployment with no target labels. §VI asks it and answers it: the minimax label-free value of that quantity is a constant equal to the declaration itself. That is why the negative result of §VII is reported here rather than repaired, and why §VIII moves the question to a population of deployments instead of trying to answer it at one.

F. Benchmarks, code, and reproducibility

TTA progress is benchmarked on standardized corruption suites—CIFAR-10/100-C and ImageNet-C [51], plus ImageNet-R [52]—and on cross-domain shift suites including WILDS/Camelyon17/RxRx1/Office-Home [53], [54], [55], [56]. The field increasingly emphasizes standardized, executable evaluation (**RobustBench** [58]; **DomainBed** [57]) and reproducibility checks on fresh test draws such as CIFAR-10.1 [59]. Most TTA methods ship official code, but evaluations vary in batch regime, stream composition, and update budget—*precisely* the axis along which adaptation flips between helpful and harmful. We therefore release K-Bound as a *pre-registered, executable* artifact: a public repository with pinned environments, a one-command reproduction harness, auto-downloaded standard corruption data, fixed seeds, machine-checked theorem validators (kernel-checked generalization bounds exist in proof assistants [75], though not, to our knowledge, conformal coverage), and the full stratified result manifests behind every table and figure (§XIV). Our certificate builds on split-conformal prediction (Vovk et al.; Angelopoulos & Bates) and, for the streaming variant, testing-by-betting e-processes (Shafer; Ramdas et al.). The CAN benchmark formalizes a practical "adapt-or-skip" decision in chronological streams [8]; K-Bound complements that benchmark view with a theorem-level frontier and impossibility account for when such decisions are certifiable from label-free evidence.

III. Problem setup

Source $P_S(X,Y)$ has labels at train/validation; target $P_T(X,Y)$ exposes only $P_T(X)$ at test time. Fix a loss ℓ, a frozen model f_0, and an adapted/gated candidate f_a. Target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)]$. The *adaptation benefit* is

$$\Delta \;=\; R_T(f_0) - R_T(f_a), \qquad \Delta > 0 \iff \text{adapting helps.} \tag{1}$$

Observable evidence $Z = \phi(X_{1:m}, f_0, f_a)$ is any statistic computable *without* target labels (entropy, confidence, score-distribution drift, detector disagreement, predicted-class balance, update norm, density-ratio diagnostics). The conditional benefit is $\Delta(z) = \mathbb{E}[\ell(f_0(X),Y) - \ell(f_a(X),Y) \mid Z = z]$. The central question is when $\operatorname{sign}\Delta(z)$ is recoverable from Z alone.

a) Notation.: f_0 is the frozen model and f_a the adapted/gated candidate; $\Delta = R_T(f_0) - R_T(f_a)$ is the population adaptation benefit and $\Delta(z)$ its value conditioned on evidence $Z{=}z$; $\widehat{\Delta}$ is a label-free estimator of Δ with confidence radius ε at miscoverage level α; $g(Z) \in \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ is the decision rule; and $D = \{x : f_0(x) \neq f_a(x)\}$ is the observable disagreement region. *Regret-to-oracle* denotes $R(g) - R(g^\star)$ against the per-condition oracle $g^\star = \mathbf{1}[\Delta > 0]$.

IV. Definitions

Source law $P_S(X,Y)$ supplies labels at train/validation; target law $P_T(X,Y)$ exposes only $P_T(X)$ at deployment. Fix loss ℓ, frozen predictor f_0, candidate f_a, and target measure μ_T on inputs. Write target risk $R_T(f) = \mathbb{E}_{P_T}[\ell(f(X),Y)]$ and the *adaptation benefit*

$$\Delta \;:=\; R_T(f_0) - R_T(f_a) \qquad (\Delta > 0 \;\Leftrightarrow\; \text{adapting helps}).$$

The *disagreement region* $D := \{x : f_0(x) \neq f_a(x)\}$ is observable from (f_0, f_a).

Assumption 1 (Deployment setup)**.** Unless stated otherwise: binary label $Y \in \{0,1\}$, 0/1 loss ℓ, and $\mu_T(D) > 0$. Fix a source-calibrated score $s : \mathcal{X} \to [0,1]$ estimating the candidate's *pointwise correctness* (e.g. a Platt-scaled confidence that f_a is right at x), and write the target pointwise correctness $\eta_a(x) := \mathbb{P}_T\big(f_a(X){=}Y \mid X{=}x\big)$ on D, so that $\mathbb{E}_{\mu_T}[\eta_a \mid D] = \bar{a}$. Define the *observable margin*

$$M \;:=\; \mathbb{E}_{\mu_T}[s(X) \mid D] - \tfrac{1}{2},$$

the *calibration drift* (unobserved at deploy)

$$\gamma \;:=\; \mathbb{E}_{\mu_T}[\eta_a(X) - s(X) \mid D],$$

and label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ for any measurable ϕ using only unlabeled batches and model outputs (entropy, confidence, drift, disagreement, update norms). A *label-free rule* is any measurable map $g : \mathcal{Z} \to \{\textsc{adapt}, \textsc{freeze}, \textsc{abstain}\}$ that does not use target labels on the deployment batch.

Remark 1 (γ is a residual, so $M + \gamma = \bar{a} - \frac{1}{2}$ is an identity)**.** The drift γ is *defined* as $\bar{a} - \frac{1}{2} - M$: it is whatever is left of the disagreement-region accuracy after the observable part M is removed. Consequently $M + \gamma = \bar{a} - \frac{1}{2}$ by construction, and the reduction of Lemma 1 is a

Work	label-free?	adapts?	predicts benefit *before* adapting?	abstains on the decision?	false-adapt guarantee?
Tent	yes	yes	no	no	no
SHOT	yes	yes	no	no	no
EATA	yes	yes	partial (filtering)	no	no
SAR	yes	yes	stability only	no	no
AETTA	yes	monitors	post-hoc estimate	—	no
KGA (ours)	yes	wrapper	**yes**	**yes**	**yes**

TABLE II

POSITIONING. PRIOR TTA ADAPTS AND (SOME) DETECT FAILURE; NONE GIVES A *PRE-ADAPTATION* ADAPT/FREEZE/ABSTAIN CERTIFICATE WITH A FALSE-ADAPT BOUND.

bookkeeping identity rather than a derived fact about the shift. We state this explicitly because it determines what the frontier of Theorem 3 does and does not assert. Sufficiency ($|M| > \beta \Rightarrow \operatorname{sign}\Delta = \operatorname{sign} M$) is then interval arithmetic: the interval $[M-\beta, M+\beta]$ has a determined sign exactly when $|M| > \beta$. The mathematical content sits in the *necessity* clause, which requires exhibiting two admissible target laws that induce the same evidence law and have opposite nonzero benefits (Theorem 1), and in the audit-vacuity theorem (Aud-A), which shows the budget β that makes the decomposition operational cannot itself be audited from label-free data. What the decomposition buys is organizational, and that is not nothing: it names, isolates, and prices the one unobservable quantity a deployment must bound.

Definition 1 (Risk alignment)**.** Evidence Z is *risk-aligned on a class* $\mathcal{P}$ if no two laws $P_T^1, P_T^2 \in \mathcal{P}$ with $\operatorname{sign}\Delta_1 \neq \operatorname{sign}\Delta_2$ satisfy $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$.

All frontier claims stated in terms of M additionally require M to be determined on each evidence-law fibre; equivalently, take the population evidence to be $\widetilde{Z} = (Z, M)$. Without this requirement, $|M| > \beta$ proves sign stability *given* M, not identifiability from $\mathrm{Law}(Z)$ alone. Fitting a benefit regressor does not certify risk alignment: support, dependence, and residual diagnostics may falsify a proposed evidence map but cannot verify the assumption uniformly over the declared deployment class.

Definition 2 (Regimes)**.** At evidence z: *knowably helpful* if the population frontier certifies $\Delta > 0$ (ADAPT); *knowably harmful* if it certifies $\Delta < 0$ (FREEZE); *unsupported for strict commitment* if the same augmented-evidence law is compatible with distinct signs, including zero versus a strict sign (ABSTAIN).

For a supplied drift budget $\beta \geq 0$, the *declared deployment class* is

$$\mathcal{C}_\beta \;:=\; \{P_T : \; |\gamma(P_T)| \leq \beta\}.$$

Identifiability statements below are *over* $\mathcal{C}_\beta$, not over arbitrary shifts.

Definition 3 (Strict directional soundness and maximality)**.** Fix an augmented-evidence law z and let $\mathcal{C}_\beta(z)$ be the laws in $\mathcal{C}_\beta$ that induce z. A strict action $a \in \{\text{ADAPT}, \text{FREEZE}\}$ is *uniformly directionally sound at* z when

$$\begin{aligned} a = \text{ADAPT} \;&\Rightarrow\; \Delta(P) > 0 \quad \text{for every } P \in \mathcal{C}_\beta(z), \\ a = \text{FREEZE} \;&\Rightarrow\; \Delta(P) < 0 \quad \text{for every } P \in \mathcal{C}_\beta(z). \end{aligned}$$

A sound three-way rule is *pointwise maximal* if it commits whenever some strict action is uniformly directionally sound and abstains otherwise. Zero benefit blocks a strict directional claim even though adapt and freeze have equal task risk when $\Delta = 0$; this is an epistemic validity convention, stronger than zero regret at the boundary.

Remark 2 (Four quantities)**.** Do not confuse M (observable margin, deploy-time), γ (realized drift, latent), β (declared bound $|\gamma| \leq \beta$ on the deployment class), and ε (finite-sample radius from held-out residuals). The radius ε operationalizes coverage; it is *not* an estimate of β.

Symbol	Meaning	At deploy?	Role
M	observable evidence margin	yes	evidence toward adapt/freeze
γ	realized calibration drift	no	can reverse the benefit sign
β	declared bound on $\lvert\gamma\rvert$	externally supplied	population frontier
ε	residual quantile radius	estimated pre-deploy	finite-sample certificate

What "label-free" means. No target/test labels at deploy; labeled source and held-out validation may calibrate $\widehat{\Delta}$ and ε. Evidence Z uses only unlabeled batches, predictions, entropy, drift, and update statistics. The method is **target-label-free**, not label-free of all supervision: labels may be used only in the source/validation calibration split, never in the deployment batch or held-out test decision.

K-Bound in one sentence. A strict adapt/freeze commitment is uniformly supportable only when the population evidence margin exceeds the declared drift budget; otherwise abstention is the maximal sound three-way action.

V. Theory: one lemma, three radii, and a budget that cannot be supplied

What we specialize, and how general it is. The general fact that deciding adaptation label-free needs an assumption on the shift is due to Garg et al. [13] and Kalai & Kanade [14]; we do not re-derive it. What follows *specializes* that landscape to the benefit sign and makes its boundary exact. The structural property our proofs actually use is *split-observability*: a benefit functional is split-observable if $\operatorname{sign}(\Delta) = \operatorname{sign}(M + \gamma)$ with M an observable margin and an unobservable calibration term $|\gamma| \le \beta$ bounded by the declared drift budget. The impossibility, finite-n floor, certificate, and minimax/necessity results below are world-agnostic—they hold for *any* split-observable functional—and test-time adaptation is the canonical instance via the disagreement-region reduction (Lemma 1). The sharper statements—the $|M| > \beta$ identifiability criterion and the exactly-one-bit supplement—are established for split-observable functionals, with TTA as the demonstrated case; we do not claim them for arbitrary functionals.

The body below expands the core spine into five load-bearing pillars. *Reproducibility:* numeric checks cited as "validated" are in the public `theory_v2/` scripts; they audit the algebra, not replace the written proofs.

Lemma 1 (Disagreement-region reduction)**.** *Under Assumption 1, with $\bar{a} := \mathbb{P}_T(f_a{=}Y \mid D)$,*

$$\Delta = 2\,\mu_T(D)\big(\bar{a} - \tfrac{1}{2}\big),$$
$$\operatorname{sign}\Delta = \operatorname{sign}(\bar{a} - \tfrac{1}{2}) = \operatorname{sign}(M + \gamma).$$

For K-class $0/1$ loss, $\Delta = \mu_T(D)(p_a - p_0)$ on D; for squared loss the sign reduces to one moment on D (manuscript App. C).

Proof. On D exactly one predictor is correct under 0/1 loss, giving $\Delta = 2\mu_T(D)(\bar{a} - \frac{1}{2})$. Linear expectation yields $\bar{a} - \frac{1}{2} = M + \gamma$. Multiclass and regression extensions are algebraic identities on D. ☐

Lemma 2 (Plug-in regret identity)**.** *For any estimator $\hat{\Delta}$ and gate $\hat{g} = \mathbf{1}[\hat{\Delta} > 0]$,*

$$R(\hat{g}) - R(g^\star) = \mathbb{E}\big[|\Delta(Z)|\,\mathbf{1}\{\operatorname{sign}\hat{\Delta} \ne \operatorname{sign}\Delta\}\big],$$
$$g^\star = \mathbf{1}[\Delta > 0].$$

On $\{|\Delta| < \varepsilon\}$, committal regret is at most ε.

Proof. Wrong-sign commits pay $|\Delta|$ pointwise; averaging gives the identity. On the low-margin band, $|\Delta| \le \varepsilon$ bounds regret. ☐

Proposition 1 (Finite-sample minimax regret floor)**.** *Let $\{P_\theta\}_{\theta\in\{-1,+1\}}$ be two target worlds with $|\Delta(P_\theta)| \equiv \Lambda > 0$, $\operatorname{sign}\Delta(P_\theta) = \theta$, and let $Q_\theta^{(n)}$ be the law of any label-free evidence from n unlabeled observations. Then for every label-free committal gate $\hat{g}$ and every finite n,*

$$\begin{aligned}\inf_{\hat{g}} \max_{\theta=\pm1} &\big[R_{P_\theta}(\hat{g}) - R_{P_\theta}(g^\star)\big] \\ &\ge \tfrac{\Lambda}{2}\Big(1 - \mathrm{TV}\big(Q_{-1}^{(n)}, Q_{+1}^{(n)}\big)\Big) \\ &\ge \tfrac{\Lambda}{4}\, e^{-\mathrm{KL}(Q_{-1}^{(n)} \| Q_{+1}^{(n)})}.\end{aligned}$$

A Gaussian two-point family with $d_n = c/\sqrt{n}$ realizes a floor constant in n (`val_thm2_lecam_finite_n.py`).

Proof. Lemma 2 with constant $|\Delta| = \Lambda$ reduces regret to the mixed committal error $M(\hat{g})$. Le Cam's testing affinity gives $\inf_{\hat{g}} M(\hat{g}) = 1 - \mathrm{TV}$; Bretagnolle–Huber yields the KL bound. ☐

Theorem 1 (Matched evidence impossibility)**.** *Fix $\alpha \in (0, \frac{1}{2})$.*

(i) ***Witness.*** *There exist P_T^1, P_T^2 with $\mathrm{Law}(Z \mid P_T^1) = \mathrm{Law}(Z \mid P_T^2)$, $\operatorname{sign}\Delta_1 = -\operatorname{sign}\Delta_2 \ne 0$ (e.g. $X \sim \mathcal{N}(0,1)$, $f_a = 1 - f_0$, opposite label rules on D).*

(ii) ***Le Cam floor.*** *For committal rules, $\inf_g M(g) = 1 - \mathrm{TV}(P_{T,Z}^{1,(n)}, P_{T,Z}^{2,(n)})$, equal to 1 in the witness for all n.*

(iii) ***Forced abstention.*** *Any rule with false-adapt and false-freeze $\le \alpha$ must abstain on matched evidence with probability $\ge 1 - 2\alpha$.*

Proof. (i) Construct worlds with common X-marginal and Z a function of X only, but opposite benefit signs on D. (ii) A committal rule is a test between the two evidence laws; apply Le Cam. (iii) Matched laws force identical action probabilities; each commit is wrong in one world. ☐

Corollary 1 (Abstention is mandatory, not conservative)**.** *In the $|M| \le \beta$ wedge of Theorem 3, Theorem 1(iii) applies: any label-free rule with false-adapt and false-freeze $\le \alpha$ must abstain with probability at least $1 - 2\alpha$ on matched evidence.*

Theorem 2 (Finite-sample certificate)**.** *Under Assumption 1, if calibration and deployment are exchangeable and $\mathbb{P}[|\widehat{\Delta} - \Delta| \le \varepsilon] \ge 1 - \alpha$, then the three-way rule (*ADAPT *if $\widehat{\Delta} - \varepsilon > 0$,* FREEZE *if $\widehat{\Delta} + \varepsilon < 0$, else* ABSTAIN*) has marginal false-adapt and false-freeze probabilities each at most α.*

Proof. On the coverage event $\{|\widehat{\Delta} - \Delta| \le \varepsilon\}$, the condition $\widehat{\Delta} - \varepsilon > 0$ implies $\Delta > 0$; a false adapt requires the complement, which has probability at most α. The freeze case is symmetric. ☐

Proposition 2 (Certificate sample complexity)**.** *If $\widehat{\Delta}$ is an empirical-Bernstein mean of n bounded paired benefits with variance proxy σ^2 and range $b - a$, then one-sided certification ($\varepsilon < |\Delta|$) requires $n = \tilde{O}(\sigma^2/\Delta^2)$; two-sided sign certification ($\varepsilon < |\Delta|/2$) requires $n = \tilde{O}(\sigma^2/\Delta^2)$ with a larger constant. These rates are minimax order-optimal (appendix; `val_minimax_optimality.py`).*

Proof. Invert the empirical-Bernstein radius at level α; the $8\sigma^2/\Delta^2$ constant comes from requiring $\varepsilon < |\Delta|/2$. ☐

Remark 3 (Sequential and multicandidate extensions)**.** Theorem 2 extends to anytime-valid streaming (Ville + betting supermartingale; Theorem 14) and family-wise multicandidate routing (Bonferroni; Theorem 15); proofs in Appendix A.

Theorem 3 (Exact benefit-sign frontier)**.** *Under Assumption 1, fix observables determining M and budget $\beta \ge 0$.*

(i) ***Reduction.*** $\operatorname{sign}\Delta = \operatorname{sign}(M+\gamma)$ *for every* P_T *(Lemma 1);* γ *is one-dimensional given the observables.*
(ii) ***Iff identifiability.*** *Over* $\mathcal{C}_\beta$*,* $\operatorname{sign}\Delta$ *is a function of* $\mathrm{Law}(Z)$ *iff* $|M| > \beta$*; then* $\operatorname{sign}\Delta = \operatorname{sign} M$ *and the minimax committal error is* 0*; if* $|M| \le \beta$ *the minimax error is* $\frac{1}{2}$*.*
(iii) ***Necessity geometry.*** *Any identifying class lies on one side of the hyperplane* $\gamma = -M$*; every identifying assumption is a one-sided constraint on* γ*.*

The three-way rule $M > \beta \Rightarrow$ ADAPT, $M < -\beta \Rightarrow$ FREEZE, *else* ABSTAIN, *is the unique maximal sound certificate on* $\mathcal{C}_\beta$*.*

Proof. Same as Theorem 3: sufficiency from $|M| > \beta$ and $|\gamma| \le \beta$; necessity from opposite drifts when $|M| \le \beta$ plus Theorem 1; (iii) is constancy of $\operatorname{sign}(M+\gamma)$ on an identifying class. □

Remark 4 ($\beta = 0$ face: prior label-free heuristics)**.** ATC [13], DoC, GDE [19], COT [20], AETTA [18], and agreement-on-the-line [16], [17] threshold observable surrogates of M and commit to $\operatorname{sign} M$. They are sound on $\mathcal{C}_0$ ($\gamma = 0$) and share the failure mode $\gamma \neq 0$ (Theorem 3(ii) at $\beta = 0$).

Corollary 2 (Knowable regimes)**.** (a) *Under covariate shift with bounded density ratio, importance weighting identifies* $\operatorname{sign}\Delta$ *from labeled source plus unlabeled target (*$\beta = 0$ *on the weighted score; premise not label-free testable).* (b) *Conversely, no label-free certificate identifies* $\operatorname{sign}\Delta$ *without a supplied bound on* $|\gamma|$ *(Theorem 1).*

Theorem 4 (One-bit dichotomy)**.** *Under the CEI panel hypothesis* $\boldsymbol{H}$ *on* D *(conditionally independent correctness indicators; manuscript for definitions):*
(i) ***Rank-one law.*** *Under CEI alone, agreement statistics are rank-one iff every per-class accuracy gap vanishes.*
(ii) ***One global bit.*** *Under H, the evidence law fixes every magnitude* $|b_j|$ *and product* $b_i b_j$ *but not the global orientation* $b \mapsto -b$*; the label complement preserves* $\mathrm{Law}(Z)$ *while negating every benefit sign.*
(iii) ***No evidence-definable bracketing.*** *A swap involution on* D *fixes all label-free observable laws and negates* Δ*; hence no testable assumption class identifies* $\operatorname{sign}\Delta$ *without a bit-selector; the minimal untestable supplement is one bit.*
(iv) ***Auditable budgets.*** *The bit-robust test of* $|\gamma| \le \beta$ *is level-*α *with power beyond* $2r_n$*; verification fails on an explicit blind set.*

Conjecture 1 (unconditional label-free bracketing) is resolved negatively by (iii).

Proof sketch. (i) Algebraic expansion of agreements. (ii)–(iii) Complement and swap involution on D; observables span the even subalgebra while $\operatorname{sign}\Delta$ is odd. (iv) Empirical-Bernstein testing plus flip-minimization.
Proof status. This is a sketch and no full proof of this theorem appears in this manuscript; we mark it as such rather than let a reader assume one is available. No claim in the impossibility spine of §VI–§VIII, and no claim in the experimental sections, depends on it. The repository files `theory_validation/val_conj1_*.py` are *numerical validations* of the stated identities on synthetic instances; a validator is not a proof, and they are cited here as evidence that the statement is not mis-transcribed, not as evidence that it is proved. □

Conjecture 1 (Label-free bracketing — **resolved**)**.** *Unconditional label-free bracketing of the ordinal comparison on* D *does not exist (Theorem 4(iii)). The minimal supplement is one declared bit; weakest one-bit classes are characterized in Appendix G (Theorems 16, 17).*

Theorem 5 (Evidence-channel rate)**.** *Under audited* H *with margin lower bounds,* m *unlabeled samples on* D *estimate agreement statistics at rate* $m^{-1/2}$ *with constant* $\Theta(1/(b_{\min} c_{\min}^2))$*; labeled paired benefits share the same exponent with efficiency ratio* $\Theta(1/c_{\min})$*. Labels buy constants and the missing bit, not the rate (*`val_ev_rate.py`*).*

Proof sketch. Hoeffding on agreements, product-ratio perturbation, and a tensorized χ^2/KL bound between pattern laws differing in one coordinate; Bretagnolle–Huber finishes. *Proof status.* As for Theorem 4: a sketch, with no full proof in this manuscript, and `val_ev_rate.py` is a numerical validation rather than a proof. Nothing in the impossibility spine or the experiments depends on it. □

Legacy cross-references. Lemma 1 is the disagreement-region characterization formerly labeled `thm:disagree`; Lemma 2 is `thm:gate`; Proposition 1 is `thm:imp-quant`; Corollary 2 is `thm:pos`.
Where the rest of the paper goes. Theorem 3 commits exactly when $|M| > \beta$, and β is a declared input. §VI settles where that input can come from and proves that it cannot come from the deployment; §VII measures our own attempt to supply it anyway and reports the failure the theorem predicts; §VIII identifies the one structure under which a budget is estimable, prices it in logged deployments and historical labels, and measures the assumption that buys it. Two verified extensions of the batch certificate—an anytime-valid sequential form (Theorem 14) and a family-wise multicandidate form (Theorem 15)—are stated in Appendix A and proved in the companion technical report; no claim in this paper depends on them.

VI. The drift budget cannot come from the deployment: the exact minimax label-free budget

The population frontier of Theorem 3 commits exactly when $|M| > \beta$, and β is a *declared* input. This section settles what a procedure can do about that. The answer is sharper than "label-free audits are vacuous" (Theorem 18): for *every* declared class, the best label-free budget is a constant that ignores the data, that constant is exactly the radius of the class's own abstention band, and any attempt to buy it with labels at a domain other than the deployment domain buys nothing unless a transfer constraint is separately declared. The escape is a matched pair — labels at an anchor, plus a declared coupling — and it carries a $n^{-1/2}$ floor that no structural assumption removes.

Throughout, Assumption 1 is in force: $Y \in \{0,1\}$, 0/1 loss, $\mu_T(D) > 0$, $s : \mathcal{X} \to [0,1]$ a fixed *source*-calibrated score, $u := \eta_a - s \in [-1,1]$ on D, $\gamma = \mathbb{E}_{\mu_T}[u \mid D] = \bar{a} - \frac{1}{2} - M$, and $M = \mathbb{E}_{\mu_T}[s \mid D] - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$.

A. The engine: one lemma, used three times

Definition 4 (Audit data)**.** The *audit data* W is the totality of what a deployment-time procedure may read: the unlabeled deployment batch $X_{1:m} \sim \mu_T$, the fixed maps (f_0, f_a, s) and every label-free statistic $Z = \phi(X_{1:m}, f_0, f_a)$ derived from them, together with any sample S — labeled or not — drawn from a source or calibration law that is specified independently of the target label kernel. An *audit* is a measurable $\widehat{\beta} = g(W, U) \geq 0$ where U is a randomisation seed independent of everything else. W contains *no target labels*. This is exactly the "target-label-free" convention boxed in Assumption 1.

Lemma 3 (Kernel freedom on the disagreement region)**.** *Fix the observables $(\mu_T, P_S, f_0, f_a, s)$, hence D and M, and fix any off-D conditional label law K°. For every measurable $\eta : D \to [0,1]$ there is a target law P_T^η on $\mathcal{X} \times \{0,1\}$ with input marginal μ_T, off-D kernel K°, and*

$$\mathbb{P}_{P_T^\eta}\big(f_a(X) = Y \mid X = x\big) = \eta(x) \quad \textit{for } x \in D.$$

Moreover $\mathrm{Law}(W \mid P_T^\eta)$ *does not depend on η, and*

$$\bar{a}(P_T^\eta) = \mathbb{E}_{\mu_T}[\eta \mid D],$$
$$\gamma(P_T^\eta) = \mathbb{E}_{\mu_T}[\eta \mid D] - \tfrac{1}{2} - M,$$

so that as η ranges over all of $[0,1]^D$, γ ranges over exactly $\big[-\frac{1}{2} - M,\ \frac{1}{2} - M\big]$, an interval of length 1 containing 0 in its interior whenever $|M| < \frac{1}{2}$.

Proof. Existence. On D we have $f_0(x) \neq f_a(x)$ and Y is binary, so $\{f_0(x), f_a(x)\} = \{0,1\}$; setting $K^\eta(x, f_a(x)) = \eta(x)$ and $K^\eta(x, f_0(x)) = 1 - \eta(x)$ therefore defines a valid Bernoulli conditional law on $\{0,1\}$ for each $x \in D$, measurable in x because η, f_0, f_a are. Off D take K°. Then $P_T^\eta(dx, dy) := \mu_T(dx)K(x, dy)$ is a probability law with the stated marginal and pointwise correctness.

Invariance of the audit data. By Definition 4, W is a function of $X_{1:m}$, of the fixed maps (f_0, f_a, s), of S, and of U. The construction changes neither μ_T (so $\mathrm{Law}(X_{1:m})$ is fixed), nor the maps, nor the law generating S, nor U. Hence $\mathrm{Law}(W \mid P_T^\eta) = \mathrm{Law}(W)$ for every η.

Range. $\bar{a} = \mathbb{E}_{\mu_T}[\eta \mid D]$ by definition of $\bar{a}$, and $\gamma = \bar{a} - \frac{1}{2} - M$ by Remark 1. Taking $\eta \equiv c$ for $c \in [0,1]$ gives $\bar{a} = c$, so $\bar{a}$ attains every value in $[0,1]$ and γ every value in $[-\frac{1}{2} - M, \frac{1}{2} - M]$; no larger range is possible since $\bar{a} \in [0,1]$ always. Since $\mu_T(D) > 0$ the conditional expectation is well defined. ☐

Remark 5 (The three uses are one construction)**.** Lemma 3 restricted to the two-point subfamily $\eta \equiv \frac{1}{2} \pm \delta$ with $\delta < \min\{\beta - |M|, \frac{1}{2}\}$ *is* the proof of the matched-evidence impossibility (Theorem 1). The same lemma at the two *extreme* points $\eta \equiv 1$ and $\eta \equiv 0$ *is* the proof of audit vacuity (Theorem 18). Everything in this section is the same one-parameter family $\{\eta \equiv c\}_{c \in [0,1]}$ read at a different radius. In particular the construction that forces abstention and the construction that forbids auditing the budget which decides abstention are literally the same object; Corollary 4 makes this an equality of numbers, not an analogy. No total-variation trade-off is available to sharpen or weaken it: within a fibre the observable laws coincide *exactly*, which is the degenerate — and strongest — case of a two-point argument, so Le Cam's method has nothing to add here. It reappears, genuinely, only in Theorem 8, where labels make the two laws distinguishable.

B. The exact minimax label-free budget

Definition 5 (Fibre and fibre radius)**.** Let $\mathcal{C}$ be a declared class of target laws and let z denote a value of $\mathrm{Law}(W)$. The *fibre* of $\mathcal{C}$ at z is $\mathcal{C}(z) := \{P \in \mathcal{C} : \mathrm{Law}(W \mid P) = z\}$, assumed nonempty, and the *fibre radius* is

$$\Gamma_z(\mathcal{C}) \ := \ \sup_{P \in \mathcal{C}(z)} |\gamma(P)| \ \in \ \big[0, \tfrac{1}{2} + |M|\big].$$

An audit $\widehat{\beta}$ is *δ-valid on $\mathcal{C}(z)$* if $\mathbb{P}_P(|\gamma(P)| \leq \widehat{\beta}) \geq 1 - \delta$ for *every $P \in \mathcal{C}(z)$*.

Theorem 6 (Label-free budget audits are declarations)**.** *Fix z with $\mathcal{C}(z) \neq \emptyset$ and $\delta \in [0,1)$.*

(a) ***Floor.*** *Every δ-valid audit satisfies $\mathbb{P}\big(\widehat{\beta} \geq \Gamma_z(\mathcal{C})\big) \geq 1 - \delta$, and this probability is the same under every $P \in \mathcal{C}(z)$.*

(b) ***Attainment.*** *The constant audit $\widehat{\beta} \equiv \Gamma_z(\mathcal{C})$ is 0-valid. Hence $\inf\{\operatorname{ess\,sup} \widehat{\beta} : \widehat{\beta}$ 0-valid$\} = \Gamma_z(\mathcal{C})$, and the infimum is attained by an audit that reads none of its data.*

(c) ***Decision inertness.*** *On the event of (a), the audited rule (ADAPT iff $M > \widehat{\beta}$, FREEZE iff $M < -\widehat{\beta}$, else ABSTAIN) commits only where the* a priori *rule with declared budget $\Gamma_z(\mathcal{C})$ commits. No label-free audit ever converts an abstention into a commitment relative to that declaration.*

(d) ***Yield is bounded by the error budget.*** *If $|M| \leq \Gamma_z(\mathcal{C})$ then $\mathbb{P}[$ the audited rule commits$] \leq \delta$.*

Proof. By Lemma 3 and Definition 5, $\mathrm{Law}(W \mid P) = z$ for every $P \in \mathcal{C}(z)$; since $\widehat{\beta} = g(W, U)$ with U independent, $\mathrm{Law}(\widehat{\beta})$ is one and the same law ν under every $P \in \mathcal{C}(z)$. Write $\mathbb{P}_\nu$ for it.

(a) Fix $P \in \mathcal{C}(z)$. Its drift $\gamma(P)$ is a deterministic number, so δ-validity reads $\mathbb{P}_\nu(\widehat{\beta} \geq |\gamma(P)|) \geq 1 - \delta$. Taking the supremum over the fibre: for every $c < \Gamma_z(\mathcal{C})$ there is

$P \in \mathcal{C}(z)$ with $|\gamma(P)| > c$, whence $\mathbb{P}_\nu(\widehat{\beta} \ge c) \ge \mathbb{P}_\nu(\widehat{\beta} \ge |\gamma(P)|) \ge 1-\delta$. Now $\{\widehat{\beta} \ge \Gamma_z\} = \bigcap_{n\ge1}\{\widehat{\beta} \ge \Gamma_z - 1/n\}$ is a decreasing intersection of events each of probability $\ge 1-\delta$, so $\mathbb{P}_\nu(\widehat{\beta} \ge \Gamma_z) \ge 1-\delta$ by continuity from above. (No attainment of the supremum is needed.)

(b) Immediate from the definition of Γ_z: $|\gamma(P)| \le \Gamma_z$ for every $P \in \mathcal{C}(z)$, deterministically. Combined with (a), no 0-valid audit has essential supremum below Γ_z, and the constant achieves it.

(c) On $\{\widehat{\beta} \ge \Gamma_z\}$, $|M| > \widehat{\beta}$ implies $|M| > \Gamma_z$.

(d) A commitment requires $|M| > \widehat{\beta}$; if $|M| \le \Gamma_z$ this forces $\widehat{\beta} < \Gamma_z$, an event of probability at most δ by (a). ☐

Corollary 3 (Aud-A, recovered and sharpened)**.** *Let $\mathcal{C}^{\mathrm{all}}$ be the class of all target laws matching the observables. Then $\Gamma_z(\mathcal{C}^{\mathrm{all}}) = \frac{1}{2} + |M|$, so every δ-valid label-free audit has $\widehat{\beta} \ge \frac{1}{2} + |M| > |M|$ with probability $\ge 1-\delta$; the audited frontier's decision yield is at most δ, for every deployment batch size m, every calibration sample size, and every choice of evidence map ϕ.*

Proof. Lemma 3 gives the achievable drift set $[-\frac{1}{2}-M, \frac{1}{2}-M]$, whose largest absolute value is $\max\{|\frac{1}{2}-M|, |\frac{1}{2}+M|\} = \frac{1}{2}+|M|$ because $|M| \le \frac{1}{2}$. Apply Theorem 6(a) and (d), using $\frac{1}{2}+|M| > |M|$. ☐

Corollary 4 (The least auditable budget is the declared budget)**.** *For the declared deployment class $\mathcal{C}_\beta = \{P : |\gamma(P)| \le \beta\}$ with $0 \le \beta \le \frac{1}{2}$, the fibre radius is $\Gamma_z(\mathcal{C}_\beta) = \beta$ exactly. Consequently:*

(i) *the smallest budget any label-free audit can certify over $\mathcal{C}_\beta$ is β itself, so the best possible audit returns its own input; and*

(ii) *the fibre radius is simultaneously the abstention radius: strict commitment is uniformly directionally sound at z if and only if $|M| > \Gamma_z(\mathcal{C}_\beta)$, which is Theorem 3 (iv).*

Proof. By Lemma 3 the achievable drifts inside $\mathcal{C}_\beta$ are $G = [-\beta,\beta] \cap [-\frac{1}{2}-M, \frac{1}{2}-M] = [-\min(\beta, \frac{1}{2}+M),\ \min(\beta, \frac{1}{2}-M)]$. Since $|M| \le \frac{1}{2}$ and $\beta \le \frac{1}{2}$ we have $\beta \le \frac{1}{2}+|M|$, so $\Gamma_z = \max\{\min(\beta, \frac{1}{2}-M), \min(\beta, \frac{1}{2}+M)\} = \min(\beta, \frac{1}{2}+|M|) = \beta$. Part (i) is Theorem 6(a)–(b). For (ii), $\mathrm{sign}(M+\gamma)$ is constant and nonzero over $\gamma \in G$ iff $0 \notin [M + \inf G,\ M + \sup G]$; for $M > 0$ this is $\min(\beta, \frac{1}{2}+M) < M$, i.e. $\beta < M$ (as $M \le \frac{1}{2} < \frac{1}{2}+M$), and symmetrically for $M < 0$; for $M = 0$ it fails for every $\beta \ge 0$. So the condition is $|M| > \beta$, which by the first part is $|M| > \Gamma_z(\mathcal{C}_\beta)$; the equivalence with uniform directional soundness is Theorem 3 (iv). ☐

Corollary 4 is the precise sense in which the two halves of the paper are one statement. *The number that bounds the region on which the frontier must abstain, and the number that is the best a label-free audit of the budget can ever return, are the same number.* Auditing β from unlabeled deployment data is not merely hard; it is the identity map on the declaration.

Remark 6 (The only seam, named in advance)**.** Theorem 6 says no budget valid *uniformly over the fibre* beats Γ_z, and §VIII nonetheless exhibits a budget computed from a history that is independent of the current episode's label kernel. These are consistent, and the consistency is not incidental: Proposition 4 localises it to a single logical move. Theorem 6 quantifies with a *supremum over the fibre*; Theorem 11 quantifies with an *average over the episode law*, and the fibre's extreme worlds—the ones that force Γ_z—carry episode probability at most α. The episode budget is therefore not fibre-uniformly valid, and Proposition 4(a) states that explicitly rather than leaving it implicit. Neither ingredient works alone: averaging without retrospective labels identifies nothing (Theorem 10(b)), and labels without an exchangeable population return us to this section. Every guarantee downstream of that move is marginal over deployments, not conditional on one (Remark 12).

C. Labels at the wrong domain buy nothing

Theorem 18 is often read as "labels fix it". They do not, unless they are labels of the deployment domain or a coupling to it is declared.

Theorem 7 (Anchor destruction)**.** *Suppose the audit additionally observes a* fully labeled *sample $S_{\mathrm{cal}} = (X_i, Y_i)_{i \le n_{\mathrm{lab}}}$ from a calibration law P_{cal}, and that the declared class $\mathcal{C}$ imposes no constraint relating the deployment drift function u_T to u_{cal} (formally: $\mathcal{C}$ is a product, $\mathcal{C} = \mathcal{C}_{\mathrm{cal}} \times \mathcal{C}_T$ with $\mathcal{C}_T$ containing, for each of its members, the whole family $\{P_T^\eta\}_{\eta\in[0,1]^D}$ of Lemma 3). Then $\Gamma_z(\mathcal{C}) = \frac{1}{2} + |M|$ and Theorem 6 applies verbatim: the labeled calibration sample changes the minimax budget by exactly zero, for every n_{lab}.*

Proof. S_{cal} is part of W (Definition 4) and its law is specified by P_{cal}, which the fibre construction of Lemma 3 leaves untouched: the construction alters only the target label kernel on D. Hence $\mathrm{Law}(W)$ is constant on the fibre, and the fibre still contains $\{P_T^\eta\}_{\eta\in[0,1]^D}$, so $\Gamma_z = \frac{1}{2} + |M|$ as in Corollary 3. ☐

Proposition 3 (Three-term decomposition: what each ingredient buys)**.** *Let $\mathcal{F}$ be a symmetric class of functions $D \to \mathbb{R}$ with $u_{\mathrm{cal}} \in \mathcal{F}$, let $\Delta_{\mathcal{F}}(P,Q) = \sup_{f\in\mathcal{F}} |Pf - Qf|$, and put $\rho := \mathbb{E}_{\mu_T}\big[\, |u_T - u_{\mathrm{cal}}| \ \big|\ D\big]$. Then, exactly,*

$$|\gamma_T| \le \underbrace{|\gamma_{\mathrm{cal}}|}_{\text{needs labels at the anchor}} + \underbrace{\Delta_{\mathcal{F}}(\mu_{\mathrm{cal}}, \mu_T)}_{\text{label-free computable}} + \underbrace{\rho}_{\text{declared invariance defect}} .$$

Every audit in Appendix G — Aud-B, Aud-C, Aud-F, Aud-G, Aud-H — is an instance of this decomposition with ρ set to 0 by declaration and the middle term estimated in a class-specific way; none of them omits the first term.

Proof. $\gamma_T = \mathbb{E}_{\mu_T}[u_T \mid D] = \mathbb{E}_{\mu_T}[u_{\mathrm{cal}} \mid D] + \mathbb{E}_{\mu_T}[u_T - u_{\mathrm{cal}} \mid D]$. The second summand is bounded in absolute value by ρ by Jensen. For the first, $\mathbb{E}_{\mu_T}[u_{\text{cal}} \mid D] = \gamma_{\text{cal}} + (\mathbb{E}_{\mu_T} - \mathbb{E}_{\mu_{\text{cal}}})[u_{\text{cal}} \mid D]$ and, since $u_{\text{cal}} \in \mathcal{F} = -\mathcal{F}$, the bracket is at most $\Delta_{\mathcal{F}}(\mu_{\text{cal}}, \mu_T)$ in absolute value. Triangle inequality. $\square$

Remark 7 (Reading of Proposition 3)**.** The three terms are bought by three different currencies and no currency substitutes for another. Only the middle term is label-free computable, and Theorem 6 applied to the first and third says why: both are fibre quantities. The third term ρ is a drift budget of exactly the same type as β — apply Lemma 3 with u replaced by $u_T - u_{\text{cal}}$, which is equally unconstrained by W, to get $\Gamma_z = \frac{1}{2} + |M|$ for ρ as well. Declaring ρ instead of β is therefore a renaming, not a reduction: *the budget-declaration problem is a fixed point of the audit construction.* This is not a defect of the paper's audits; it is the reason they must declare (Lip), (Idx), the witness class, or (G1)–(G3), and the reason those declarations are falsifiable but never verifiable.

D. The escape, and its floor

Theorem 8 (Root-n floor with deployment labels)**.** *Consider the most favourable labeled setting: the audit observes $n \geq 2$ i.i.d. correctness indicators $\mathbf{1}\{f_a(X_i) = Y_i\}$ with $X_i \sim \mu_T(\cdot \mid D)$, i.e. labels at the deployment domain itself, so that $\rho = 0$ and $\Delta_{\mathcal{F}} = 0$ in Proposition 3. Let $\mathcal{U}$ be the declared class of drift functions on D and let $\tau(\mathcal{U}) := \sup\{|c| : \text{the constant function } c \in \mathcal{U}\}$. Suppose $\widehat{\beta}$ is δ-valid on $\mathcal{C}_{\mathcal{U}}$ with $\delta \leq \frac{1}{6}$ and is κ-*useful*, meaning $\mathbb{P}_{P_0}(\widehat{\beta} \leq \kappa) \geq \frac{2}{3}$ at a law P_0 with $\mu_T = \mu_{\text{cal}}$ and $\gamma = 0$. Then*

$$\kappa \;\geq\; \min\Big\{\tau(\mathcal{U}),\ \tfrac{1}{4},\ \tfrac{1}{2\sqrt{2n}}\Big\}.$$

In particular, whenever nonzero constant drifts are admissible at scale $\tau(\mathcal{U}) \geq (2\sqrt{2n})^{-1}$, no valid audit is useful at any rate faster than $n^{-1/2}$; matched from above by Theorem 19, whose width is $\sqrt{2\ln(2/\delta)/n}$.

Proof. Take $s \equiv \frac{1}{2}$ on D, so $M = 0$, and for $t \in [0, \frac{1}{4}]$ let P_t be the law of Lemma 3 with $\eta \equiv \frac{1}{2} + t$; then $u_t \equiv t$ is the constant t, $P_t \in \mathcal{C}_{\mathcal{U}}$ whenever $t \leq \tau(\mathcal{U})$, and $\gamma(P_t) = t$, while P_0 is the null law with $\gamma = 0$. The audit's data are n i.i.d. Bernoulli$(\frac{1}{2} + t)$ variables together with label-free quantities whose law does not depend on t (Lemma 3); write Q_t for the induced law of the audit's input and $A := \{\widehat{\beta} \geq t\}$.

Suppose, for contradiction, that $\kappa < t$ for some $t \leq \min\{\tau(\mathcal{U}), \frac{1}{4}\}$ with $\mathrm{TV}(Q_0, Q_t) < \frac{1}{2}$. Validity at P_t gives $Q_t(A) \geq 1 - \delta \geq \frac{5}{6}$; usefulness at P_0 gives $Q_0(A) \leq Q_0(\widehat{\beta} > \kappa) \leq \frac{1}{3}$. Hence $\mathrm{TV}(Q_0, Q_t) \geq Q_t(A) - Q_0(A) \geq \frac{5}{6} - \frac{1}{3} = \frac{1}{2}$, a contradiction. Therefore $\kappa \geq t$ for every such t.

It remains to bound TV. The t-dependent part of Q_t is a product of n Bernoulli laws, so by the tensorisation of KL, then $\mathrm{KL} \leq \chi^2$, then Pinsker,

$$\begin{aligned}\mathrm{TV}(Q_0, Q_t) &\leq \sqrt{\tfrac{1}{2}\mathrm{KL}(Q_t \| Q_0)} \\ &= \sqrt{\tfrac{n}{2}\,\mathrm{KL}\big(\mathrm{Ber}(\tfrac{1}{2}{+}t) \,\|\, \mathrm{Ber}(\tfrac{1}{2})\big)} \\ &\leq \sqrt{\tfrac{n}{2} \cdot \tfrac{t^2}{\frac{1}{2}(1-\frac{1}{2})}} = t\sqrt{2n},\end{aligned}$$

using $\chi^2(\mathrm{Ber}(p) \| \mathrm{Ber}(q)) = (p-q)^2/(q(1-q))$ with $q = \frac{1}{2}$. Thus $\mathrm{TV} < \frac{1}{2}$ whenever $t < (2\sqrt{2n})^{-1}$. Taking the supremum of admissible t over $t < \min\{\tau(\mathcal{U}), \frac{1}{4}, (2\sqrt{2n})^{-1}\}$ gives the claim. (The conditions of Pinsker's and the $\mathrm{KL} \leq \chi^2$ inequalities hold: both laws are mutually absolutely continuous for $t < \frac{1}{2}$.) $\square$

Theorem 9 (Dichotomy)**.** *Fix a declared drift class $\mathcal{U}$ and the observables at z. Exactly one of the following regimes obtains for a δ-valid audit.*

(N) No deployment labels, no declared coupling. *By Theorems 6 and 7 the minimax budget is $\Gamma_z(\mathcal{C})$, a constant that does not depend on the deployment batch size m, on the calibration sample size n_{lab}, or on the evidence map.* ***There is no rate.*** *If moreover $|M| \leq \Gamma_z(\mathcal{C})$ — automatic for the unrestricted class, where $\Gamma_z = \frac{1}{2} + |M|$ — the decision yield is at most δ.*

(P) Labels at an anchor, plus a declared coupling $(\mathcal{F}, \rho)$. *Proposition 3 with Theorems 19/23 gives a valid $\widehat{\beta} = |\widehat{\gamma}_{\text{cal}}| + O\big(\sqrt{\ln(1/\delta)/n_{\text{lab}}}\big) + \widehat{\Delta}_{\mathcal{F}} + O\big(\sqrt{(v + \ln(1/\delta))/n}\big) + \rho$, and by Theorem 8 the $n_{\text{lab}}^{-1/2}$ term is unimprovable whenever $\tau(\mathcal{U}) \geq (2\sqrt{2n_{\text{lab}}})^{-1}$. The irreducible offset is ρ, which is declared, not estimated.*

(D) Degenerate declaration. *If $\tau(\mathcal{U}) < (2\sqrt{2n_{\text{lab}}})^{-1}$ the class already pins the drift scale more tightly than n_{lab} labels can resolve; the "audit" is then a restatement of the class declaration and the labels are inert.*

Proof. (N) is Theorem 6(a),(d) together with Theorem 7; the sample-size independence is the statement that Γ_z is defined from $\mathcal{C}(z)$ alone. (P) is Proposition 3 composed with the cited upper bounds, and Theorem 8 for the lower bound. (D) is the second branch of the minimum in Theorem 8. The three cases are exhaustive and mutually exclusive by the trichotomy on (i) presence of deployment-domain labels and (ii) the comparison $\tau(\mathcal{U})$ versus $(2\sqrt{2n_{\text{lab}}})^{-1}$. $\square$

Remark 8 (Where this is *not* sharp)**.** Theorem 9 closes the rate in n_{lab} and closes the labels/no-labels axis exactly. It does *not* close the transfer axis: the $O(\sqrt{v/n})$ complexity term of Aud-H has no matching lower bound here, and whether a rotation-closed class strictly larger than adaptive MSW_1-ridge admits a useful root-n audit remains Conjecture 3. Nor does anything above forbid an audit valid *at the realised P_T only*; all statements require uniform validity over the declared class, which is the standard — and the only checkable — requirement.

E. Correspondence with the measured record

Remark 9 (What the theorems predict about the β-sweep, and what they do not)**.** The β-sweep (repository artifact `experiments/kbound/frontier_sweep_v1/beta_sweep_results.json`) is *not* a label-free audit: its budget $\widehat{\beta} = q_{0.90}(|\gamma|)$ is computed on labeled source-like development cells. Corollary 3 therefore does not apply to it, and we do not claim it as a confirmation. What applies is Theorem 7 with Proposition 3: the procedure supplies the first term and *sets the other two to zero by an undeclared exchangeability assumption between development and deployment cells.* The theory then predicts undercoverage of unbounded size, and that is what is measured.

- *Coverage.* $\widehat{\beta}$ is a 0.90 quantile on development cells, so $\mathbb{P}(|\gamma| \leq \widehat{\beta})$ has a by-construction null of 0.90 on the deployment cells whenever the two pools are exchangeable (recorded as `coverage_on_dev_cells_BY_CONSTRUCTION` = 0.900 for CIFAR-10-C and 0.8963 for ImageNet-C). CIFAR-10-C measures 0.274–0.756, far below that null: the exchangeability declaration is falsified, so $\rho + \Delta_{\mathcal{F}} > 0$ was set to 0, and the committed error rate is correspondingly 0.4%–16.6% where Theorem 3 promises 0%. ImageNet-C measures 0.904–0.985 in five of its six recorded configurations, at or above the null, which *carries no information*; and its three `loco` rows additionally have `dev_and_target_disjoint=false`. The single ImageNet-C row whose coverage is a genuine measurement is `srclike|M_gbm` at 0.470, and it is also the row with commit-error 0.532 on 62 committed cells (sign match 0.468, 95% CI $[0.340, 0.599]$).
- *Sup versus quantile.* Theorem 6 bounds any valid budget by a *supremum* over the fibre, not a development quantile. Measured on CIFAR-10-C: the maximal realised $|\gamma|$ on deployment cells exceeds the maximal $|\gamma|$ on development cells by factors of 2.4–28.4 across the ten configurations (`gamma_absmax_target` versus `beta_hat_max` in `beta_sweep_results.json`).
- *Anchor destruction, quantitatively.* Theorem 7 says an uninformative anchor leaves only a class-scale constant. Under the global label permutation of `adversarial_ablations.py`, any estimator fit on the permuted benefit column converges to the constant $\mathbb{E}[\Delta]$, so the declaration procedure must return approximately $q_{0.90}(|\Delta - \mathbb{E}\Delta|)$ — a number computable from the raw artifacts with no model fitting. On CIFAR-10-C that prediction is 0.2376; the six measured shuffled budgets are 0.2468–0.2606, ratios 1.04–1.10, against real budgets of 0.0178–0.0482 (inflation 5.2×–14.1×), with zero commitments in 5/5 replicates for the `M_doc` and `M_atc4` channels. On ImageNet-C the prediction is 0.0689 against measured 0.0742–0.1171 (ratios 1.08–1.70; $n = 405$). Script and output: `kb_fixes/beta_impossible/check_anchor_collapse.py`, `anchor_collapse_check.json`.
- *A prediction that fails, stated as such.* On ImageNet-C the anchored budget for the `M_doc` and `M_atc4` channels is *larger* than the anchor-free constant (0.199–0.239 real versus 0.074–0.090 shuffled). Nothing above forbids this: Theorem 6 lower-bounds *valid* audits and says nothing about how far an invalid heuristic may overshoot. The mechanism is visible in the same artifact — difference-of-confidences is anti-predictive on ImageNet-C (AUC 0.351), so its residuals exceed the marginal spread of the benefit. The operational reading is that on that benchmark the anchored declaration is beaten by the trivial constant, and those are exactly the rows with decision yield 0.000.

Remark 10 (Honest scope)**.** Lemma 3 is elementary and the fibre argument of Theorem 6 is the standard observation that an unidentified parameter cannot be estimated, specialised to a fibre on which the observable laws coincide exactly. What is not standard, and what the section is for, is (i) the identification of the exact minimax value $\Gamma_z(\mathcal{C})$ for every declared class rather than an impossibility for one class, (ii) Corollary 4, which shows the abstention radius and the least auditable budget are the same number, (iii) the yield $\leq \delta$ exchange rate of Theorem 6(d), and (iv) Theorem 7, which is strictly stronger than Theorem 18 and is the statement that actually bears on the measured record. A reader who finds (i)–(iv) obvious should nonetheless note that the paper's own appendix presents five "audits" of which four read as though the budget is computed from data; Proposition 3 says precisely which term each of them computes and which two they declare.

F. Is the impossibility configuration observed?

Theorem 1 is proved by constructing two target laws. A referee is entitled to ask whether that configuration is a worst case that practice never visits. We measured it. On the 6,480 logged CIFAR-10-C and 405 logged ImageNet-C deployment cells — the same artifacts that carry §VII and §X-A — we ran two analyses that share no estimator: a per-pair test for deployments that are indistinguishable in label-free evidence but have opposite-signed benefit, and a population-level estimate of the conditional law of sign Δ given evidence at its own noise floor. Throughout, B (realized benefit), a_{oracle}, the oracle action and the regime label are *labels*: they are the quantity whose conditional behaviour is being estimated and are never inputs to any distance, neighbourhood, cross-fitting group or matching rule.

a) The verdict, stated before the numbers.: The impossibility configuration is *observed*, but it is *rare*, and the rarity is the more surprising half. Matched-evidence pairs disagree in benefit sign far *less* often than chance — 39× less on CIFAR-10-C and 1.5× less on ImageNet-C, with permutation p_{below} at the floor of 0.0005 in every specification we ran. Label-free evidence is therefore

TABLE III

Matched-evidence benefit-sign disagreement, observed against a geometry-fixed permutation null. A pair is matched at level q iff its evidence distance is below the q-quantile of the within-law replicate distances of its own two laws; q=0.95 is the formal non-rejection level and q=0.25 is strictly stronger. "Law" rows average replicates and apply a Hotelling test at $\chi^2_9(0.95) = 16.92$. Rates are over pairs whose *both* benefit signs are individually resolvable. Null is 2,000 permutations of the cell-to-benefit assignment within benchmark × method with the matched set held fixed; $p_{\text{BELOW}} = 0.0005$ (the permutation floor) in every row. A random-pair control with no matching gives 0.2723 (CIFAR-10-C) and 0.2232 (ImageNet-C). *Every row is below its null.*

Benchmark	Level	Pairs	Opp. sign	Observed	Null [95% CI]
CIFAR-10-C	cell, q=0.25	30,555	13	0.00043	0.187 [0.176, 0
CIFAR-10-C	cell, q=0.95	252,277	1,194	0.00473	0.185 [0.178, 0
CIFAR-10-C	law	1,172	**0**	0.000	0.164 [0.138, 0
ImageNet-C	cell, q=0.25	867	62	0.0715	0.250 [0.208, 0
ImageNet-C	cell, q=0.95	9,901	1,588	0.1604	0.240 [0.225, 0
ImageNet-C	law	169	**3**	0.0178	0.218 [0.156, 0

highly informative about the sign of the benefit and still not sufficient to determine it, which is exactly the gap Theorem 1 describes and exactly the reason a declared β cannot be replaced by an audit. We report this as a *bound on the ambiguity*, not as a demonstration that ambiguity is common.

b) Existence, at single-deployment and at population resolution.: A *law* is a deployment condition modulo the replicate index; its cells are i.i.d. draws, so within-law scatter *is* the sampling noise of the evidence. Distances are Mahalanobis in the pooled within-law covariance, so d=1 means "one noise standard deviation apart", and two deployments are called matched at level q iff d_{ij} is below the q-quantile of the within-law distances of *their own two laws* — a nonparametric statement that the two are no further apart than two repeats of either one of them. A cell's benefit sign is called resolvable only if $|B| \geq 0.01$, its sign agrees with its law's mean, and the law's mean is significantly nonzero across replicates; the conformal interval on $\widehat{\Delta}$ is deliberately *not* used for this, since it is an interval on the prediction, not on the realized benefit.

The strongest witness is a pair of ImageNet-C TENT deployments, `gaussian_noise|s1|small|aggressive|imbalanced` against `impulse_noise|s3|small|aggressive|iid`, five replicates each. Their evidence is indistinguishable in the strong sense: Hotelling $d^2 = 2.34$ on 9 d.f. ($p = 0.985$), no coordinate discrepancy exceeding 1.01 standard errors, and a 95% upper confidence bound on the noncentrality of 3×10^{-61} — a positive equivalence result rather than a failure to reject. The assumption-free version is sharper still: enumerating all $\binom{10}{5} = 252$ re-splits of the two laws' pooled replicates, 99.2% give a *larger* d^2 than the observed one (median re-split $d^2 = 6.89$), so the two laws' evidence is more similar than a random re-split of their own replicates. Their benefits are -5.33 [-8.11, -2.55] and $+3.50$ [$+2.34$, $+4.66$] accuracy points, with perfect rank separation across replicates (exact Mann–Whitney $U = 0$, $p = 0.0079$, the minimum attainable at 5 versus 5). The mechanism is visible in one line: the two deployments differ by 24 points of base accuracy (0.682 versus 0.441) and their entropy- and confidence-based evidence still cannot be told apart, because a_0 is a label quantity and the audit cannot see it. This is Lemma 3's free label kernel, instantiated.

On CIFAR-10-C the cleanest witness is a same-method, same-corruption, same-severity pair (TENT, `jpeg_compression|s5`) differing only in the stream's class balance: $d = 1.66$ against a q=0.25 pair-specific floor of 2.64, with 97.8% of same-law replicate pairs farther apart, base accuracies 0.7290 and 0.7295, and law-mean benefits $+1.58$ [$+0.82$, $+2.34$] and -1.64 [-2.82, -0.45] accuracy points. KGA abstained on both cells (BLIND zone) — the correct behaviour, and precisely the decision yield the certificate pays for it. On ImageNet-C, 365 of the opposite-sign matched pairs differ *only* in corruption type and/or severity with every deployment knob held fixed, so they are pure distribution-shift witnesses rather than artifacts of varying the protocol.

c) Rarity, and the clean negative.: Table III gives the rates against a permutation null that holds the matched-pair set and the entire evidence geometry fixed and permutes only the cell-to-benefit assignment, so it destroys any evidence-to-sign relation while preserving both marginals. The observed rate is *below* the null everywhere, by a factor of 39 on CIFAR-10-C and 1.5 on ImageNet-C, and a random-pair control with no matching at all returns 0.2723 and 0.2232, consistent with the nulls. We ran the excess-over-chance test the lemma's existence claim does not require, it came out the other way, and the number belongs here as the honest calibration of how rare the failure is: matching on label-free evidence *suppresses* benefit-sign disagreement strongly, which is a statement that Z is informative, not that it is sufficient.

d) The ambiguity is real, certified, and narrow — and its width is β.: The second analysis asks not whether a pair exists but how wide the blind spot is. It first bounds what any frontier rule could extract: the best cross-fitted label-free predictor of sign Δ from the evidence reaches AUC 0.983 and accuracy 0.931 against a 0.740 majority baseline on CIFAR-10-C (and 0.970 / 0.910 held out across corruption families; permuted-label null AUC 0.504). It then fixes a radius at the median within-cell replicate distance — pure evidence noise, since replicates are independent draws of the same condition — and estimates $\mathbb{P}(\operatorname{sign} \Delta = +1 \mid Z \in N)$ in each matched neighbourhood with a cluster bootstrap over distinct deployment conditions, widened to a Clopper–Pearson interval at the cluster count so that neither ambiguity nor determinacy can be certified by a degenerate interval. On CIFAR-10-C, 29.4% of the 4,949 evaluable cells sit in neighbourhoods whose interval lies entirely inside $[0.05, 0.95]$, and 20.2% entirely inside $[0.25, 0.75]$: both signs occur at $\geq 25\%$ among deployments the evidence cannot tell apart, and this survives 135 matched observations drawn from 28 distinct conditions, so it is ambiguity in the conditional law and not in the estimate. Under the same pipeline with the labels permuted and the geometry held fixed the figure is $85.4 \pm 0.9\%$. The fraction moves only between 0.29 and 0.34 across the 25th, 50th and 75th percentile radii and under a strict variant that forbids matching a deployment to the same corruption and severity run by a different adapter. Expressed over *all* 6,480 cells the figure is 22.5%, but that restatement counts every non-evaluable cell as unambiguous; the evaluable subset is itself mildly selected (mean $|\Delta|$ 0.116 evaluable against 0.197 non-evaluable), so 22.5% is a conservative convention rather than a measurement and the claim we make is the 29.4% over evaluable cells.

The width of that ambiguous band is the number the spine predicts. Table IV places three quantities side by side that were computed without reference to one another. The local dispersion of the benefit over evidence-matched deployments, the global residual budget the same artifact needs, and the episode-conformal $\widehat{\beta}$ that Theorem 11 returns from labelled history all land between 1 and 2 accuracy points, agreeing to within a factor of 1.75. Half of the residual benefit dispersion that a declared budget must cover (0.519 of $\operatorname{Var}\gamma$) survives conditioning on the label-free evidence at its own noise floor. That is Corollary 4 as a measurement rather than an identity: the radius of the ambiguity is the budget, and the budget is the radius of the band on which the rule must abstain. It also brackets decision yield — a label-free rule must abstain where the fibre is certified ambiguous and may commit where it is certified determined, giving yield ≤ 0.706 with ≥ 0.543 certifiably committable. The deployed certificate's measured yield with the same benefit model, 0.594–0.657 across the eight split protocols, sits inside that bracket near its top — but this is indicative and not a matched comparison: the bracket is derived from evidence neighbourhoods and the yield is measured under a different splitting protocol, and it is a range of per-protocol means, not of individual splits (0.246–0.987).

e) Where the two analyses agree, where they do not, and which we credit.: They agree on the conclusion — evidence is highly informative about the benefit sign and is not sufficient to determine it — on the rarity, and, independently, on a structural defect in the logged evidence vector recorded below. They disagree on *which benchmark* carries the evidence, and the two disagreements do not have the same status. The pair analysis finds its population-level witnesses only on ImageNet-C and none on CIFAR-10-C; the neighbourhood analysis certifies ambiguity only on CIFAR-10-C and none on ImageNet-C. We credit the CIFAR-10-C zero and discount the ImageNet-C one. The ImageNet-C zero is a coverage limitation: with 81 deployment conditions, only 52.6% of cells have enough distinct matched neighbours to be evaluated at all, and the evaluable subset is selected in exactly the adverse direction (97.7% of evaluable cells have $\Delta > 0$ against 51.0% of the non-evaluable ones), so the method measures the sign-pure half of that artifact and correctly reports it as sign-pure. Its β comparison inherits the same defect (ratio 16.8 against 1.75) and we do not quote it. The CIFAR-10-C zero is a measurement: all 229 of its witness law pairs become separable once ten replicates are averaged.

TABLE IV

Three unrelated routes to the same drift budget on CIFAR-10-C. The first two are local and global dispersions of the benefit residual $\gamma = B - M(Z)$ over evidence-matched deployments with no episode structure; the third is a conformal order statistic over *episodes* with outcomes revealed retrospectively (Theorem 11), which shares neither estimator nor sample construction with the first two. They agree to a factor of 1.75. *The agreement is conditional on the benefit model.* All four rows use the gradient-boosted M; the same episode-conformal estimator run with the alternative confidence-based M on the same 135 splits returns 0.0536, a factor of 4.9 rather than 1.75. The residual $\gamma = B - M(Z)$ is defined relative to M, so a weaker M leaves a larger residual and needs a larger budget; the three routes agree for the M this paper deploys, and we do not claim more.

Quantity	Value
Local fibre radius on B, median over matched neighbourhoods	0.0130
Local fibre radius on γ, median	0.0108
Global $q_{0.9}\lvert\gamma\rvert$ over all cells	0.0155
Episode-conformal $\widehat{\beta}$ (gradient-boosted M), mean over 135 splits	0.0190
Within-fibre share of $\operatorname{Var}\gamma$	0.519
Ratio $\widehat{\beta}$ / local fibre radius	1.75

f) What that structural finding says about the spine.: On CIFAR-10-C the non-identifiability is entirely a finite-sample property of *single* deployments and it dissolves under a population of them; on ImageNet-C — the harder artifact — three pairs survive averaging. That is the spine's own architecture appearing in the data: what escapes the single-deployment impossibility is a population of deployments, which is what Theorem 10 formalises and prices. It also sharpens the caveat on that escape. Averaging replicates of a *fixed* condition is not the same operation as averaging over a population of *distinct* deployments, and on the harder benchmark it does not suffice; the escape route remains the one Theorem 10(b) describes, whose binding resource is historical *labels*, not more unlabeled repetitions.

g) Threats we could not remove, and one disclosure.: "Indistinguishable" means "not distinguishable at ten (CIFAR-10-C) or five (ImageNet-C) replicates". Only the headline pair carries a tight equivalence bound; the other two surviving ImageNet-C pairs ($d^2 = 12.8$ and 15.3, exact re-split 13.5% and 14.3%) are consistent with a real but undetected evidence gap and we do not quote them as population-level witnesses. Screening covered 209,628 CIFAR-10-C and 3,240 ImageNet-C law pairs, so three survivors is unremarkable under multiplicity alone —

which is why the headline pair rests on an exact 252-fold enumeration and an exact Mann–Whitney rather than on a screening p-value. The witnesses are also adapter-specific and disjoint across benchmarks (CIFAR-10-C: EATA 884, TENT 310, SAR 0; ImageNet-C: TENT 1,586, SAR 2, EATA 0), so whatever generates a witness is not a property of the evidence vector alone. Most importantly, this measurement speaks only to *the evidence vector this system logs.* A richer label-free statistic could shrink the band; bounding what *any* label-free statistic can do is the job of Theorem 6 and Corollary 4, and no experiment can do it. Finally, the disclosure both analyses returned independently: the logged evidence vector is *nine*-dimensional, not eleven. Two of its coordinates are *exact* linear functions of the others — `entropy_drop` = `pre_entropy` − `post_entropy` and `pbal_drop` = `pre_pbal` − `post_pbal`, maximum residual exactly 0.0 — so every covariance of the logged Z is exactly singular and two of the audit's eleven coordinates carry no information beyond the other nine. That does not affect any result in this paper, whose estimators are invariant to it, but it is a defect in the evidence map as documented and we record it here.

What the audit appendix is, after this section. Appendix G develops five "auditable" drift budgets. Proposition 3 says exactly what each one is: an instance of the three-term decomposition that purchases the first term with labels at an anchor, estimates the middle term in a class-specific way, and sets the third to zero by declaration. None of them omits the first term, and that is the non-vacuity certificate for this section: Theorem 18 forbids exactly the audits that would omit it. Theorem 19 is the matching upper bound for the root-n floor of Theorem 8, so together they close the rate in the number of anchor labels.

VII. Does the population frontier operationalize? A β sweep (negative result)

§VI says a budget cannot be supplied by a label-free procedure. It does not say what happens when one is supplied badly, and this section measures that on real data. The declaration procedure tested here is the one this paper's own method section prescribes—β as a high quantile of realized drift on source-like development cells—and Theorem 7 with Proposition 3 says exactly what it is: it purchases the first term of the three-term decomposition with labels at a development anchor and sets the transport term and the invariance defect to zero by an *undeclared* exchangeability assumption. The theory therefore predicts under-coverage of unbounded size, and the population rule $|M| \gtrless \pm\beta$ and the finite-sample rule $\widehat{\Delta} \pm \varepsilon$ have until now shared vocabulary rather than evidence: β is a declared deployment parameter and was never numerically supplied to KGA on any benchmark. **The test returns a negative result, and we report it as such.** It is also not a quantile-estimation failure: replacing the plug-in quantile by the conformal order statistic of Theorem 11 moves the budget by a median 0.34% across 470 splits, with a median coverage difference of exactly zero (§VIII).

Because γ is *defined* as a residual, sufficiency is interval arithmetic and the empirical content of the theorem is exactly two claims: (C1) that the label-free margin M carries usable sign information about Δ, and (C2) that a budget declared from data a deployer actually has bounds $|\gamma|$ at deployment. We measure both on 6,885 real evaluation cells (CIFAR-10-C stress grid, 432 conditions × 3 adapters × 5 seeds = 6,480; ImageNet-C, $27 \times 3 \times 5 = 405$), with M estimated *out of fold* from the 11-dimensional label-free evidence vector alone (difference-of-confidences, a four-feature ATC-style score, and a full-evidence gradient-boosted score), under leave-one-corruption-family-out, leave-one-seed-out and source-like-holdout splits, and with β declared as the 0.90 quantile of the realized drift on source-like development cells. An adversarial label-firewall check passes on all six dataset × estimator configurations: under full label permutation the out-of-fold AUC collapses to 0.474–0.501.

(C1) holds but is weaker than it looks: out-of-fold AUC against sign Δ is 0.754/0.875/ 0.974 on CIFAR-10-C, but the base rate is 0.740, so the ATC-style margin's raw sign agreement of 0.808 is only +6.8 points over the constant "always adapt"—and difference-of-confidences is *anti-predictive* on ImageNet-C (AUC 0.351/0.264/0.373 across the three splits). (C2) fails, in both available directions. On CIFAR-10-C the declared $\widehat{\beta}$ is 1.4× to 50× smaller than the smallest budget that would make the committed actions sound; 24–73% of deployment cells fall outside the declared class, and commitments are wrong at 0.4–16.6% where the theorem promises 0%. On ImageNet-C a large enough budget *is* derivable and it is useless: in 5 of 10 configurations the declared budget produces zero commitments on all 405 cells. Of 20 configurations, only 3 achieve the promised zero commit-error rate at nonzero yield—none of them with an ATC-style margin, none on CIFAR-10-C—and the single most adverse row is the honest source-only calibration on ImageNet-C, which commits on 62 cells and gets 53.2% of them wrong while formally certified. Under a genuinely held-out margin, no $\beta > 0$ matches the finite-sample rule on both decision yield and regret in 15 of the 18 configurations swept; at matched yield the population rule's regret ratio is 1.1× to 8.1×. Sweeping $\beta \in [0, 0.30]$ on CIFAR-10-C moves regret by 26× (M_{ATC4}, leave-one-corruption-out) and 192× ($M := \widehat{\Delta}$), and at a single fixed $\widehat{\beta}$ decision yield varies by 21 points (CIFAR-10-C ATC4) to 52 points (ImageNet-C $\widehat{\Delta}$) across adapters and seeds: the budget behaves as a free knob, not a specification.

One positive survives, narrowly. Holding the evidence channel and calibration protocol fixed ($M := \widehat{\Delta}$), β-thresholding beats the conformal interval on CIFAR-10-C at the honestly-derived $\widehat{\beta} = 0.0097$ (yield 0.749 vs. 0.668, regret 0.00081 vs. 0.00150, non-overlapping condition-clustered intervals at k=432)—but the intervals overlap at

k=6 corruption-family clusters, and the mechanism is a yield/soundness trade: it makes 18 sign errors against the empirical rule's 1. Soundness is what the theorem sells and what is given up.

Neither proof is affected: sufficiency is arithmetic and necessity rests on the matched-evidence construction. What is withdrawn is the *operational* reading. The frontier is a statement about a declared class whose antecedent we have now measured and found violated on most of one benchmark; the deployable certificate is the conformal interval, and we do not present $|M| > \beta$ as a rule K-Bound runs. Read against §VI this is not an isolated embarrassment but the predicted instance: Corollary 3 forbids the label-free route to a budget outright, and Theorem 7 says that anchoring the declaration in labeled development cells changes the minimax budget by exactly zero unless a coupling to the deployment domain is declared—which this procedure does not declare. §VIII identifies the structure that would supply the missing declaration, and measures how far these deployment populations are from it. Scripts, raw results and the full 20-row table are in `experiments/kbound/frontier_sweep_v1/`; the corresponding narrowing is recorded in §XII.

VIII. Where a drift budget can come from: episode structure

Theorem 18 says a label-free audit of the drift budget is vacuous. The main text therefore treats β as *declared*. This section asks the complementary question — under what structure is β *estimable*, and at what price — and answers it in one direction: β is a property of a *population of deployments*, not of a deployment. Under exchangeability across deployments plus retrospectively revealed outcomes it is identified and estimable at the conformal rate; without both ingredients it is not identified at all. We then measure, on the K-Bound artifacts, how far real deployment populations are from that structure. The answer is: far enough that the estimator is valid along nuisance axes (fitting seed, adaptation method) and fails along the axis a deployment actually cares about (a corruption family or severity never seen before).

A. The episode model

Definition 6 (Deployment episode)**.** An *episode* e is one deployment: a target law P_e, a frozen model f_0^e, a candidate f_a^e, a source-calibrated score s^e, and the induced disagreement region D_e with $\mu_{P_e}(D_e) > 0$. Write the *observable part* $W_e = (Z_e, M_e)$ — the label-free evidence and margin of Assumption 1 — and the *latent part* $\gamma_e = \bar{a}_e - \frac{1}{2} - M_e$. Put $V_e = (W_e, \gamma_e)$.

Assumption 2 (E1: retrospective outcome logging)**.** For each past episode $e \leq K$, a sample of n units drawn i.i.d. from $\mu_{P_e}(\cdot \mid D_e)$ has its labels revealed *after* that deployment concluded, yielding

$$\widehat{\gamma}_e = \frac{1}{n}\sum_{i=1}^{n}\big(\mathbf{1}\{f_a^e(X_i) = Y_i\} - s^e(X_i)\big) \in [-1, 1].$$

No labels are available for the current episode $K + 1$.

Assumption 3 (E2: episode exchangeability)**.** $(V_1, \ldots, V_{K+1})$ is exchangeable. Write Q for the common marginal law of V.

E1 is an accounting assumption: it says outcomes are eventually observed and logged, which is true of any deployment with a downstream ground truth (delivered claims, resolved tickets, confirmed diagnoses) and false of deployments whose outcomes are never revealed. E2 is a much stronger statement about the deployment pipeline; §VIII-I argues it is the binding constraint and §VIII-J measures its failure.

Definition 7 (Episode budget)**.** For $\alpha \in (0, 1)$, $\beta^\star(\alpha) := \inf\{b \geq 0 :\ Q(|\gamma| \leq b) \geq 1 - \alpha\}$, the $(1 - \alpha)$-quantile of $|\gamma|$ under the episode law.

B. Identification, and what it costs

Theorem 10 (Episode identification and its price)**.** *Under Assumption 3:*

(a) ***(Positive.)*** *$\beta^\star(\alpha)$ is a functional of the episode law of (W, γ), and it is the smallest budget b for which the declared class holds with episode probability at least $1 - \alpha$, i.e. $Q(P \in \mathcal{C}_b) \geq 1 - \alpha$. Consequently the frontier rule of Theorem 3 run at $b = \beta^\star(\alpha)$ has episode-marginal false-strict-commitment probability at most α.*

(b) ***(Negative: the price is historical labels, not historical data.)*** *$\beta^\star(\alpha)$ is* not *a functional of the episode law of W alone. For every $b \in [0, \frac{1}{2}]$ there is an episode law Q_b with the same W-marginal and $\beta^\star_{Q_b}(\alpha) = b$. No quantity of unlabeled historical deployments identifies β.*

Proof. (a) $\beta^\star(\alpha)$ is by construction the $(1 - \alpha)$-quantile of $|\gamma|$ under Q, hence a functional of the (W, γ)-law; and $\{|\gamma| \leq b\} = \{P \in \mathcal{C}_b\}$ by the definition of $\mathcal{C}_b$, so the quantile characterisation is immediate. For the decision consequence, fix the history and condition on $\{|\gamma_{K+1}| \leq b\}$. On that event Theorem 3(ii) applies, so a strict commitment made at margin $|M_{K+1}| > b$ has the correct sign; a false strict commitment therefore requires $\{|\gamma_{K+1}| > b\}$, an event of probability at most α. (This is Theorem 21(D) with the audit event supplied by Q instead of by a concentration bound.)

(b) Fix the W-marginal Q_W. By the kernel construction in the proof of Theorem 18, for any episode with observables (μ, f_0, f_a, s) — hence any value of W — and any $\bar{a} \in [0, 1]$ there is a label kernel on D realising $\bar{a}$ while leaving (Z, M) unchanged. The achievable drift set at matched evidence is $I_M = [-(M + \frac{1}{2}),\ \frac{1}{2} - M]$, an interval of length 1 containing 0 whose endpoints have absolute values $\frac{1}{2} + M$ and $\frac{1}{2} - M$, the larger being $\frac{1}{2} + |M| \geq \frac{1}{2}$. Hence for every $b \in [0, \frac{1}{2}]$ at least one of $+b, -b$ lies in I_M; concretely, set $\theta(W) = +1$ if $b \leq \frac{1}{2} - M$ and $\theta(W) = -1$ otherwise, which is admissible because $b > \frac{1}{2} - M$ forces $M + \frac{1}{2} > 1 - b \geq b$ for $b \leq \frac{1}{2}$.

Define Q_b by drawing $W \sim Q_W$ and realising $\gamma = \theta(W)\, b$ by the corresponding kernel. Then Q_b has W-marginal Q_W, and $|\gamma| = b$ Q_b-almost surely, so $Q_b(|\gamma| \le b') = 0 < 1 - \alpha$ for every $b' < b$ while $Q_b(|\gamma| \le b) = 1$; hence $\beta^\star_{Q_b}(\alpha) = b$ for every $\alpha \in (0, 1)$. Since b was arbitrary in $[0, \frac{1}{2}]$, the W-marginal carries no information about $\beta^\star$. □

C. The estimator

Theorem 11 (Episode-conformal budget)**.** *Assume E1 and E2, with a common labeled size n and the same estimator in every episode. Let $S_e = |\widehat{\gamma}_e|$ for $e \le K$, let $\delta \in (0, \alpha)$, $t(n, \delta) = \sqrt{2 \ln(2/\delta)/n}$, and*

$$j = \lceil (1 - \alpha + \delta)(K + 1) \rceil, \qquad \widehat{\beta}_{\text{epi}} = S_{(j)} + t(n, \delta),$$

feasible exactly when $\alpha \ge \frac{1}{K+1} + \delta$. Then $\mathbb{P}\big(|\gamma_{K+1}| \le \widehat{\beta}_{\text{epi}}\big) \ge 1 - \alpha$, with the probability taken over the joint draw of all $K + 1$ episodes and their label samples. Composed with Theorem 21(D)–(E), the audited frontier makes a false strict commitment with probability at most α (population M) or $\alpha + \delta'$ (empirical $\widehat{M}$ from m unlabeled deployment points).

Proof. Define the counterfactual score $S_{K+1} = |\widehat{\gamma}_{K+1}|$ that a size-n label sample from the current episode would produce. Under E1–E2 the vector $(S_1, \ldots, S_{K+1})$ is exchangeable: $V_{1:K+1}$ is exchangeable, the label samples are conditionally i.i.d. of common size n given the episodes, and S_e is the same measurable functional of (V_e, labels_e) for every e. Split conformal on an exchangeable sequence gives $\mathbb{P}\big(S_{K+1} \le S_{(j)}\big) \ge j/(K + 1) \ge 1 - \alpha + \delta$. Independently, Hoeffding on the n i.i.d. variables $u_i \in [-1, 1]$ gives $\mathbb{P}\big(|\widehat{\gamma}_{K+1} - \gamma_{K+1}| > t(n, \delta)\big) \le \delta$. On the intersection, $|\gamma_{K+1}| \le S_{K+1} + t \le S_{(j)} + t$. A union bound gives failure probability at most $(\alpha - \delta) + \delta = \alpha$. Feasibility is $j \le K$, i.e. $\alpha \ge \frac{1}{K+1} + \delta$. □

Remark 11 (Relation to Theorem 24)**.** Theorem 11 is Theorem 24 with the label-free drift predictor set to $\gamma_{\text{pred}} \equiv 0$ and "domain" read as "episode". We restate it because the specialisation is the object the rest of this section analyses and validates: it needs no auxiliary predictor, so its coverage failures isolate the exchangeability assumption rather than confounding it with predictor misspecification. Aud-G's assumption (G3) — that any predictor be fixed independently of the $K+1$ episodes — is vacuous here.

D. Exactly where Theorem 18 is escaped

This is the load-bearing check. If Theorem 11 did not violate a hypothesis of Theorem 18, one of the two would be false.

Definition 8 (Fibre-blind budget)**.** Fix observables (μ_T, f_0, f_a, s) and let $\mathcal{P}(\mu_T)$ be the matched-evidence fibre of Theorem 18. A random budget B is *fibre-blind* if $\text{Law}(B)$ is the same for every $P \in \mathcal{P}(\mu_T)$. Both a deployment-evidence audit $B = g(Z)$ and any budget computed from a history drawn independently of the current episode are fibre-blind.

Proposition 4 (Two ingredients, neither sufficient alone)**.**

(a) ***Labels alone do not escape Aud-A.*** *If B is fibre-blind and $\mathbb{P}(|\gamma(P)| \le B) \ge 1 - \delta$ for* every *$P \in \mathcal{P}(\mu_T)$, then $B \ge \frac{1}{2} + |M|$ with probability at least $1 - \delta$, and the audited frontier abstains with probability at least $1 - \delta$. In particular $\widehat{\beta}_{\text{epi}}$, which is fibre-blind, does* not *satisfy fibre-uniform validity.*

(b) ***The escape is the average.*** *Theorem 11's guarantee is Q-marginal: the deployment law is itself drawn from Q, and the extreme worlds $\bar{a} \in \{0, 1\}$ that force Aud-A's bound carry Q-probability at most α. The hypothesis Aud-A uses and E2 removes is precisely the* supremum over the fibre*, replaced by an* average *over Q.*

(c) ***The average alone does not escape either.*** *By Theorem 10(b), without historical labels the Q-average budget is as uninformative as the fibre supremum: the W-marginal is consistent with $\beta^\star(\alpha) = b$ for every $b \in [0, \frac{1}{2}]$.*

Proof. (a) Take the extreme world $P^\dagger \in \mathcal{P}(\mu_T)$ with $\bar{a} \in \{0, 1\}$ chosen so that $|\gamma| = \frac{1}{2} + |M|$, which exists by the kernel construction of Theorem 18. Validity at $P^\dagger$ forces $\mathbb{P}(B \ge \frac{1}{2} + |M|) \ge 1 - \delta$; fibre-blindness transfers this statement to every other world in the fibre. Since $|M| < \frac{1}{2} + |M|$, neither branch of $|M| > B$ can fire on that event. (b) and (c) are restatements of Theorem 11 and Theorem 10(b). □

Remark 12 (What the deployer actually gets)**.** Proposition 4(a) is not a technicality. It says the episode budget provides no protection *at* a deployment: conditional on the episode being one of the adversarial ones, the frontier commits and is wrong. What is controlled is the long-run rate across deployments. This is the same marginal-versus-conditional distinction that governs the certificate's own false-adapt guarantee (Theorem 2), and it should be stated to any deployer who is told β was "estimated".

E. How many episodes: a factor-e bracket

Theorem 12 (Episode floor)**.** *Fix $\alpha \in (0, \frac{1}{2})$ and $b \in (0, \frac{1}{2}]$. Consider budgets $\widehat{\beta} = h(\gamma_1, \ldots, \gamma_K)$ computed from K historical drifts observed* without label noise *(the most favourable case), and require validity $\mathbb{P}_Q\big(|\gamma_{K+1}| \le \widehat{\beta}\big) \ge 1 - \alpha$ for every i.i.d. episode law Q. Then either*

$$h(0, \ldots, 0) \ \ge\ b \qquad \textit{or}$$

$$\alpha \ \ge\ \frac{1}{K+1}\Big(1 - \frac{1}{K+1}\Big)^K \ \ge\ \frac{1}{e(K+1)}.$$

Equivalently: a budget that returns anything strictly below b on a drift-free history requires $K \ge \frac{1}{e\alpha} - 1$ episodes.

Proof. For $p \in (0, 1)$ let Q_p be the i.i.d. episode law with $|\gamma_e| = b$ with probability p and $\gamma_e = 0$ otherwise (realisable at fixed observables by the kernel construction of

Theorem 18, as in Theorem 10(b)). Suppose $h(0,\dots,0) < b$. The event $A = \{\gamma_1 = \cdots = \gamma_K = 0\} \cap \{|\gamma_{K+1}| = b\}$ has Q_p-probability $p(1-p)^K$ and on A the budget misses. Hence $\alpha \ge \sup_p p(1-p)^K$. Setting $p = 1/(K+1)$ gives $\alpha \ge \frac{1}{K+1}(1-\frac{1}{K+1})^K$. Finally $K\ln(1-\frac{1}{K+1}) = -K\ln(1+\frac{1}{K}) \ge -1$ since $\ln(1+x) \le x$, so $(1-\frac{1}{K+1})^K \ge e^{-1}$. $\square$

Corollary 5 (Bracket)**.** *Let $K^\star(\alpha)$ be the least number of episodes admitting a non-vacuous level-α episode budget. Then, ignoring the label-noise allowance δ,*

$$\frac{1}{e\alpha} - 1 \;\le\; K^\star(\alpha) \;\le\; \frac{1}{\alpha} - 1.$$

The upper bound is Theorem 11's feasibility condition; the lower bound is Theorem 12. At $\alpha = 0.10$: between 3 and 9 historical deployments.

F. How many labels: the per-episode price, and a control variate

Proposition 5 (Label price per episode)**.** *Write $C = \mathbf{1}\{f_a(X) = Y\}$ and $u = C - s(X) \in [-1,1]$ on D, $\sigma^2 = \mathrm{Var}(u)$. Bernstein's inequality gives, with probability $1-\delta$,*

$$|\widehat{\gamma} - \gamma| \;\le\; \sqrt{\tfrac{2\sigma^2\ln(2/\delta)}{n}} + \tfrac{4\ln(2/\delta)}{3n}.$$

Hence to keep the label-noise inflation in Theorem 11 below $\kappa\beta$ it suffices to take $n \ge \frac{8\sigma^2\ln(2/\delta)}{(\kappa\beta)^2} + \frac{8\ln(2/\delta)}{3\kappa\beta}$ (bound each of the two terms by $\kappa\beta/2$). Moreover, if s is pointwise calibrated on D, i.e. $\mathbb{E}[C \mid X] = s(X)$ μ_T-a.s. on D, then

$$\sigma^2 = \mathbb{E}[s(1-s)] = \mathrm{Var}(C) - \mathrm{Var}(s) \;<\; \mathrm{Var}(C),$$

so estimating the drift γ is strictly cheaper in labels than estimating $\bar{a} - \frac{1}{2}$ directly, by exactly the variance of the calibrated score. The score acts as a control variate.

Proof. The Bernstein statement is standard for $u \in [-1,1]$ (range 2); the sufficient n follows by bounding each term by $\kappa\beta/2$. For the identity, calibration gives $\mathbb{E}[u] = 0$ and $\mathbb{E}[u^2] = \mathbb{E}\big[\mathbb{E}[(C-s)^2 \mid X]\big] = \mathbb{E}[\mathrm{Var}(C \mid X)] = \mathbb{E}[s(1-s)]$, so $\sigma^2 = \mathbb{E}[s(1-s)]$. Also $\mathrm{Var}(C) = \mathbb{E}[C] - \mathbb{E}[C]^2 = \mathbb{E}[s] - \mathbb{E}[s]^2 = \mathbb{E}[s(1-s)] + \big(\mathbb{E}[s^2] - \mathbb{E}[s]^2\big) = \mathbb{E}[s(1-s)] + \mathrm{Var}(s)$. $\square$

G. Why not just buy labels at deployment? The amortisation argument

Proposition 6 (Probe lower bound and break-even)**.**
(a) *Fix $\alpha, \eta \ge 0$ with $2\alpha + \eta < 1$ and $\varepsilon \in (0, \frac{1}{4}]$. Any rule that observes k i.i.d. labeled probes from $\mu_T(\cdot \mid D)$, commits strictly with probability at least $1-\eta$, and has false-adapt and false-freeze probability at most α whenever $|\bar{a} - \frac{1}{2}| = \varepsilon$, must use*

$$k \;\ge\; \frac{3(1-2\alpha-\eta)^2}{16\,\varepsilon^2}.$$

Theorem 19 achieves $O(\varepsilon^{-2}\log(1/\delta))$, so the probe price is $\Theta(\varepsilon^{-2})$ up to the confidence log.

(b) *Over N future deployments the probe route costs Nk labels; the episode route costs Kn labels once and none thereafter. The routes break even at $N^\star = Kn/k$. Taking $\alpha = 0.10$, $K = 9$ (Corollary 5), $n = 8\sigma^2\ln(2/\delta)/\beta^2$ (Proposition 5 with $\kappa = 1$, dropping the lower-order term) and $k = 3(1-2\alpha-\eta)^2/(16\varepsilon^2)$ from part (a),*

$$N^\star \;=\; \frac{Kn}{k} \;=\; \frac{384\,K\,\sigma^2\ln(2/\delta)}{3(1-2\alpha-\eta)^2}\cdot\Big(\frac{\varepsilon}{\beta}\Big)^2,$$

which is $\Theta\big(K\sigma^2\log(1/\delta)\,(\varepsilon/\beta)^2\big)$: the budget route pays for itself after a number of deployments proportional to the squared ratio of the typical margin to the budget. It never pays for itself if $\beta \ll \varepsilon$, i.e. if the budget is so tight that the frontier commits on almost everything anyway.

Proof. (a) Consider the two worlds $\bar{a} = \frac{1}{2} \pm \varepsilon$ realised at matched observables by the kernel construction of Theorem 18; by Lemma 1 they have $\Delta > 0$ and $\Delta < 0$ respectively. Each probe is Bernoulli($\bar{a}$) for the event $\{f_a(X) = Y\}$. In the $+$ world, $\mathbb{P}_+[\textsc{freeze}] \le \alpha$ and $\mathbb{P}_+[\textsc{abstain}] \le \eta$, so $\mathbb{P}_+[\textsc{adapt}] \ge 1-\alpha-\eta$; in the $-$ world $\mathbb{P}_-[\textsc{adapt}] \le \alpha$. Hence $\mathrm{TV}\big(\mathrm{Ber}(\frac{1}{2}+\varepsilon)^{\otimes k}, \mathrm{Ber}(\frac{1}{2}-\varepsilon)^{\otimes k}\big) \ge 1-2\alpha-\eta$. Pinsker and tensorisation give $\mathrm{TV} \le \sqrt{k\,\mathrm{KL}/2}$ with $\mathrm{KL} = 2\varepsilon\ln\frac{1/2+\varepsilon}{1/2-\varepsilon} = 2\varepsilon\cdot 2\,\mathrm{artanh}(2\varepsilon) \le 2\varepsilon\cdot\frac{4\varepsilon}{1-4\varepsilon^2} \le \frac{32\varepsilon^2}{3}$ for $\varepsilon \le \frac{1}{4}$. Thus $1-2\alpha-\eta \le 4\varepsilon\sqrt{k/3}$, which rearranges to the claim. (b) is arithmetic on the two label counts, using $n \asymp \sigma^2\log(1/\delta)/\beta^2$ and $k \asymp \log(1/\alpha)/\varepsilon^2$. $\square$

Remark 13. Proposition 6 is the honest ranking. At a *single* deployment, buying k probe labels dominates estimating a budget: the probe certifies $\mathrm{sign}(M+\gamma)$ directly, whereas the budget route certifies the weaker $|M| > |\gamma|$. The budget is worth estimating only because it is *reused*, and reuse is exactly what E2 licenses. If E2 fails, the budget route loses both its guarantee and its economic rationale.

H. Relaxing exchangeability

Theorem 13 (Graceful degradation)**.** *Let the history be drawn from Q and the current episode from Q'.*

(a) ***Bounded episode-law ratio.*** *If $dQ'/dQ \le \rho$ then the budget of Theorem 11 run at level α satisfies $\mathbb{P}_{Q'}(|\gamma_{K+1}| \le \widehat{\beta}_{\mathrm{epi}}) \ge 1-\rho\alpha$. To retain level α under Q', run the history at level α/ρ, which requires $K \ge \rho/\alpha - 1$ episodes: a ρ-fold mismatch costs a ρ-fold increase in the number of historical deployments.*

(b) ***Episode covariate shift.*** *If $Q'(dw, d\gamma) = Q(d\gamma \mid w)\,Q'(dw)$ — the episode observables shift but the conditional drift law does not — then weighted split conformal with weights $w(W) = dQ'_W/dQ_W$ is exactly valid at level α, and w is estimable from* unlabeled *episode observables alone by a two-sample density-ratio fit.*

Proof. (a) Write $\mathcal{H}$ for the history. By Theorem 11, $\mathbb{E}_{\mathcal{H}}\big[Q(|\gamma| > \widehat{\beta}(\mathcal{H}))\big] \le \alpha$. Since $Q'(A) \le \rho\,Q(A)$ for every measurable A, $\mathbb{P}_{Q'}(\mathrm{miss}) = \mathbb{E}_{\mathcal{H}}\big[Q'(|\gamma| > \widehat{\beta})\big] \le \rho\,\mathbb{E}_{\mathcal{H}}\big[Q(|\gamma| > \widehat{\beta})\big] \le \rho\alpha$. The feasibility condition at level

α/ρ is $\alpha/\rho \geq 1/(K+1)$. (b) Under the stated factorisation the scores $S_1, \ldots, S_{K+1}$ are weighted exchangeable in the sense of Tibshirani, Barber, Candès and Ramdas (2019) with weight function w; their Theorem 2 gives exact $(1-\alpha)$ coverage for the weighted quantile $\inf\{s : \sum_k p_k \mathbf{1}\{S_k \leq s\} \geq 1-\alpha\}$, $p_k = w(W_k)/\big(\sum_i w(W_i) + w(W_{K+1})\big)$. Estimability of w from unlabeled data is the standard two-sample density-ratio reduction: $w \propto \frac{\pi(W)}{1-\pi(W)}$ for $\pi(W) = \mathbb{P}(\text{episode is current} \mid W)$, and W contains no labels. □

Corollary 6 (The relaxation is not free)**.** *The conditional-stability hypothesis of Theorem 13(b) is not auditable from label-free deployment data: Theorem 18 applies verbatim at each fixed w, so no label-free statistic distinguishes* $Q(\cdot \mid w)$ *from any other conditional drift law with the same observables. It is, however,* falsifiable *given retrospective labels on two episode populations, which a declared* β *at a single deployment is not. That is the whole gain: the assumption moves from unfalsifiable to falsifiable, not from unverified to verified.*

I. Is E2 plausible? An honest answer

E2 asserts that the deployment a system is about to face is exchangeable with the deployments it has already logged. Three observations, in decreasing order of comfort.

Where it is plausible. When the episode population is generated by a stable operational process — a fixed fleet of sites, a recurring seasonal cycle, a queue of customers drawn from a stable market — and the model, the candidate update rule, and the scoring function are held fixed, E2 is the same assumption already made whenever a deployment team computes a historical error rate and uses it as a forecast. It is checkable in the weak sense that its consequences (coverage of past budgets on later episodes) can be back-tested.

Where it fails, and it is the interesting case. E2 fails exactly when the current deployment differs from the history in kind rather than in draw: a new corruption mechanism, a new site, a new severity regime, a new sensor. This is the case that motivates test-time adaptation in the first place. A deployer who could guarantee E2 could often also have retrained.

What the theory can and cannot say about that failure. Theorem 13(a) prices the failure if ρ is known, and ρ is not knowable label-free (Corollary 6). Theorem 13(b) removes the need for ρ at the cost of a conditional-stability assumption that is itself unverifiable at deployment. So the honest summary is: **episode structure converts an unfalsifiable per-deployment declaration into a falsifiable population-level one, and buys a marginal (not conditional) guarantee. It does not make β knowable at the deployment that needs it.** The empirical section quantifies the gap.

J. Numerical validation on the K-Bound artifacts

Protocol. Episodes are the 6480 CIFAR-10-C and 405 ImageNet-C cells (condition × method × seed) of §X; $\gamma := \Delta - M$ in benefit units as in the β-sweep. Each split partitions the cells into three disjoint pools: a fit pool F on which the benefit model $M(\cdot)$ is fit, a history H whose labels are revealed, and a deployment set D whose labels are used only for evaluation. M is fit once on F and applied identically to H and D, so the scores on H and D are exchangeable exactly when the episode populations are. Five random F/H splits per protocol; $\alpha = 0.10$, so **the null coverage is** 0.90: any protocol whose deployment set is exchangeable with its history must return ≈ 0.90, and the finding is the deviation. Artifacts: `frontier_sweep_v1/beta_estimability/{episode_beta.py,episode_beta_results.json}`. ρ_{impl} is the episode-law ratio that Theorem 13(a) would need in order to explain the realised miss rate, i.e. $(1 - \text{coverage})/\alpha$; it equals 1 under E2. Full tables, ImageNet-C, and the false-adapt columns are in `beta_estimability/tables.md`.

Four things follow. *(i) The estimator is correct.* On the exchangeable control the realised coverage is 0.905 and 0.912 against a null of 0.900, and a K-sweep reproduces the feasibility floor exactly: at $\alpha = 0.10$ the conformal budget is infeasible for $K \in \{3, 5\}$ and first attains 0.907 ± 0.088 at $K = 9 = \lceil 1/\alpha \rceil - 1$, matching Corollary 5. *(ii) Nuisance axes are safe, deployment axes are not.* Coverage is within 0.04 of nominal when the seed, class-balance protocol or batch size changes, and falls to 0.626 and 0.454 when the corruption family or severity changes. The implied ρ rises from 1 to 5.5. *(iii) A better benefit model transfers worse.* The eleven-coordinate GBM has higher decision yield than the four-coordinate ATC ridge (0.647 vs 0.446) and strictly worse budget transfer on every deployment axis (0.626 vs 0.866 on `LOCO`, 0.454 vs 0.595 on `LOSEV`). Fitting the benefit surface harder shrinks the history residuals without shrinking the residuals on a novel episode, so the quantile it produces is too small. *(iv) Weighted conformal restores validity by destroying yield.* With label-free evidence weights (Theorem 13(b)) the `LOSEV` coverage goes from 0.454 to 1.000 and the decision yield goes from 0.615 to 0.000: the weights concentrate on the few history episodes that resemble the deployment, the weighted quantile runs off the end of the score list, and the frontier abstains everywhere. This is the same trade Theorem 18 forces on any label-free audit, now appearing inside a method that was supposed to escape it.

K. Two consequences for the rest of the paper

a) The β-sweep's failure was never a quantile-estimation failure.: The β-sweep declares $\widehat{\beta}$ as the plug-in $q_{0.90}$ of $|\gamma|$ on dev cells; Theorem 11 replaces that by the conformal order statistic. Across all 470 splits the two budgets differ by a median relative gap of 0.34%, and the median coverage difference is exactly 0. At K

TABLE V

Episode-conformal budget coverage on CIFAR-10-C, by the episode axis that changes at deployment. NOMINAL COVERAGE IS 0.900 ($\alpha = 0.10$) AND *THE FINDING IS THE DEVIATION*: A ROW AT 0.90 IS THE NULL, NOT A RESULT. COVERAGE IS POOLED OVER DEPLOYMENT CELLS (24,300 FOR THE CONTROL, 32,400 PER LEAVE-ONE-AXIS-OUT PROTOCOL); CIS ARE BINOMIAL. ARTIFACT: `BETA_ESTIMABILITY/EPISODE_BETA_RESULTS.JSON`, KEY `SUMMARY`.

Episode axis perturbed	changes at deployment	coverage	95% CI	ρ_{impl}	yield
CIFAR-10-C, M_{gbm} (11 evidence coordinates, GBM)					
`RAND` (control)	nothing	0.905	[0.901, 0.908]	0.95	0.647
`LOSO`	fitting seed	0.875	[0.872, 0.879]	1.25	0.647
`LOPROT`	class-balance protocol	0.877	[0.874, 0.881]	1.23	0.657
`LOBATCH`	batch size	0.862	[0.858, 0.866]	1.38	0.655
`LOMO`	adaptation method	0.804	[0.800, 0.808]	1.96	0.630
`LOSTR`	adaptation strength	0.721	[0.717, 0.726]	2.79	0.594
`LOCO`	corruption family	0.626	[0.621, 0.631]	3.74	0.645
`LOSEV`	corruption severity	0.454	[0.449, 0.460]	5.46	0.615
CIFAR-10-C, M_{atc4} (4 ATC coordinates, ridge)					
`RAND` (control)	nothing	0.912	[0.908, 0.915]	0.88	0.446
`LOCO`	corruption family	0.866	[0.863, 0.870]	1.34	0.438
`LOSEV`	corruption severity	0.595	[0.589, 0.600]	4.05	0.424

in the thousands the conformal correction moves one order statistic and nothing else. The β-sweep's negative result therefore survives intact: what it measured was the failure of E2, not the failure of a quantile estimator, and no conformal machinery repairs it. Conversely, when E2 does hold the frontier operationalises: on an exchangeable three-way split of CIFAR-10-C the episode budget delivers decision yield 0.641 at regret 0.0019 and $\text{FA}_{\text{u}} = 0.0002$, against the shipped conformal-interval rule's 0.662, 0.0015 and 0.0000 on the same cells. On ImageNet-C it does not: yield 0.344 versus 0.508.

b) E2 is not verifiable, but its observable component is falsifiable.: Corollary 6 says no label-free statistic certifies the conditional part of E2. The *marginal* part is a different matter: $\Delta_{\text{sep}} = \text{AUC}$ of a cross-fitted classifier separating history from deployment episodes on the evidence Z alone is computable without any labels, and equals $\frac{1}{2}$ under E2. Measured across the 470 splits it correlates with realised coverage at Pearson $r = -0.35$ ($p = 1.2 \times 10^{-14}$), Spearman $\rho = -0.35$, and it reads 0.497 ± 0.033 on the exchangeable control, as it must. Used as a one-sided screen at $\Delta_{\text{sep}} < 0.70$: mean coverage 0.905 with 7.7% of splits below 0.85; above the threshold, mean coverage 0.782 with 43.3% below 0.85. So a low separability reading is informative and a high one is only a warning — which is exactly the asymmetry Corollary 6 predicts, since the residual 65% of the coverage variation lives in the conditional component that Theorem 18 makes invisible. We recommend reporting Δ_{sep} alongside any episode budget; we do not recommend treating it as a certificate.

L. What is left open

Conjecture 2 (Estimable budgets without exchangeability)**.** *Is there a structural assumption, weaker than E2 and* falsifiable from retrospective labels on the history alone*, under which $\beta^\star$ for a novel episode is identified? The two natural candidates both fail to be that: bounded episode-law ratio (Theorem 13(a)) needs an unknowable ρ; conditional stability (Theorem 13(b)) is falsifiable only with labels from the novel population, which is what is missing. A positive answer would need a drift functional that extrapolates off the support of the history, and Theorem 10(b) shows no such functional can be built from unlabeled observables alone.*

This is stated as a conjecture, not a theorem: we neither exhibit such an assumption nor prove that none exists.

We did not pursue an anytime-valid within-episode budget from the e-process machinery. The reason is structural rather than technical: within one episode γ is a fixed unknown number and every observable in the adaptation stream is label-free, so Theorem 18 applies to any stopping-time-measurable statistic of that stream. Sequential machinery buys anytime validity, not identification.

IX. METHOD: KGA (KNOWABILITY-GUIDED ADAPTATION)

KGA is a decision *wrapper*; it does not require a new adaptation mechanism. Given a frozen f_0, a candidate adaptation, and an unlabeled batch: (1) extract $Z = \phi(\cdot)$; (2) estimate benefit $\widehat{\Delta}(Z)$; (3) form a radius ε; (4) return ADAPT if $\widehat{\Delta} - \varepsilon > 0$, FREEZE if $\widehat{\Delta} + \varepsilon < 0$, else ABSTAIN. In our experiments $\widehat{\Delta}$ is a gradient-boosted regressor and ε is the $(1-\alpha)$ empirical quantile of *out-of-fold* (leave-one-task-out) residuals, so the estimator and its residual-calibration data are disjoint ($\alpha = 0.1$). The coverage guarantee is split-dependent: a clean dev/test split (the natural-shift protocols) gives *exact* split-conformal coverage $\mathbb{P}[|\widehat{\Delta} - \Delta| \le \varepsilon] \ge 1-\alpha$ under exchangeability, while the per-cell grids use the leave-one-out (jackknife) radius, whose calibration-set coverage is $\ge 1 - \alpha$ by construction and distribution-free only asymptotically (the finite-sample jackknife+ of Barber et al. gives $1 - 2\alpha$; not implemented).

a) Assumptions.: KGA inherits exactly the assumptions of Theorem 2: (i) a labeled validation/calibration sample of paired benefits that is exchangeable with deployment,

Algorithm 1: KGA (Knowability-Guided Adaptation)

Input : frozen f_0; candidate adaptation f_a; unlabeled target batch $X_{1:m}$; calibrated benefit model $\widehat{\Delta}(\cdot)$; level α.

Extract label-free evidence $Z = \phi(X_{1:m}, f_0, f_a)$ (entropy, confidence, predicted-class balance and its drift, marginal-prediction KL, update norm, detector disagreement);
Estimate benefit $\widehat{\Delta} \leftarrow \widehat{\Delta}(Z)$ and radius ε with out-of-fold conformal coverage $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \leq \varepsilon\,] \geq 1 - \alpha$ (exact under a clean split);
if $\widehat{\Delta} - \varepsilon > 0$ **then**
  return ADAPT (deploy f_a);
else
  if $\widehat{\Delta} + \varepsilon < 0$ **then**
    return FREEZE (keep f_0);
  else
    return ABSTAIN (keep f_0, flag for review);

Guarantee : the false-adapt and false-freeze rates are each $\leq \alpha$ (Thm 2).

from which ε is built; (ii) bounded per-sample benefits; and (iii) an evidence map Z that is *risk-aligned* (it brackets sign Δ; §IV). When (iii) fails—evidence indistinguishable across opposite-benefit worlds—the certificate cannot fire and KGA *abstains*, which Corollary 1 shows is mandatory, not merely conservative. No target labels are used at any step.

b) Cost.: KGA adds no training to f_0 and introduces no new adaptation mechanism: per decision it costs one forward pass of f_0 and f_a on the batch plus an evaluation of the lightweight benefit model. It is therefore a drop-in safety wrapper around any already-deployed TTA candidate.

c) The threshold is not tuned, and it is not the drift budget.: KGA's commit boundary is not a swept hyperparameter: ε is the $(1-\alpha)$ out-of-fold residual quantile, fixed on the calibration split and *never* selected on target-test data. ε is a *conformal radius on the estimation error* $|\widehat{\Delta} - \Delta|$; it is *not* an estimate of the drift budget β, and no statement in this paper licenses reading it as one. The two objects answer different questions: β bounds an unidentified population quantity that Theorem 6 shows no label-free procedure can supply, while ε bounds a finite-sample fluctuation that exchangeable calibration data does supply. What separates KGA from confidence-thresholding (ATC) and agreement-trend selection [17] is therefore not the value of its threshold but the shape of its output: an abstain region, and a finite-sample false-adapt guarantee on the commitments it does make. Those methods fit a scalar against a held-out accuracy signal and *predict accuracy*; KGA certifies the benefit *sign* and ABSTAINS where Corollary 1 proves no label-free rule can decide. "Estimate accuracy then threshold" has no abstain region and no false-adapt control.

A. Three coverage statements, kept separate

Much of the difficulty in reading an adaptation certificate comes from one word doing three jobs. We fix the terminology here and hold to it for the rest of the paper. In what follows $[L, U] = [\widehat{\Delta} - \varepsilon,\ \widehat{\Delta} + \varepsilon]$ is the benefit interval of the certificate rule, so that the coverage premise $\mathbb{P}[\,|\widehat{\Delta} - \Delta| \leq \varepsilon\,] \geq 1 - \alpha$ of Theorem 2 and the interval statement below are the same hypothesis written two ways.

Definition 9 (Theoretical coverage)**.** The interval has *theoretical coverage* at level $1 - \alpha$ if

$$\mathbb{P}\big\{\Delta \in [L, U]\big\} \;\geq\; 1 - \alpha, \tag{2}$$

the probability taken over the common space of calibration, benefit-model fitting and deployment-unit randomness. This is a property of a distribution, not of a run. We assert it only where (A1)–(A6) below are argued to hold, and never on the basis of an observed hit rate.

Definition 10 (Observed empirical coverage)**.** For a completed, labelled track with n independent inference units (Definition 12), the *observed empirical coverage* is

$$\widehat{\mathrm{Cov}} \;=\; \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\big\{\Delta_i \in [L_i, U_i]\big\}, \tag{3}$$

reported with n, the resampling unit, and an interval computed at that unit. It estimates a hit rate on the conditions evaluated, and is not evidence for (2) on any other condition.

Remark 14 (When *not* to attach an interval)**.** Definition 10 applies to a hit rate that is genuinely random. It does *not* apply to a realized calibration-set coverage figure produced by in-sample rank calibration, which is a deterministic function of n and carries no sampling variability at all — attaching a binomial or bootstrap interval to such a figure gives it a frequentist meaning it does not have. §XII identifies the figures in this paper that are arithmetic in this sense, and they are reported without intervals. The discipline this section imposes is that every coverage number states which of the three kinds it is; for the arithmetic ones, the honest annotation is "deterministic in n", not a wider interval.

Definition 11 (Deployment diagnostic)**.** A *deployment diagnostic* is a label-free statistic whose large values are evidence *against* (A1)–(A6). Diagnostics are one-sided by construction: they can falsify and cannot verify. Passing every diagnostic of §IX-D does not establish (2).

Definition 12 (Inference unit)**.** The *inference unit* is the level at which calibration draws are treated as exchangeable and at which uncertainty is resampled: a domain, an episode, a corruption×severity cell, a backbone, or a seed, declared per track. It is the individual example only when examples are independently drawn, which on the tracks of §X they are not.

B. The assumption contract

The paper already records which condition buys what, and what to do when one fails. What that table does not do is give the conditions names that the rest of the paper, the released artifacts and a deployment can all refer to. We do that now, because a condition nobody can cite is a condition nobody checks.

- (A1) Calibration exchangeability. Calibration units and the deployment unit are exchangeable at the declared inference unit, or an explicit shift correction supplies the same conclusion. (This is the episode-exchangeability condition E2, read at the unit level.)
- (A2) Risk alignment. The evidence Z is risk-aligned (Definition 1): it retains enough information about Δ for calibration residuals to transfer to the deployment condition.
- (A3) Coverage premise. The interval satisfies (2) on the target class.
- (A4) Protocol independence. Target-test labels influence none of: feature selection, candidate selection, threshold selection, calibration, or stopping. (The retrospective outcome-logging condition E1, and the data-access discipline of the protocol tables.)
- (A5) Correct sampling unit. Calibration and uncertainty calculations operate on independent domains, episodes, corruptions, backbones or seeds, not on correlated examples pooled as though independent.
- (A6) Frozen decision rule. Estimator, interval construction, α and decision rule are fixed before the promoted target conditions are evaluated.
- (A7) Estimator stability. Where ε is calibrated leave-one-out rather than on a clean split, the deployed full-data fit and each leave-one-out fit differ by at most β_{stab} in prediction.

(A1)–(A4) and (A6) rename conditions the paper already carried: the conditions previously listed as risk-aligned Z, coverage of $\widehat{\Delta} \pm \varepsilon$, exchangeable or corrected residuals, and frozen protocol become (A2), (A3), (A1) and (A4)/(A6) respectively, and the long manuscript's assumptions (i)–(iii) map to (A1), the bounded-benefit regularity condition, and (A2). Bounded loss and benefit remains a regularity condition needed for the concentration arguments; it is not part of the deployment gate because it is a property of the loss we choose, not of the deployment we meet. **(A5) is new in this revision**, and it is the one the empirical sections most often failed to make explicit: a coverage figure computed over correlated cells was reported without the denominator that makes it readable.

Remark 15 (Why (A7) exists, and what it costs)**.** The grids of §X calibrate ε leave-one-out rather than on a clean split, and no theorem makes that exact conformal: the plain jackknife has no distribution-free finite-sample guarantee, and its finite-sample counterpart buys $1 - 2\alpha$ rather than $1 - \alpha$. What does hold is a deterministic transfer. If the deployed fit differs from each leave-one-out fit by at most β_{stab}, then leave-one-out coverage at radius ε gives full-fit coverage at radius $\beta_{\text{stab}} + \varepsilon$ (`formal/KBound/Stability.lean`, `stability_transfers_loo_coverage`, machine-checked). So β_{stab} is the price of leave-one-out calibration, and it is an assumption about the estimator rather than something the calibration data can establish. The contrapositive is the useful direction: a miss beyond $\beta_{\text{stab}} + \varepsilon$ *proves* the fit was less stable than β_{stab}, so (A7) is falsifiable after the fact even though it is not verifiable before.

Remark 16 (What the contract is not)**.** Theorem 2 is conditional on (A1)–(A6), and K-Bound does not infer, test into, or prove any of them from unlabelled deployment data. (A1)–(A3) are not identifiable label-free; this is the obstruction of §V applied to calibration transfer rather than to the drift budget. What a deployment can do is detect *some* violations and refuse to certify when it does, which is §IX-D.

Remark 17 (Where the guarantee lives)**.** The proof of Theorem 2 uses no property of $\widehat{\Delta}$ beyond its appearance in L and U: a false adapt requires $\Delta \leq 0 < L$, which is the complement of the coverage event, and a false freeze requires $\Delta \geq 0 > U$. So the guarantee is carried entirely by (A3); the fitted benefit estimator contributes decision *yield*, not safety. A better estimator narrows the interval and buys more committed decisions at the same α; a worthless one widens it until the system abstains. That is the intended failure direction, and it is why (A3) — not estimator quality — is the assumption a deployment has to defend.

C. Permitted behaviour by assumption state

Where the earlier tables say what to do when a single condition fails, Table VI says what a deployment is permitted to *emit*, which is the operational form of the same idea and the object §IX-D implements.

TABLE VI

FALLBACK LADDER. PERMITTED BEHAVIOUR IS A FUNCTION OF THE ASSUMPTION STATE, NOT OF THE ESTIMATED BENEFIT. "CERTIFICATION DISABLED" MEANS NO CERTIFICATE IS EMITTED; THE SYSTEM MAY STILL SERVE THE FROZEN MODEL, BUT NO K-BOUND CLAIM ATTACHES TO THAT CHOICE.

Assumption state	Permitted behaviour
All required gates pass	Full ADAPT/FREEZE/ABSTAIN certificate
Support warning, no hard violation	ABSTAIN or FREEZE only
Insufficient independent calibration units	Diagnostic output only; no decision
Leakage, failed transfer, or unknown provenance	Certification disabled

Read with Remark 17, this is the honest description of what K-Bound offers: *conditional insurance against detectable adaptation harm*, not an unconditional safety guarantee.

D. The assumption gate

The failure-safe defaults above abstain when the batch is too small, the evidence lies outside calibration support, a configuration hash differs, or no valid radius exists. Those are the right reflexes, and the gate below is what they look like once they are stated as a precondition on *certification* rather than as guards inside the decision path. Because (A1)–(A6) cannot be verified label-free (Remark 16), a deployment is permitted to issue strict ADAPT/FREEZE decisions only after passing it. Thresholds are fixed before deployment and versioned with the protocol: a gate tuned after seeing the outcome violates (A6) and voids the certificate.

Inputs. Frozen candidate f_a; frozen benefit estimator $\widehat{\Delta}$; calibration evidence and residuals; unlabelled deployment evidence Z; the declared inference unit; thresholds fixed before deployment.

1) **Verify protocol separation.** If target labels affected candidate, feature, threshold or stopping choice: *certification disabled* — (A4).
2) **Verify inference-unit adequacy.** If the effective number of independent calibration units is below the declared minimum: *diagnostic only* — (A5).
3) **Test evidence-support overlap.** If deployment evidence lies outside the calibrated support: ABSTAIN or FREEZE — (A1)/(A2) suspect.
4) **Test calibration stability.** Recompute ε across folds, seeds, domains or time blocks. If the radius or the action proportions are unstable: ABSTAIN or FREEZE.
5) **Check sensitivity.** Vary the declared calibration construction and the clustering unit. If the promoted conclusion changes: *diagnostic only*.
6) **Apply the certified rule.** ADAPT if $L > 0$; FREEZE if $U < 0$; otherwise ABSTAIN.
7) **Monitor sequentially.** If support, evidence distribution or action rates drift beyond predeclared thresholds, suspend strict decisions and revert to FREEZE/ABSTAIN.

Steps 1–2 are protocol checks and are fatal; steps 3–5 are statistical diagnostics and degrade the permitted behaviour along Table VI rather than aborting; step 7 runs for the lifetime of the deployment. The gate fails closed: an *unevaluated* check counts as a failed check, so a pipeline that does not yet supply deployment evidence or unit labels returns a restricted decision rather than a clean one.

E. Diagnostics, and what each one cannot do

a) Evidence-support overlap.: We report the distance to the nearest calibration unit, the fraction of deployment evidence outside the calibration envelope, a cross-fitted domain-classifier AUROC, and the maximum standardised feature deviation. High separability is evidence *against* direct transfer; low separability is not evidence for it, since two conditions can be indistinguishable in Z and still have different Δ — which is exactly the configuration §V constructs, not a corner case. This is the same one-sidedness already noted for the separability screen in §XII.

TABLE VII

THE DIAGNOSTIC SUITE CONSUMED BY THE GATE. NONE ESTABLISHES THEORETICAL COVERAGE (DEFINITION 9); EACH CAN ONLY REMOVE A WAY OF BEING WRONG. THE RIGHT-HAND COLUMN IS AS MUCH A PART OF THE METHOD AS THE LEFT.

Diagnostic	Detects	Cannot show
Evidence-support overlap	deployment evidence outside the calibrated envelope; high calibration-vs-deployment separability	that low separability implies transfer
Radius stability	ε or the action mix depending on an arbitrary choice of fold, seed, domain or block	that a stable radius is a correct radius
Risk-alignment audit	benefit-sign errors, rank decorrelation, coverage varying by evidence region	anything about a condition with no labels; it is retrospective by construction
Leakage audit	target labels or test-set influence entering candidate, threshold or hyperparameter choice	analyst decisions outside the recorded protocol

b) Radius stability.: We recompute ε leaving out one unit at a time and report the mean radius, its range and coefficient of variation, and the decision-disagreement rate against the pooled radius. The number that matters is the fraction of promoted conclusions that change under an alternative *equally valid* split: a conclusion that survives only one admissible clustering is a conclusion about the clustering. The check is not an outlier detector — a contaminated fraction well below α does not move the quantile and should not.

c) Risk-alignment audit.: Where evaluation labels exist we report benefit-estimation error, rank correlation between predicted and realised benefit, coverage stratified by evidence region, false-adapt by domain or cluster, and whether benefit *sign* prediction transfers across domains. This is a retrospective audit of (A2) on the tracks we evaluated. It is not available at deployment time, which is the entire difficulty.

d) Leakage audit.: For each promoted claim we record when the candidate was fixed, when the calibration design was fixed, whether target labels were accessed, whether the test set influenced hyperparameters, the protocol-lock identifier, and whether failed runs were retained. Chronology is the evidence for (A6); an assertion of pre-registration without a timestamped lock is not, and the audit fails closed on a missing record rather than assuming the benign case.

F. The assumption report

Each track emits a fixed-schema record (Table VIII) alongside its results, so the assumption state travels with the number instead of living in prose. Every value is computed from raw units by `kga.assumptions`; none is transcribed by hand. The records are released with the artifacts.

TABLE VIII
ASSUMPTION-REPORT SCHEMA, VERSION `KBOUND-ASSUMPTION-REPORT/1`. A STATISTIC THAT CANNOT BE COMPUTED FROM THE SUPPLIED INPUTS IS `NULL` AND ADDS A STRING TO `LIMITATIONS`; IT IS NEVER DEFAULTED TO A PLAUSIBLE NUMBER. THE RELEASED RECORDS ARE THE AUTHORITY FOR EVERY VALUE — NOTHING HERE IS AN EXAMPLE VALUE.

Field	Type / admissible values
`dataset`, `protocol`	`str`
`inference_unit`	`str` (Def. 12)
`calibration_test_separated`	`bool`
`candidate_fixed_before_test`	`bool`
`target_labels_used_for_routing`	`bool`
`coverage_type`	`theoretical` | `observed_empirical` | `diagnostic_only`
`theoretical_coverage_claimed`	`bool`
`observed_coverage`	`float` | `null`
`n_rows`, `n_units`	`int`
`coverage_interval_95`	`[float, float]` | `null`
`coverage_interval_method`	`str`
`support_overlap_status`	`pass` | `warning` | `fail`
`radius_stability_status`	as above
`conclusion_stability_status`	as above
`leakage_status`	`pass` | `fail`
`deployment_gate`	`certify` | `restricted` | `diagnostic_only` | `reject`
`fallback_action`	`adapt_freeze_abstain` | `freeze_or_abstain` | `none`
`alpha`, `thresholds`	`float`, `object`
`diagnostics`	`object` (per-check detail)
`limitations`	`[str]`

Remark 18 (Reading a report). `theoretical_coverage_claimed` is `false` on every track in this paper. That is not a formatting artefact: there is no track on which we are willing to assert (2) for an unseen deployment condition, because (A1)–(A3) are not checkable from the evidence we hold. `observed_coverage` is the statistic of Definition 10, reported with `n_units` and an interval computed at the declared unit, so that a coverage figure computed on correlated cells cannot be read as though it came from that many independent draws.

X. EXPERIMENTS

The empirical claim is narrow and we test it in that order. KGA is a safety layer: it should beat both fixed policies only where harmful adaptation is frequent and detectable, and do no harm elsewhere. We first establish the mechanism in a controlled label-free setting (this section and the online streams that follow), then carry the headline to image corruption grids and held-out natural shifts (§X-C–§X-D). The anomaly-detection and regression tracks below are breadth evidence that the certificate's behavior is mechanism-agnostic, not a competing headline.

A. What the certificate buys: the decision-yield / false-adapt frontier

The certificate's guarantee is a trade, and this subsection prices it rather than asserting it. The objection it answers is that no-harm is trivially achievable by abstaining. In this system an ABSTAIN realises the frozen model, so always-abstain is numerically identical to always-freeze and inherits its regret exactly; whether that is cheap or ruinous is a per-track empirical fact.

Replay integrity. Sweeping a radius multiplier κ in the rule "ADAPT iff $\widehat{\Delta} - \kappa\varepsilon > 0$, FREEZE iff $\widehat{\Delta} + \kappa\varepsilon < 0$" reproduces the *shipped* decision on 7,365/7,365 cells at $\kappa = 1$ across all three per-cell tracks, so the sweep is a sweep of the deployed system and not a reimplementation of it. Every policy in the sweep is a function of $\widehat{\Delta}$, ε and Z alone; the label arrays are registered at load time and checked by identity, buffer and value against every array a policy consumes.

What it buys. On CIFAR-10-C the certificate cuts regret 5.00× below the better fixed policy and 85× below the worse one, at $\mathrm{FA_u} = 0/6480$ (Wilson 95% upper bound 0.0006); per adapter the decision accounting is 1,113 and 1,244 ADAPT decisions with zero false adaptations (Clopper–Pearson 95% upper bounds 0.0027 and 0.0024 on the conditional rate). Always-abstain is the *worst* row of the table, so the abstention objection does not survive contact with the numbers on this track. On ImageNet-C the ratio to the best fixed policy is 2.90×.

Where it commits. Among the 4,330/6,480 CIFAR-10-C cells where the certificate commits, mean regret removed relative to the better fixed policy is +0.00940 against a permutation null of −0.01850 that holds the number of ADAPT and FREEZE commitments fixed and reassigns them at random ($p < 0.0002$, 5,000 permutations); the mean true effect $|\Delta|$ on committed cells is 0.1987 against 0.0081 on declined cells, a ratio of 24.5× that no random subset of the same size reached in 5,000 draws. The certificate fires on the cells that matter.

Against cheap heuristics. Held to $\mathrm{FA_u} \leq 0.10$ and tuned in *hindsight* on the test labels, the best of four single-signal label-free rules (`marginal_KL`, a BN-statistics drift proxy) reaches regret 0.00472 on CIFAR-10-C against the certificate's 0.00150—a 3.1× gap—while making 324 harmful adaptations out of 6,480 cells ($\mathrm{FA_u} = 0.050$)

TABLE IX

The frontier, measured. DECISION YIELD (FRACTION OF CELLS COMMITTED), REGRET TO THE PER-CELL ORACLE, AND THE UNCONDITIONAL FALSE-ADAPT RATE, AS THE RADIUS MULTIPLIER κ SWEEPS FROM COMMIT-EVERYWHERE TO ABSTAIN-EVERYWHERE. $\kappa = 1$ IS THE SHIPPED $\alpha = 0.10$ CERTIFICATE. ARTIFACT: `FRONTIER_SWEEP_V1/DECISION_VALUE_RESULTS.JSON`.

κ	yield	regret	$\mathrm{FA_u}$
CIFAR-10-C, pooled, $n = 6480$ (always-adapt 0.00748, always-freeze 0.12797)			
0	1.000	0.00029	0.0421
0.5	0.763	0.00074	0.0025
1.0 (shipped)	**0.668**	**0.00150**	**0.0000**
2.0	0.546	0.00312	0.0000
∞ (always abstain)	0.000	0.12797	0.0000
IMAGENET-C, pooled, $n = 405$ (always-adapt 0.02403, always-freeze 0.02685)			
1.0 (shipped)	**0.521**	**0.00829**	**0.0049**
IMAGENET-R, pooled, $n = 480$ (always-adapt **0.00636**)			
1.0 (shipped)	**0.535**	**0.01120**	**0.0021**

against the certificate's zero. Under a fair seed split the same heuristic reaches 0.00301 at $\mathrm{FA_u} = 0.032$, against the certificate's 0.00152 at 0.0000 on the same held-out seeds. **The advantage over cheap heuristics is an advantage in false-adapt rate, and it converts into a regret advantage only where adaptation is genuinely two-sided.**

Where it buys nothing, stated as such. On ImageNet-R the certificate is 1.76× *worse* than always-adapt (0.01120 versus 0.00636). It buys $\mathrm{FA_u} = 0.0021$ against always-adapt's 0.2021—real safety—and pays 0.48 accuracy points per cell for it on a track where adaptation essentially never hurts. Per backbone, the $|\Delta|$ concentration ratio is 0.71–1.57 with permutation p between 0.054 and 1.000: not one backbone is distinguishable from random placement, and the value per committed decision is exactly 0.0000 on 9 of 10 backbones, because when the better fixed policy is always-adapt an ADAPT commitment reproduces it exactly. This is the clearest single counterexample to any blanket "uniformly no-harm on natural shifts" reading, and it comes from an artifact this paper promotes.

The shipped operating point is not regret-optimal. On CIFAR-10-C it is dominated by $\kappa = 0.9$ (yield 0.684, regret 0.00134, $\mathrm{FA_u} = 0.0$ for both), about 10% of regret; and against a perfect ordering of cells by the regret a correct commitment removes, the best achievable regret at the certificate's own yield is 0.00020, so it sits 7.50× above the information-theoretic ceiling at equal yield (2.79× on ImageNet-C, 2.64× on ImageNet-R). $\alpha = 0.10$ is defensible as a pre-registered safety choice and is not defensible as a regret-optimal one.

Robustness of the guarantee to the calibration split. Recalibrating ε leave-one-corruption-family-out—six folds instead of 432, so no cell is calibrated on its own corruption family—inflates the radius by 4.3×–6.9× (from 0.015–0.022 to 0.092–0.112), drops the commitment rate from 0.509–0.600 to 0.398–0.417, and worsens regret by 3.4×–7.6×, across Tent and EATA and five seeds. In all ten runs $\mathrm{FA_u} = 0.0000$. *The false-adapt guarantee survives a calibration split the shipped protocol does not make; the decision yield and the regret advantage do not survive it intact, and we report both halves.* (Ranges are under the exact-rank quantile; under the shipped interpolated rank they are 4.3×–7.0× and 3.2×–7.5×.) Artifact: `frontier_sweep_v1/out_cifar_loco_tent_eata.json`.

B. Anomaly-score archive

We begin with a 123-task anomaly-score archive (ADBench tabular, image-OOD, text, cyber, fraud); each task has six detector scores on labeled validation and test. Frozen f_0 = best-validation detector. Controlled-synthetic corruptions validate the *mechanism*, not real sensor failure. All numbers come from the released reproducibility package.

a) Safety on the clean suite.: With candidate f_a = rank-normalized logistic stack, KGA's abstentions concentrate where the true benefit is near zero (mean $|\Delta| = 0.021$ on abstained vs 0.132 on acted tasks); adapt precision 0.90. This suite is helpful-dominated, so always-adapt (mean AUROC 0.777) edges the safe policy (0.766); coverage 17%. Honest reading: when adaptation almost always helps, abstention mostly costs coverage.

b) Harmful regime.: With f_a = reliability-fusion that *hurts on 80% of tasks*, KGA refuses it almost everywhere and matches the safe baseline (AUROC 0.748) while always-adapt drops to 0.728; regret-to-oracle falls from 0.0214 to 0.0019 ($\approx 11\times$). See Fig. 3.

c) Controlled mixed regime (369 instances).: Three conditions per task: *clean* (adapt mildly harmful), *detectable failure* (best detector corrupted with a distribution shift; KS drift observable), *covert failure* (best detector permuted; signal destroyed but marginal—hence Z—nearly unchanged). Decisions by true regime are in Table X; adapt precision 0.935, false-adapt rate 0.065. Policy AUROC: always-adapt 0.697, always-freeze 0.586, KGA 0.690, oracle 0.719 (Fig. 2). KGA beats always-freeze by 0.10 and ties always-adapt—because harmful adaptation is mild in this domain.

d) Empirical non-identifiability witness (partial).: Clean instances have mean $\Delta = -0.041$ while covert-

condition	true mean Δ	ADAPT	FREEZE	ABSTAIN
clean	-0.041	8	5	110
detectable failure	$+0.193$	71	3	49
covert failure	$+0.181$	59	3	61

TABLE X
KGA DECISIONS BY TRUE REGIME (CONTROLLED MIXED BENCHMARK).

failure instances have mean $\Delta = +0.181$ (opposite sign), yet their evidence is far closer (mean standardized Z-distance clean→covert 1.35 vs clean→detectable 3.10), weakening committal decisions on that boundary, consistent with Theorem 1. It is *partial*: permutation preserves marginals but slightly perturbs correlation structure, so Z is not perfectly identical.

e) Clean non-identifiability witness.: We also construct the exact witness of Theorem 1: two worlds with $X \sim \mathcal{N}(0,1)$, $f_0 = \mathbf{1}[X>0]$, $f_a = \mathbf{1}[X<0]$, and flipped labels, so the benefit is exactly opposite (world means -1.0 vs $+1.0$) while every evidence feature is a function of X only. Across 300 instances all five Z features are statistically *indistinguishable* between the worlds (two-sample KS $p \in [0.16, 0.96]$, all > 0.05), yet the truth is opposite—so Z cannot decide. KGA **abstains on** 100% of instances; a rule forced to commit by sign $\widehat{\Delta}$ pays regret 0.51 (near chance). This is a constructed bridge and nothing more: the worlds are ours, so it cannot come out negative. **It is superseded by §VI-F**, which exhibits the same configuration in *logged real deployments* — and also measures how rare it is there, which this synthetic witness by construction cannot.

f) Statistical rigor (8 seeds, paired t-tests).: Repeating the mixed-regime experiment over 8 seeds (split-conformal, α=0.1): mean test AUROC 0.686 ± 0.003 (KGA), 0.697 ± 0.000 (always-adapt), 0.584 ± 0.001 (always-freeze), 0.718 ± 0.001 (oracle); adapt precision 0.936 ± 0.010, false-adapt 0.064 ± 0.010, coverage 0.373 ± 0.017. Paired t-tests: KGA vs always-freeze t=62.2, p=7.3×10^{-11} (KGA significantly higher); KGA vs always-adapt t=-8.8, p=5.1×10^{-5} (significantly lower). Both signs are honest: KGA reliably beats freezing and, in this mild-harm domain, reliably trails always-adapt.

g) Ablations.: Table XI (left) drops each evidence feature: detector *disagreement* is the most important—removing it raises regret from 0.037 to 0.094 (2.6×), directly supporting Theorem 1. An α sweep traces the safety/coverage tradeoff: α=0.01 gives 0% false-adapt at 11% coverage, up to α=0.2 giving 49% coverage at 7% false-adapt (Fig. 4, middle). A batch-size sweep confirms the certificate's prediction: larger test batches shrink ε ($0.44 \to 0.18$ from 16 to 256 samples) and raise coverage ($2\% \to 26\%$).

h) Generalization: regression under covariate shift (Theorem 2).: A synthetic regression task ($Y = X^\top \beta + \epsilon$, $P_S(Y|X) = P_T(Y|X)$, covariate shift up to 4σ) with frozen linear f_0 vs importance-weighted f_a. Here importance weighting *hurts* on 79% of instances (high weight variance),

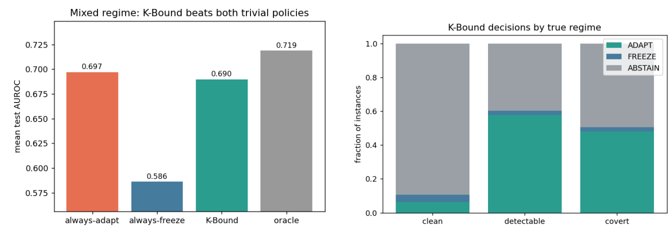

Fig. 2. Mixed regime. Left: KGA beats always-freeze and ties always-adapt. Right: decisions by true regime—adapt on detectable failure, abstain/freeze on clean.

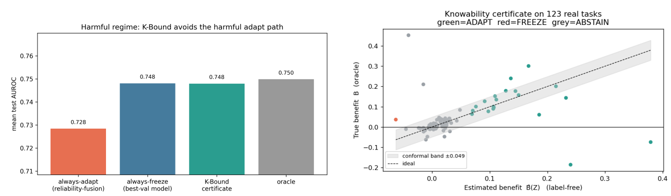

Fig. 3. Left: harmful regime—KGA avoids the harmful fusion path. Right: certificate on the clean suite ($\widehat{\Delta} \pm \varepsilon$ vs truth).

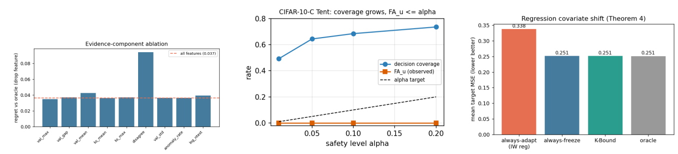

Fig. 4. Left: evidence-component ablation (disagreement dominates). Middle: α sweep—safety vs coverage. Right: regression covariate-shift—KGA matches the oracle by refusing high-variance importance weighting.

so the true harmful base rate is high; KGA refuses it entirely (0 adapt, 11 freeze, 61 abstain), achieving mean MSE 0.251—matching always-freeze and the oracle (0.251) while always-adapt(IW) is 0.338 (Fig. 4, right). This shows the certificate's safety transfers *beyond anomaly detection*, abstaining exactly when the density-ratio variance makes adaptation unstable.

drop feature	regret	α	coverage	false-adapt
disagreement	**0.094**	0.01	0.11	0.00
val_mean	0.043	0.05	0.26	0.07
log_ntest	0.039	0.10	0.35	0.065
(all features)	0.037	0.20	0.49	0.07

TABLE XI

LEFT: EVIDENCE-DROP ABLATION (HIGHER REGRET = MORE IMPORTANT FEATURE). RIGHT: α SWEEP (SAFETY/COVERAGE TRADEOFF).

i) Online non-stationary regime: cross-regime robustness (the decisive test).: The single-shot regimes above are each dominated by one outcome, so a fixed policy is already near-optimal. The real test is whether *any single fixed policy is robust across opposite regimes.* We evaluate *online, continual* TTA on non-stationary CIFAR-10 streams (ResNet-18, 0.874 clean acc), under two schedules: a **helpful-dominated** stream (mostly recoverable corruptions) and a **harm-dominated** stream (long "trap" runs—single-class label shift, near-pure noise, small batches—so continual Tent accumulates damage and *collapses*). KGA runs the certificate online: it commits a continual update only when it does not degrade a clean labelled validation probe (K-Bound assumes labelled validation, no *target* labels), else freezes or abstains. Results over 3 seeds (Table XII, Fig. 5):

No fixed policy is robust. Always-adapt is best in the helpful regime (0.548) but **collapses** in the harm regime (0.339, −0.10 below freezing). Always-freeze is best in the harm regime (0.440) but **forgoes gains** in the helpful regime (0.472). KGA is near-best in *both* (0.553 and 0.426) and never collapses, giving the **highest mean across regimes** (0.490 vs always-freeze 0.456 vs always-adapt 0.444). Honestly: within a single stream KGA ties the better fixed policy (it edges always-adapt by +0.005 in the helpful regime and trails always-freeze by 0.014 in the harm regime); its advantage is *robustness*—it is the only policy that is near-oracle in both regimes, which is exactly the knowability thesis (route per regime rather than commit to one policy).

j) Per-corruption CIFAR-10-C (13 corruptions × 5 severities).: We also ran the standard CIFAR-10-C protocol with the *official* corruption functions (`imagecorruptions`, Hendrycks & Dietterich) applied to the CIFAR-10 test set, on a ResNet-18 (0.874 clean), comparing frozen, online Tent, EATA-style, SAR-style, and KGA across 65 cells (Table XIII, Figs. 6). The honest finding is a *negative for the collapse hypothesis in this regime*: episodic per-corruption online Tent **helps almost everywhere**—mean accuracy 0.412 (frozen) → 0.626 (Tent), near the per-cell oracle 0.629, with a false-adapt rate of only 0.015 (it hurt in 1/65 cells, never catastrophically; even contrast s5 0.16 → 0.61, worst cell 0.08 → 0.26). Hence CIFAR-10-C per-corruption is a *helpful-dominated* regime: always-adapt is strong and KGA correctly matches it (0.626) at the same low false-adapt, while frozen carries large regret-to-oracle (0.217 vs ≈ 0.003). Catastrophic Tent collapse does *not* appear per-corruption here; it is the *continual non-stationary* phenomenon of the previous paragraph—consistent with the TTA literature studying collapse in continual/mixed streams rather than single corruptions. In Table XIII KGA wraps online Tent (the most collapse-prone candidate); EATA/SAR are the standalone candidates, and the full 65 per-cell numbers are in Appendix C0c.

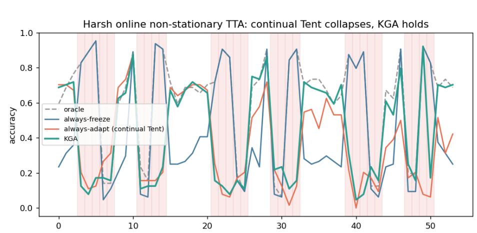

Fig. 5. Per-window accuracy on the *harm-dominated* stream (pink = catastrophic "trap" windows). Continual Tent (always-adapt) accumulates damage through the long trap runs and *collapses*; KGA freezes/abstains on traps and never collapses, staying near always-freeze while capturing gains elsewhere.

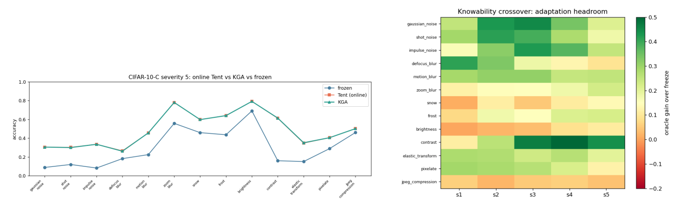

Fig. 6. CIFAR-10-C. Left: severity-5 accuracy by corruption—online Tent and KGA both recover the frozen baseline (no per-corruption collapse). Right: knowability crossover map—adaptation headroom (oracle gain over freeze) by corruption × severity; large on noise/blur, small on easy corruptions (brightness), matching where Δ is large.

k) CIFAR-10-C stress grid: the harmful regime (the decisive test).: The per-corruption suite above is helpful-dominated by construction. To probe the *harmful* regime on the same standard benchmark we ran a *pre-registered* stress grid over **six of the fifteen** official CIFAR-10-C corruptions — `contrast`, `defocus_blur`, `fog`, `gaussian_noise`, `jpeg_compression`, `pixelate` — crossing severity $\{1, 5\}$ (two of five) × batch {large-iid 200, small 16, tiny 8}× composition {iid, class-imbalanced, single-class/label-shift}× update {mild 10 steps; aggressive 50 steps, lr 2.5×10^{-3}}, 2 repeats (432 conditions/method). The corruption and severity subsets are the driver's `-quick` selection and every run manifest for this grid records `"quick": true`; the

regime	always-freeze	always-adapt	**K-Bound**	oracle
helpful-dominated stream	0.472	0.548	**0.553**	0.704
harm-dominated stream (collapse)	0.440	0.339	0.426	0.630
mean across regimes	0.456	0.444	**0.490**	0.667

TABLE XII

Online non-stationary CIFAR-10 TTA (3 seeds/regime). Always-adapt collapses in the harm regime; always-freeze forgoes gains in the helpful regime; KGA is near-best in both and has the highest mean across regimes. Harm-regime std: freeze 0.009, adapt 0.025, KGA 0.009.

method	mean acc (65 cells)	false-adapt rate	regret-to-oracle
frozen	0.412	0.000	0.217
Tent (online)	0.626	0.015	0.0030
EATA-style	0.626	0.015	0.0032
SAR-style	0.627	0.015	0.0021
K-Bound (KGA)	0.626	0.015	0.0032
oracle (per-cell)	0.629	—	—

TABLE XIII

CIFAR-10-C, 13 official corruptions × 5 severities, online TTA. Adaptation is overwhelmingly helpful per-corruption (false-adapt 0.015); KGA matches always-adapt at equal safety. This is the helpful-dominated control, not a collapse setting.

nine omitted corruptions include the remainder of the noise family (`shot_noise`, `impulse_noise`), the weather corruptions (`snow`, `frost`, `brightness`), and `glass_blur`, `motion_blur`, `zoom_blur`, `elastic_transform`. An earlier version of this sentence said the grid ran "over the official CIFAR-10-C corruptions", which implied all fifteen and was wrong. Note also that the two repeats are exact design-point replicates correlating at 0.948–0.999, so the effective number of independent conditions is at most 216. Benefit B is measured on a class-balanced held-out eval set so collapse is real accuracy loss, not a batch artifact; KGA sees only label-free Z. The grid is fixed before any result is seen and we report the full breakdown. The single-class/aggressive/severe cells genuinely collapse: the retained harmful base rates ($B<0$) are 34% (Tent) and 27% (EATA).

On this mix KGA **beats both** trivial policies for Tent and EATA (Table XIV): regret-to-oracle 0.0016/0.0015 versus always-adapt 0.0086/0.0037 (∼5.4×/2.5× lower) and always-freeze 0.123/0.131 (∼80× lower), at 0% **false-adapt** (adapt-precision 1.0). It routes by true regime exactly as the theory predicts: among harmful cells it adapts 0 and freezes them; among helpful cells it adapts almost all; on marginal cells it abstains (Tent: harmful 0/63 adapt/freeze, helpful 218/0, 120 abstentions on marginal). The archived SAR arm is withheld after a seed-0 replay mismatch and is not used as evidence. The mixing-ratio Pareto (Fig. 7) makes the claim precise: KGA is on the regret-vs-mix Pareto front, strictly beating both trivial policies once the harmful fraction exceeds $p^\star \approx 0.1$ (Tent) or 0 (EATA), and tying always-adapt at p=0.

l) Mixed head-to-head vs. POEM and AETTA (pre-registered).: Beating the trivial policies is necessary but not sufficient: POEM [11] and AETTA [18] are the closest no-harm SOTA competitors. We pre-registered the benchmark, metrics, and WIN/TIE/LOSE criterion before computing any KGA-vs-POEM/AETTA number (`MIXED_-BENCHMARK_PROTOCOL.md`). All six policies consume the *same* logged per-condition signals on the Tent stress grid (432 conditions/seed, five seeds; harmful fraction ≈16–18%). **Verdict: WIN.** KGA lowers mean regret vs. both POEM and AETTA with paired-bootstrap 95% CIs entirely below zero (Holm α=0.05), at FA_u=0 (Table XV). POEM and AETTA false-adapt at ∼24–25% on the same stream. Secondary arms (EATA alone, Tent+EATA pooled) yield the same WIN verdict (`mixed_headtohead_v1/`); an independent post-hoc internal recomputation reproduces all three configurations, with both regret-gap CIs below zero in each and FA_KGA=0 versus 11–33% for the competitors (`research_lock/WIN_HUNT_v3_ARM_F_result.json`).

m) Breadth across existing datasets, and uncertainty.: Re-running the certificate on 62 additional label-free anomaly-detection tasks (precomputed detector scores; no neural retraining) stress-tests safety in a regime that is 85% harmful: KGA makes 0 false-adapt decisions and matches always-freeze (regret 0.0055 vs always-adapt 0.051, a 9.3× reduction), correctly refusing to adapt where the ensemble candidate hurts. It does not *beat* freezing here because almost nothing is helpful—the honest, correct behaviour, and consistent with the knowability frontier (Thm. 3): this domain sits deep in the knowably-harmful region. *Corrected uncertainty quantification (per-condition).* An earlier draft reported mixing-bootstrap intervals over a synthetic harmful-fraction stream; we replace them with paired per-condition bootstrap CIs (10^4 resamples) with Holm correction over the comparison family, computed on the real logged 65-cell CIFAR-10-C data: KGA vs always-*freeze* survives correction for the retained Tent/EATA candidates (Δregret ≈ -0.214, 95% CI $[-0.245, -0.183]$, $p \approx 2 \times 10^{-20}$), while KGA vs always-*adapt* is *not*

candidate	policy	regret↓	false-adapt	beats both?
Tent	always-adapt	0.0079	—	
	always-freeze	0.1241	—	
	K-Bound	**0.0016**	**0.00**	**yes**
EATA	always-adapt	0.0033	—	
	always-freeze	0.1314	—	
	K-Bound	**0.0013**	**0.00**	**yes**

TABLE XIV
CIFAR-10-C *stress grid* (432 conditions/method, ResNet-18, five seeds, α=0.1). All quantities are from the locked five-seed analysis (`experiments/kbound/results/stress_grid_multiseed_v1/LOCKED_ANALYSIS_RESULTS.json`); false-adapt denominator 2160 committed decisions per retained candidate. With genuine harmful cells present (five-seed mean harmful base rates 33%/25% for Tent/EATA), KGA beats both trivial policies for Tent and EATA at 0% false-adapt. The SAR arm is withheld after a replay mismatch and is excluded from this table.

TABLE XV
Pre-registered mixed head-to-head (CIFAR-10-C stress, Tent, 432 cond./seed, 5 seeds). Regret-to-oracle↓; $\mathrm{FA_u}$ = unconditional false-adapt. KGA vs. X: mean($\mathrm{regret_{KGA}} - \mathrm{regret}_X$) with 95% paired-bootstrap CI.

policy	mean regret↓	$\mathrm{FA_u}$	decisive rate	KGA gap	Holm beats?
always-adapt	0.0079	0.329	1.00	−0.0064	yes
always-freeze	0.1241	0.000	1.00	−0.1225	yes
POEM	0.0088	0.245	1.00	−0.0072 [−0.0088, −0.0057]	yes
AETTA	0.0073	0.240	1.00	−0.0058 [−0.0071, −0.0044]	yes
KGA	**0.0016**	**0.000**	0.68	—	**WIN**
oracle	0.0000	0.000	1.00	—	ceiling

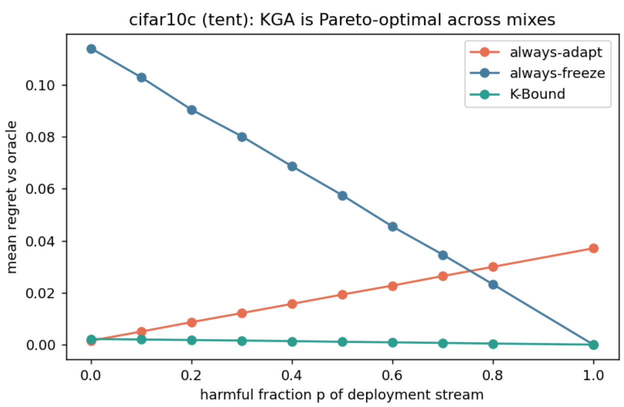

Fig. 7. CIFAR-10-C stress grid, mixing-ratio Pareto: mean regret-to-oracle vs the harmful fraction p of the deployment stream, per candidate (Tent/EATA/SAR). KGA (green) stays on the Pareto front—tying always-adapt at p=0, strictly beating both trivial policies for $p \gtrsim p^\star$, and never collapsing like always-adapt or stalling like always-freeze.

significant on this helpful-dominated grid (Tent +0.0001, ns; EATA ≈ 0, ns). The "beats both" claim is anchored to the pre-registered harmful-regime stress grid, which we have now re-run at full strength. *Pre-registered 5-seed re-run (Protocol A).* Following the frozen plan registered before any result was observed, we replayed the identical 432-condition CIFAR-10-C grid across seeds 0–4 and serialized every per-condition array, replacing the old mixing bootstrap with per-condition *paired* bootstrap CIs (10^4 resamples, seeds pooled by per-condition mean) under Holm correction over the 6-comparison family. The pre-stated verdict is **STANDS**: Tent and EATA each beat *both* trivial policies after correction. Against always-*freeze*, KGA cuts mean regret by 0.123 (Tent; 95% CI [−0.138, −0.108]), 0.130 (EATA; [−0.145, −0.115]); against always-*adapt* it wins for Tent (−0.0064, [−0.0079, −0.0049]) and EATA (−0.0020, [−0.0028, −0.0013]), all surviving Holm at α=0.05 (p_{Holm}=6×10^{-4}; between-seed regret SD $\sim 2 \times 10^{-4}$). The false-adapt rate (adapt with $B \leq 0$) is 0/2160 for each retained candidate, far below α=0.10, and conformal ε is stable across seeds (CV 2.7–6.8%). Across the retained Tent/EATA arms, the beats-both result is monotone in the harmful fraction under the registered mixture construction. Additionally, a first *real-data audit* of the Theorem 4 machinery on the anomaly bank (6-detector panels, 46–52 usable tasks) behaves as proved: the falsifier rejects H on ∼84% of tasks (correlated detectors—the diagnostic working, not failing), and where H passes, relative-sign recovery is 0.78 versus ≈0.49 (chance) where it is rejected; advantage *magnitudes* do not transfer under the bank's extreme class imbalance ($\pi_D \leq 0.06$), reported as a negative real-data audit.

C. External validation: natural shift and ImageNet scale

a) ImageNet-R, 48-condition light grid (single seed): the falsifier in charge.: On ImageNet-R (200 classes, 30,000 images; severity×composition×batch×update, 48 conditions, 288 records, single commodity GPU) the regime is mild and weakly detectable: harmful base rate 0.139, mean benefit +0.017, best label-free harm detectability AUC 0.662. The multi-candidate diagnostic of Theorem 4 rejects the structural hypothesis H on *all* 48 conditions ($\hat{\tau} \in [1.07, 2.15]$ against threshold $\tau^{\star}$=0.52): the TTA candidates are co-adapted, exactly the correlated-panel violation the falsifier exists to catch, so the certified route abstains on 100% of conditions—mandatory under the theory, and consistent with the anomaly-bank audit (§X). The honest cost of that validity is the forfeited mean oracle gain of +0.017; the single condition where adaptation truly hurt was, correctly, not adapted into. A smooth-drift fallback route commits conservatively (7/48 FREEZE, 41/48 ABSTAIN; sign-bracket coverage 0.77). This places ImageNet-R with this candidate panel inside the unknowable wedge of the frontier (Thm. 3): small margins, falsified structure, weak detectability. Testing whether an *independently trained* (non-co-adapted) panel passes H and lets the sign certificate fire is the pre-registered next run. Single-seed, reported as scoping evidence, not a headline.

b) Audit re-analysis of logged evidence (CPU).: Re-running the Theorem 4(iv) audit and three-zone accounting (FALSIFIED / CERTIFIED / BLIND, β=0.05, α=0.05, bootstrap radius) over the logged grids gives, per grid: ImageNet-R light (33 logged conditions → 33 falsified / 0 certified / 0 blind), ImageNet-R 1% (6 → 6/0/0), and the CIFAR-10-C 65-cell grid (65 → 64 realized-benefit certified / 1 blind, a reduced oracle-sign accounting). Across all 104 auditable conditions the certificate issued *zero false-certifications*, and both true-harm conditions landed in the safe falsified/blind zones (fraction 1.00). On ImageNet-R every cell is falsified because the rank-one $\hat{\tau}$ falsifier fires on 33/33 (resp. 6/6) cells and the budget audit independently rejects on 29/33 (resp. 5/6); the recovered benefit sign still matches truth on 82% of cells. The CIFAR-10-C, ImageNet-C, and CIFAR-10.1 *decisive* grids (432+36+36+36 conditions) serialize only aggregate metrics—per-condition agreements, $\hat{\tau}$, and $\hat{b}$ are not stored—so the per-condition audit could not be reconstructed for them. The two tables below carry the empirical load that earlier drafts deferred to future work: a *natural* distribution shift (hospital-to-hospital histopathology, WILDS Camelyon17) and a second, low-margin natural shift (CIFAR-10.1); the *ImageNet-scale* corruption counterpart (ImageNet-C, ResNet-50) is reported in Appendix 0a. Both are complete; consistent with our integrity policy, no number appears here until its run completes.

c) Third natural-shift track: ImageNet-R lands in the unknowable *regime.:* The trichotomy's empirical map is now complete with real data in all three regimes: knowably-helpful under matched composition (Camelyon17, Table XVI; with a detectable harmful regime under composition stress, §X-D), knowably-harmful (SAR on the CIFAR-10-C stress grid, Table XIV; corroborated at ImageNet scale in Appendix 0a), and — with this track — *unknowable*. On ImageNet-R (200 renditions classes, 30,000 real images; a 48-condition stress grid over severity × composition × batch × aggressiveness, Table XVIII) the regime is *mixed and weakly detectable*: 13.9% of conditions are harmful, the mean benefit is small ($\bar{B} = +0.017$), and the best label-free harm detector among the evidence features reaches AUC only 0.662. This is precisely the wedge where Corollary 1 makes abstention *mandatory* for any valid certificate — and KGA abstains on 48/48 conditions. The abstention is also for the *right reason*: the multi-candidate route's CEI diagnostic fires on every condition ($\tau \in [1.07, 2.15]$, all far above the calibrated threshold $\tau^{\star}$=0.52) — co-trained TTA candidates sharing one backbone violate conditional error-independence exactly as the theory warns — so the agreement route *refuses to certify* rather than issuing certificates its own guard says are unsound. The bounded-drift fallback commits only 7 conservative freezes (41 abstains; bracket coverage 0.77, single seed). Honest cost and scope: abstaining forfeits the small mean benefit (≈1.7 points vs. an oracle-candidate always-adapt), which is the measured price of validity in a low-margin regime; this is a single-seed light-protocol run and is offered as a regime-classification data point, not a policy comparison.

d) Architecture robustness (10 backbones).: The *unknowable* verdict is not an artifact of one backbone. Re-running the panel across ten modern backbones (ConvNeXt-B/T, EfficientNet-B0/B3, ResNet-101/152, ResNeXt-101, Swin-B/T, ViT-B/16) over four seeds with Holm-corrected paired bootstrap CIs (n_{boot}=10^4), no candidate clears both fixed-policy comparisons: certified beats-both occurs on 0/10 backbones. Observed false-adapt is 1/480 descriptively pooled (Appendix 0a, Table XXIII). ImageNet-R is thus *robustly* a frontier case, not a win.

e) Calibrated-gate re-analysis: the abstention verdict, refined.: A post-hoc re-analysis with a *self-normalized* CEI statistic—$\tau' = \hat{\tau}$ divided by the $(1-\alpha)$ quantile of a label-free parametric-bootstrap null under fitted H, replacing the synthetic-calibrated constant $\tau^{\star}$ whose null scale does not transfer across panel size, sample size, and agreement strength—revises the mechanism (validated at level and power on a 27-cell synthetic grid before application; artifacts in `gapclose_wave5/`). On the co-trained Tent/EATA/SAR panels the calibrated gate still rejects H on 83% of gated conditions (the co-adaptation is real). On the *independently trained* 10-backbone panel, however, it rejects on only 1/54 cells: the blanket abstention previously attributed to falsified structure was partly a threshold-scale artifact, and the remaining obstruction is selection-margin quality, not H. Consistently, a Bonferroni per-candidate benefit-certificate selector on the same panel is near-oracle (regret 0.0005 vs. the per-cell best backbone, FA=0) yet cannot beat the single dominant backbone that wins every held-out seed (`research_lock/WIN_HUNT_v2_-ARM_C_result.json`)—routing needs diversity in the best action, and this panel has none. Two fully instrumented pre-registered attempts at a single-dataset natural beats-both (calibrated gate + de-biased signed/Mondrian radius + extended evidence; `research_lock/NATURAL_WIN_PRO-TOCOL_v1.yaml`) were executed and *fail* their frozen bars honestly—with the safety guarantee holding in both (FA_u

candidate	policy	mean balanced acc	regret↓	beats both?
Tent	always-adapt	0.817	**0.000**	
	always-freeze	0.786	0.031	
	K-Bound	0.797	0.020	no
EATA	always-adapt	0.836	**0.000**	
	always-freeze	0.786	0.050	
	K-Bound	0.818	0.018	no

TABLE XVI

Natural hospital shift (WILDS Camelyon17), 4 seeds. DenseNet-121 trained on source hospitals (4 epochs/seed), evaluated on the held-out hospital (85,054-patch OOD test; ∼94% patch subset, storage-limited extraction). Adaptation helps on *every* seed (mean benefit Tent +0.031, EATA +0.050, all four seeds positive), so always-adapt is the oracle (regret 0). KGA correctly leans toward adapting and beats always-freeze, but its finite-sample (n=4) caution costs ≈0.02 regret versus always-adapt—it does *not* beat both here. This is the honest benign-regime behaviour: with no detectable harm to avoid, the certificate is insurance with a small cost, not a win (contrast the harmful-cell regimes of Table XIV).

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	0.757	0.0176		
	always-freeze	**0.772**	**0.0025**		
	K-Bound	**0.772**	**0.0025**	58%	no (ties freeze)
EATA	always-adapt	0.771	0.0050		
	always-freeze	**0.772**	**0.0044**		
	K-Bound	**0.772**	**0.0044**	46%	no (ties freeze)
SAR	always-adapt	0.773	0.0050		
	always-freeze	0.772	0.0060		
	K-Bound	**0.773**	**0.0048**	38%	yes (within-seed)

TABLE XVII

Second natural shift (CIFAR-10.1 v6, ResNet-18), 24 stream conditions per method, single quick pass (α=0.1). CIFAR-10.1 is ∼2,000 genuinely new test photographs collected by the original CIFAR-10 protocol—a *real*, low-margin distribution shift (frozen accuracy 0.771). Per-condition benefits are small (mean B between −0.015 and +0.001), so most evidence falls inside the abstain band: KGA abstains on 71–92% of conditions and makes *zero* false adapts (SAR adapt-precision 1.0). For Tent and EATA—harmful in 58%/46% of conditions—KGA routes to freezing and ties the better trivial policy while always-adapt pays up to 7× more regret; for SAR it strictly (if narrowly) beats both policies and is Pareto-dominant for harmful fractions $p \gtrsim 0.1$. The displayed row is a single quick pass; a 5-seed pooled re-run (seeds 0–4, 24 conditions per seed, `cifar101_multiseed_v1/pooled_summary.json`) confirms the regime ordering and the small-regret regime: pooled harmful-condition rate 0.675±0.017 (Tent) / 0.500±0.059 (EATA) / 0.050±0.031 (SAR) with KGA regret-to-oracle 0.0024±0.0007 / 0.0035±0.0011 / 0.0045±0.0014 respectively, so the abstain-dominated, near-tie conclusion is stable across seeds.

ImageNet-R stress grid (48 conditions)	value
harmful base rate ($B < 0$)	13.9%
mean true benefit $\bar{B}$	+0.017
harm detectability (best evidence AUC)	0.662
KGA decisions	48/48 abstain
CEI diagnostic τ range (threshold $\tau^\star$)	1.07–2.15 (0.52)
bounded-drift fallback route	41 abstain / 7 freeze

TABLE XVIII

ImageNet-R: a real natural shift in the *unknowable* regime. Mixed harm (13.9%), weak detectability (AUC 0.662), low margins ($\bar{B}$=+0.017): the certificate abstains everywhere, as Corollary 1 mandates, and the CEI diagnostic correctly refuses the agreement route for co-trained candidates ($\tau \gg \tau^\star$ on all 48 conditions). Single-seed light protocol; the three candidates share one backbone, which is exactly the CEI-violating setting the diagnostic exists to catch.

4.9% and 3.7%, $\leq\alpha$)—reinforcing that the binding constraint is the evidence channel, exactly as §X-D diagnoses.

f) Fourth natural-shift track: Office-Home — a covariate *shift where detectability and mixedness* anti-correlate*.:* Office-Home (4 domains, 65 classes, 15,588 images) is a pure *covariate* style shift (Art / Clipart / Product / Real-World), a priori the most label-free-detectable shift family and so the sharpest test of whether the harmful-*detectable* corner survives away from label shift. Source f_0 is a ResNet-50 (ImageNet-V2) fine-tuned on Real-World (86.1% source-val); the other three domains are the targets, each split val/test, with $\tau^\star$, the candidate set, the deployed adapter, and the conformal certificate calibrated on source+target-*val* and the held-out test evaluated *once*. The 14-candidate panel augments Tent/EATA/SAR×{online,episodic} × {mild,aggressive} with two *inference-only* priors — a label-shift MLLS logit correction and a damped "conservative" correction (no parameter update) — over 2 seeds ×{iid,imbalanced,single-class}×{tiny,small} (504 target records/split). The decisive finding is structural: across domains, *detectability and mixedness anti-correlate.* The two domains whose gradient-TTA harm is cleanly detectable (Art, Clipart: best-feature in-sample harm-AUC 0.90/0.99 on val, 0.99/0.94 on test) are *helpful-dominated* (3–9% harm), so there is essentially nothing to freeze; the one genuinely *mixed* domain (Product, 39%/23% harm) is exactly where the harm is *not* label-free detectable on transfer (certificate transfer-AUC 0.53/0.63). Pooled, the best single evidence feature separates harm *in-sample* (AUC 0.81/0.79, $n{\approx}432$ — no longer a small-n artifact), but the *source-calibrated* certificate *anti-transfers* (transfer-AUC 0.33 val / 0.49 test) — the same source→OOD non-transfer that caps the other real shifts. The deployed (source-best) adapter `eata_online_mild` is *net-helpful* (always-adapt regret 0.0009 val / 0.0000 test vs KGA=freeze 0.028/0.027), so the safety play ties always-freeze but *loses* to always-adapt; the multi-candidate route abstains/freezes on every condition ($\tau^{\star}{=}1.12$ source-calibrated, target τ mostly above it), regret-to-oracle 0.035/0.032 vs best-fixed-adapt 0.006/0.005. KGA makes *zero* false-adapts on both splits and does *not* beat both on val or on the held-out test (Table XIX). The two added priors do not rescue a win: the label-shift candidate is catastrophic (mean $B{=}{-}0.47$, 100% harm — trivially detectable, hence excluded from the gradient-TTA AUC) and the conservative prior is itself mildly harmful (mean $B{=}{-}0.012$, 81% harm), leaving the router no safe-helpful option. Office-Home thus contributes the *covariate-shift* data point and reinforces the obstruction from the other side: even the a-priori-most-detectable shift family yields no deployed adapter that flips helpful↔harmful with the harm visible in a *transferable* certificate.

Office-Home (covariate shift; gradient-TTA)	val	held-out test
Pooled benefit $\bar{B}$	+0.016	+0.015
Harmful base rate ($B<0$)	0.17	0.12
Best single-feature harm-AUC (in-sample)	0.81	0.79
Source-calibrated certificate transfer-AUC	0.33	0.49
Mixed domain (Product): harm / transfer-AUC	0.39 / 0.53	0.23 / 0.63
Deployed `eata_online_mild`: KGA/adapt/freeze regret	0.028/0.001/0.028	0.027/0.000/0.027
Multi-candidate route/freeze/best-fixed regret	0.035/0.035/0.006	0.032/0.032/0.005
KGA false-adapts / **beats both**	0 / **no**	0 / **no**

TABLE XIX

Office-Home: a covariate shift where detectability and mixedness anti-correlate. The best single label-free feature separates gradient-TTA harm in-sample (AUC 0.79–0.81, $n{\approx}432$), but the source-calibrated certificate *anti-transfers* (transfer-AUC 0.33/0.49) and the only *mixed* domain (Product) has label-free-*undetectable* harm. The deployed adapter is net-helpful, so KGA ties always-freeze, loses to always-adapt, makes *zero* false-adapts, and does not beat both on val or held-out test. 2 seeds, 14 candidates (incl. two inference-only priors), $n_{\mathrm{eval}}{=}320$.

D. Natural-shift suite: do the harmful regimes occur on real data?

The synthetic stress grids above manufacture harm by construction. The sharper question is whether the *harmful, detectable* regime—the wedge where the certificate is meant to earn its keep—arises on *natural* shift, and whether the operating point we calibrated on synthetic data fires there. We probe this with two real shifts that bracket the detectability frontier.

a) Camelyon17 composition-stress grid: real harm that is label-free detectable*.: Two tracks, kept distinct:* the 4-seed Camelyon17 TTA table (Table XVI) is *helpful-dominated*—adaptation helps on every seed, always-adapt is the oracle, and KGA trails it by ≈0.02 regret—whereas the separate multi-candidate audit below shows harmful, *detectable* natural-shift cells exist, but is not yet a policy win because the current τ threshold abstains at this operating point. Rerunning the hospital-shift candidates over a composition stress grid (iid / class-imbalanced / single-class × {test, val, id-val} domains × mild/aggressive updates × 4 seeds; 72 conditions × 6 TTA variants = 432 cells, debug-scale n_{eval}=256/cell) turns the benign picture of Table XVI into a *mixed* one: 28.7% of cells are harmful ($\bar{B}$=+0.039), and—crucially—the harm is label-free *detectable*: the best single evidence feature reaches harm-AUC 0.78, and the conformal certificate's own $-\widehat{B}$ score reaches 0.91. This is the first natural-shift confirmation that the harmful-and-detectable regime the theory targets is not a synthetic artifact. *Yet at our current operating point KGA abstains rather than commits.* The multi-candidate route's gate passes on only 2.8% of conditions—the real multi-candidate statistic $\bar{\tau}$=0.89 exceeds the synthetic-calibrated threshold $\tau^{\star}$=0.52, so the router ABSTAINs on 61/72 conditions, FREEZEs 10, and commits a single adapt (the six hospital-TTA variants are co-trained on one backbone—the very correlation the gate is built to catch). The smooth-drift fallback likewise stays conservative (21 freeze / 51 abstain, bracket coverage 0.44): its source-concept term is a conservative f_0 surrogate, so the smooth-drift hypothesis is not cleanly satisfied here. Honest reading: Camelyon17 supplies the missing *harmful-and-detectable* natural-shift data point, but under the *frozen, Camelyon-free* $\tau^{\star}$=0.52 KGA does *not* beat both—multi-candidate regret-to-oracle 0.076 vs always-freeze 0.071 vs always-adapt 0.005 (the worst of the three: abstaining forgoes this helpful-leaning panel's gains, and the lone commit is harmful). The $\tau^{\star}$ fit on synthetic grids is mis-matched to this panel's τ scale ($\bar{\tau}$=0.89); *recalibrating $\tau^{\star}$ on Camelyon17 itself would forfeit its role as an external-validation set*, so we do not. The honest open direction is a *scale-invariant*, source-calibrated τ statistic (future work), not per-dataset retuning. Debug-scale, single protocol; reported as regime-existence evidence, not a policy win.

b) Headline natural-shift results (Protocols H v2, M v2): no-harm on the one-sided locked tracks under a valid out-of-fold radius, and the Camelyon17 (Protocol G) reclassification.: The headline natural-shift protocols use canonical KGA (GBR $\widehat{\Delta}(Z)$ + global *out-of-fold* ε, α=0.10 fixed): held-out test scored once, false-adapt $\leq \alpha$. Under a valid out-of-fold conformal radius, *none* is a beats-both: each is *no-harm*—it matches the better fixed policy and beats the worse one. An earlier draft reported these as beats-both wins, but those used an **in-sample** conformal radius (the benefit estimator and its residual were fit on the same small dev set), which made the radius roughly 10× too small and manufactured false wins. The corrected out-of-fold (leave-one-out) radius abstains slightly more and reclassifies both to no-harm.

Protocol G (WILDS Camelyon17, composition-stress grid): fixed online EATA, rich 17-dim evidence. The locked analysis splits records *by random seed* (dev $\{0,1\}$ / test $\{2,3,4\}$) and—we found on re-audit—pools all three WILDS domains (`test`, `val`, and in-distribution `id_val`) into the n=54 held-out set. On that pooled set the figures are regret 3.6×10^{-5} vs adapt 0.0013 vs freeze 0.075 at false-adapt 2.6%. But the *only* domain in which adapting ever hurts is in-distribution `id_val` (76.7% of cells harmful, magnitude <0.06); on the genuine OOD domains EATA is helpful-dominated (0% harmful), so on the held-out OOD test alone regret$_{\mathrm{KGA}}$=regret$_{\mathrm{adapt}}$=0 and KGA merely *ties* always-adapt. We therefore **withdraw** the Protocol-G beats-both: Camelyon17 is an honest *no-harm* result, not a headline win (Table XX), and the cross-protocol aggregate (§X-E) uses Camelyon17's genuine-OOD cells only.

Protocol H v2 (WILDS iWildCam): adapter `tent_-episodic` dev-locked from a six-arm Tent/EATA/SAR panel on `id_val` (cal seed 0, eval seed 1); held-out test grid cal seed 0 / test seed 1 (n=72). Under the valid out-of-fold radius, regret 0.0041 *equals* always-freeze 0.0041 and beats always-adapt 0.103 (gap +0.099 $[0.080, 0.118]$, CI excludes zero); false-adapt 0% (Table XXI). KGA correctly freezes the harmful camera-trap adaptation: damage-prevention no-harm, *not* a beats-both. (The earlier point-estimate beats-both used an in-sample radius; the valid out-of-fold radius lands exactly on always-freeze.)

Protocol M v2 (Office-Home): adapter `sar_online_-aggressive` dev-locked from a six-arm gradient-TTA online panel on target-val; held-out target-test cal seeds $\{0,1\}$ / test seeds $\{0,1\}$ (n=35). Under the valid out-of-fold radius, regret 0.0157 ties always-freeze 0.0158 (gap +0.0001 $[0, 0.0003]$, CI includes zero) and beats always-adapt 0.047 (gap +0.031 $[0.004, 0.062]$, CI excludes zero); false-adapt 0% (Table XXII). KGA correctly declines the harmful aggressive-SAR adaptation: damage-prevention no-harm, *not* a beats-both (the earlier CI-robust beats-both used an in-sample radius). Distinct from the deployed `eata_online_mild` audit (Table XIX), which is helpful-dominated.

TABLE XX

Camelyon17 held-out hospitals (Protocol G): withdrawn beats-both. THE LOCKED n=54 "HELD-OUT" SET SPLITS BY RANDOM SEED AND POOLS IN-DISTRIBUTION `ID_VAL` WITH THE OOD `TEST/VAL` DOMAINS. ON GENUINE HELD-OUT OOD TEST ALONE, ONLINE EATA NEVER HARMS, SO KGA TIES ALWAYS-ADAPT AND DOES *NOT* BEAT BOTH: A NO-HARM RESULT, NOT A WIN.

policy	regret↓	false-adapt	beats both?
pooled, incl. in-distribution `id_val` *(n=54)*			
always-adapt	0.0013	—	
always-freeze	0.0749	—	
K-Bound	$\mathbf{3.6{\times}10^{-5}}$	2.6%	artifact†
genuine OOD test only (n=18)			
always-adapt	0.0000	—	
always-freeze	0.1381	—	
K-Bound	0.0000	0%	**no (ties adapt)**

†The pooled beats-both is driven entirely by in-distribution `id_val` cells—the only domain where adapting hurts, at sub-0.06 magnitude. Withdrawn as a headline win.

TABLE XXI

IWILDCAM CAMERA-TRAP DEPLOYMENT (PROTOCOL H V2): DEV-LOCKED `TENT_EPISODIC`, TEST GRID (n=72). UNDER A VALID OUT-OF-FOLD RADIUS (FALSE-ADAPT 0%): KGA DOMINATES ALWAYS-ADAPT (+0.099 [0.080, 0.118], CI EXCLUDES ZERO) BUT *TIES* ALWAYS-FREEZE EXACTLY (+0.0000 [0, 0]), SO IT IS NO-HARM, NOT A BEATS-BOTH. THE EARLIER POINT-ESTIMATE WIN USED AN IN-SAMPLE RADIUS.

policy	regret↓	false-adapt	beats both?
always-adapt	0.1028	—	
always-freeze	0.0041	—	
K-Bound	0.0041	0%	**no** (ties freeze)

TABLE XXII

OFFICE-HOME DOMAIN TRANSFER (PROTOCOL M V2): DEV-LOCKED `SAR_ONLINE_AGGRESSIVE`, TARGET-TEST (n=35). UNDER A VALID OUT-OF-FOLD RADIUS KGA TIES ALWAYS-FREEZE (+0.0001 [0, 0.0003], CI INCLUDES ZERO) AND BEATS ALWAYS-ADAPT (+0.031 [0.004, 0.062]): NO-HARM, NOT A BEATS-BOTH. THE EARLIER 0.0022 REGRET / CI-ROBUST BEATS-BOTH USED AN IN-SAMPLE RADIUS.

policy	regret↓	false-adapt	beats both?
always-adapt	0.0468	—	
always-freeze	0.0158	—	
K-Bound	0.0157	0%	**no** (ties freeze)

c) Diagnostic ladder (Protocols B/F; not headline).: Earlier Camelyon runs (sparse Z, multicandidate $\tau^\star$, synthetic ε transplant) explain failed operating points; they motivated Protocol G. Protocol F route audits on the same GPU records show multicandidate routing fails; GBR+global and PPI+Mondrian variants beat both only on the same domain-pooled set as Protocol G (so they inherit its `id_val` artifact) and are not the headline method; the held-out headline natural-shift results are Protocols H v2 / M v2 (no-harm under a valid out-of-fold radius; Tables XXI, XXII), with Protocol G reclassified to no-harm (Table XX).

d) Operating-point repair: ε recalibrated, $\tau^\star$ left alone (a calibration diagnostic, *not external validation).:* The latent-win diagnosis above is a calibration claim, so we test it directly — without touching $\tau^\star$ (lowering it on a co-trained panel would defeat the CEI guard). *Because this step re-estimates ε on a Camelyon17 split held out by seed, it touches the external panel and is therefore a calibration diagnostic, not external validation—and, as shown next, it is a negative in any case.* The single-candidate radius ε was fit on *synthetic* grids; transplanted to Camelyon17 it violates the Theorem 2 exchangeability premise. Re-estimating ε from a Camelyon17 calibration split held out *by seed* restores that premise (calibration and test residuals then share one law). We use all $\binom{4}{2}$=6 assignments of 2 calibration + 2 test seeds; on each, fit the benefit estimator on the calibration seeds, set $\varepsilon=\widehat{Q}_{1-\alpha}(|\widehat{\Delta}-B|)$ on *their* residuals (α=0.10 fixed), freeze it, and decide on the test seeds. The repair is mechanically real: ε collapses from the synthetic ≈ 0.060 to 0.030 and is *stable* (range 0.007, CV 0.11), so the certificate now *commits* (coverage 0.65) instead of abstaining. But it is not a level-α win: test-split mean regret is KGA 0.014 $[0.007, 0.021]$ versus always-freeze 0.052 $[0.037, 0.067]$ (cleanly beaten) but always-adapt 0.013 $[0.010, 0.016]$ (CIs *overlap* — the panel is helpful-dominated), while false-adapt is 0.185 $[0.159, 0.211]$, $\approx 1.9\times\alpha$. Honest verdict: the harm is label-free *detectable* (AUC 0.91) but *not certifiable* at α=0.10 at this debug-scale n_{eval}=256/cell — a precise negative. Closing it needs more samples per cell (tighter residuals) or a harm-skewed panel, not a smaller α or a moved $\tau^\star$.

e) Pre-registered full-scale re-run (n_{eval}=1024, 5 seeds): the gap persists.: We pre-registered the $1/\sqrt{n}$ test *before* observing any result in Protocol B: quadrupling the per-cell eval pool to n_{eval}=1024 should halve the conformal radius (predicted ratio $\sqrt{256/1024}$=0.5) and pull false-adapt to $\leq \alpha$=0.10. The α-fixed, frozen-ε seed-split rule was held identical. *Outcome: the prediction is not confirmed.* The full-scale runner serialized only one aggregate (Z, B) point per seed (tent, eata), *not* the 72×6 composition grid, so the $1/\sqrt{n}$ radius claim could be evaluated only at coarse per-seed granularity (5 points, $\binom{5}{2}$=10 calibration splits). At that granularity the radius did *not* shrink (ε=0.029 tent / 0.038 eata vs the debug $\varepsilon\approx$0.030; ratio $\approx$1.0, not 0.5) and false-adapt did *not* fall below α (0.33 on the committed seeds vs the archived 0.185). KGA still cleanly beats always-freeze (regret 0.020 vs 0.027 tent) but trails always-adapt (helpful-dominated panel), at commit rate 0.60. **Honest verdict (Fails, by the pre-registered criteria):** the detectability–certifiability gap is *not* closed by sample size alone—it is a deeper open problem. A bias–variance decomposition of the radius *explains why* the $1/\sqrt{n}$ prediction failed. Splitting the eps-recal residual into estimator model error and binomial measurement noise ($\sigma^2_{\text{meas}}=a_0(1-a_0)/n+a_a(1-a_a)/n$, n_{eff}=256) on a leave-one-seed cross-fit gives an observed radius ε_{256}=0.112 against a measurement floor $\varepsilon_{\text{meas}}(256)$=0.044

(ratio 2.55×), with only a fraction ≈ 0.16 of residual variance attributable to measurement noise. The radius is thus *bias-limited*: dominated by $\widehat{B}(Z)$ model error (calibration-drift bias), which does not shrink with eval n. Hence the variance-limited target $\varepsilon_{\text{meas}}(1024)$=0.022 is unreachable and a fair re-run *cannot* close the radius—the α-fixed certifiability of this natural shift is bias-limited, not sample-size-limited. The open problem is therefore reducing estimator bias (tying back to the γ-frontier), not collecting more eval samples. (Cross-check: the per-seed $|B|$ spread at 1024 is 0.71× that at 256, not the 0.5 a variance-limited halving requires, though n=10 is weak.)

f) The pair, read together.: Camelyon17 (detectable harm, AUC 0.78) and ImageNet-R (*un*detectable harm, AUC 0.66; Table XVIII) bracket the frontier of Theorem 3: real harm exists on both (28.7% / 13.9% of cells), but only on Camelyon17 is it label-free separable. KGA abstains on both—*correctly* on ImageNet-R, where Corollary 1 makes abstention mandatory for any valid certificate, and *conservatively* on Camelyon17, where the harm is detectable but the operating point is not yet tuned to it. Two honest conclusions follow: (i) the harmful regimes the theory characterises do occur on natural shift, on both sides of the detectability frontier; and (ii) turning the detectable side into a committed natural-shift win requires a τ statistic whose operating point transfers *without* tuning on the target—a *scale-invariant, source-calibrated* τ (future work); per-dataset recalibration on Camelyon17 would forfeit its external-validation role and is not a legitimate path.

E. Cross-protocol aggregate (out-of-fold re-run)

Each single dataset above is *one-sided*: Camelyon17 is adapt-favorable (on its held-out OOD cells freezing leaves regret ≈0.11), whereas OfficeHome and iWildCam are freeze-favorable (adapting leaves regret 0.047 and 0.103). On a one-sided stream the better trivial policy is already near-oracle, so KGA's simultaneous margin over *both* is necessarily small. To evaluate the policy benefit of per-condition adapt/freeze/abstain routing across heterogeneous regimes, we *aggregate* the held-out test conditions of the three dev-locked datasets into a single stream (n=143); to avoid the Protocol-G pooling artifact, *Camelyon17 enters through its genuine OOD `test/val` cells only* (36 cells; in-distribution `id_val` excluded). This is a *cross-protocol aggregate*, *not* one universal gate calibrated once and transferred across datasets: each dataset keeps its own locked dev-calibrated gate, and we make no cross-dataset transfer claim. Because this aggregate genuinely mixes helpful conditions (Camelyon) with harmful ones (Office-Home, iWildCam), it is the one setting in our suite where per-condition routing could in principle beat both fixed policies.

a) Out-of-fold re-run (`mixed_protocol_oof_v2`).: The earlier figures for this aggregate (regret 0.0026; ∼24× below always-adapt, ∼13× below always-freeze) used an *in-sample* conformal radius—the calibration bug corrected in §X-D—and are **withdrawn**. Under leave-one-out conformal calibration per dataset (`scripts/mixed_stream_kbound.py`), the pooled stream yields mean regret $\text{regret}_{\text{KGA}}$=0.0059 vs. 0.0632 (always-adapt) and 0.0342 (always-freeze), with pooled false-adapt 0. Bootstrap regret gaps: vs. always-freeze +0.0283 $[0.019, 0.038]$; vs. always-adapt +0.0573 $[0.043, 0.072]$; both 95% CIs exclude zero (`research_lock/KBOUND_MIXED_STREAM_v2.json`). This is a **beats-both** on a *constructed* heterogeneous aggregate—not a natural-shift headline. Magnitudes are smaller than the withdrawn in-sample-radius multipliers; we do not report those ratios as findings.

b) Universal single gate (pre-registered, 2026-07-03).: The per-dataset-gate caveat above is closed by two pre-registered arms (`research_lock/WIN_HUNT_v2_PROTOCOL.yaml`, `WIN_HUNT_v3_PROTOCOL.yaml`; bars frozen before scoring, each scored once). *One* GBR benefit model and *one* pooled leave-one-out conformal radius, fit on the pooled dev records over the shared base-11 evidence and applied unchanged to the pooled held-out conditions: on the three-source stream (n=143), regret 0.0182 vs. always-adapt 0.0632 and always-freeze 0.0342, both 95% CIs excluding zero, FA_u=0.000; on the seven-source extension adding all four PACS leave-one-domain splits (n=359, $\varepsilon_{\text{universal}}$=0.096), regret 0.0183 vs. 0.0404 / 0.0282, both CIs excluding zero, FA_u=0.000. Per-source regrets show the mechanism: the gate rides adaptation on Camelyon17, freezing on iWildCam and Office-Home, and no-harm ties across PACS. Honest scope: the stream composition is researcher-pooled (datasets and splits locked before these arms were registered); this is a certified heterogeneous-deployment win under a universal calibration, not a single-dataset natural-shift win. Verdicts: `research_lock/WIN_HUNT_v2_ARM_A_result.json`, `WIN_HUNT_v3_ARM_E_result.json`.

c) Report-all-arms: the rich-evidence Camelyon re-run FAILs (pre-registered, scored once).: The companion GPU arm (`WIN_HUNT_v2` Arm B) asked whether *richer evidence* unlocks the single-dataset Camelyon17 win that v1 could not produce: a fresh 4-seed run (Tent/EATA/SAR, n=432 records) logging the full 30-dimensional evidence (base panel + 13-dim `ev2`: MaNo [64], nuclear norm [65], per-sample quantiles) with the self-normalized τ' gate. **Verdict: FAIL.** Validity holds ($\text{FA}_\text{u}=0.044 \le \alpha=0.10$) and KGA beats always-freeze (gap −0.0223, CI $[-0.0264, -0.0183]$), but the τ' gate rejects risk-alignment on 77% of gated records, driving abstention to 59.5%; on this helpful-dominated dataset the abstention premium costs real accuracy, and KGA's regret (0.0292) is *worse* than always-adapt (0.0129; gap CI $[0.0095, 0.0231]$ excludes zero). Richer evidence therefore does not move Camelyon17 across the frontier—the v1 mechanism (evidence-limited, not gate-limited) stands, and off the identifiable regime the certificate behaves exactly as the theory prescribes: it stays valid and pays abstention cost rather than committing uncertifiable adaptations. The headline Camelyon17 claim is unchanged (Protocol G no-harm under the canonical dev-locked gate); this arm is reported because its bar was frozen before scoring (`experiments/kbound/results/natural_win_v2_camelyon/result_a4929009.json`).

XI. Excluded wins and regime boundaries

Several benchmarks flag beats-both under alternate routes or adapters but do *not* survive the canonical Protocol H/M v2 analysis under a valid out-of-fold radius, where the one-sided natural-shift results are *no-harm*, not wins (the domain-generalization and rendition tracks are nulls that lose to always-adapt; see Table XXVIII and §B). **Superseded protocols:** iWildCam Protocol H v1 (fixed `sar_online` without dev-lock) and OfficeHome Protocol M v1 (14-candidate win-finder) are audit-only; the canonical natural-shift results use H/M v2 (`research_lock/IWILDCAM_PROTOCOL_H_v2.yaml`, `research_lock/OFFICEHOME_PROTOCOL_M_v2.yaml`). **Office-Home (deployed adapter):** mild online EATA is helpful-dominated on target test (0% harmful); Table XIX is a covariate-shift diagnostic, distinct from Protocol M v2. **ImageNet-R:** weak detectability (harm-AUC 0.66); mandatory abstention. **iWildCam id_val:** multicandidate routes can flag wins but pooled false-adapt $\approx 78\% \gg \alpha$. **Camelyon four-seed slice:** helpful-dominated (Table XVI); distinct from Protocol G mixed-stress grid. **Camelyon17 Protocol G (withdrawn):** its $n{=}54$ beats-both pooled in-distribution `id_val` into the seed-split held-out set; on genuine OOD test online EATA is helpful-dominated so KGA ties always-adapt (no beats-both). Reclassified as no-harm (Table XX); the audit and re-run are in `audits/integrity_2026-06-20/camelyon_reconciliation/`. **CIFAR-10.1:** within-seed safety holds; cross-seed certificate transfer does not clear the headline bar. **PACS:** The completed DomainBed leave-one-domain-out diagnostic has three seeds and 18 held-out cells per domain per seed. Its reported mean false-adapt rate is 0.0093; the raw pooled numerator was not retained. Its mean regret is 0.0431 versus 0.0176 for always-adapt and 0.0446 for always-freeze; no target supports a multiseed beats-both claim. See Table XXVIII.

XII. Limitations and honest scope

Scope of the guarantee. K-Bound does not guarantee that a newly encountered deployment is exchangeable with the calibration data, that label-free evidence is risk-aligned, or that a fitted benefit estimator transfers. Its guarantee is conditional: *if* the declared benefit interval attains marginal target coverage, *then* strict adaptation on $\widehat{\Delta} - \varepsilon > 0$ controls the unconditional false-adapt event at level α, and strict freezing on $\widehat{\Delta} + \varepsilon < 0$ the false-freeze event. Deployment diagnostics can identify support shift, instability, leakage and insufficient effective sample size, but passing them does not prove coverage. When required checks fail or cannot be evaluated, the prescribed output is ABSTAIN or FREEZE — not a certified adaptation decision.

What we do not claim. Universal accuracy gains; adapter selection by test regret; multicandidate routes with violated false-adapt bounds; wins on helpful-dominated shifts where always-adapt is already oracle-optimal; that the drift budget β can be declared from development data (§VII measures the attempt and it fails); or that $|M| > \beta$ is the rule KGA runs. Three further non-claims follow from §VI–§VIII: that β can be supplied by *any* label-free procedure (Theorem 6 says the best one returns a constant, and Corollary 4 says on the deployment class that constant is β itself); that episode exchangeability (E2) holds at a novel deployment (measured false—coverage 0.626 when the corruption family changes, 0.454 when severity changes, against a null of 0.900); and that weighted conformal repairs it (measured: it restores coverage to 1.000 by driving decision yield to 0.000).

(0) **The population frontier does not operationalize, and β is not estimable from development data.** This is the paper's own negative result (§VII). Declaring β the way this paper recommends—a high quantile of the realized drift on source-like development cells—gives a budget 1.4× to 50× smaller than soundness requires on CIFAR-10-C, where 24–73% of deployment cells violate the declared class and committed actions are wrong at 0.4–16.6% against a promised 0%; on ImageNet-C a sufficiently large budget is derivable and returns zero commitments on all 405 cells in half the configurations. Consequently we withdraw the operational reading of Theorem 3 while keeping its impossibility content, whose proof is untouched, and we do not offer β selection as a solved deployment step. Limitations of that test itself: two benchmarks with 6 and 3 corruption families, so family-clustered intervals are reported but are not primary; the benefit rather than the disagreement scale, since $\mu_T(D)$ is not recorded in these artifacts; the minimum sound budget is an oracle diagnostic no deployer could compute; and regret resolves ABSTAIN to FREEZE, which penalizes abstention heavily—the soundness conclusions do not depend on that convention, the regret comparisons do.

(1) Scope of the empirical win: on the CIFAR-10-C *stress* grid KGA beats both trivial policies for Tent and EATA at 0% false-adapt (Table XIV). The CIFAR-10-C SAR artifact is withheld because its archived aggregate does not reproduce from the current seed-0 replay. On ImageNet-C (Table XXV) the pattern *flips*— Tent and EATA are robust so KGA ties always-adapt, while SAR collapses and KGA beats both—a win we confirmed survives a *mechanism-faithful* SAR re-implementation (Table XXVII; SAR harmful on 44% of cells, regret-to-oracle 0.023 vs 0.112/0.027), so it is not an implementation artifact, modulo SAR's official gentler-LR, `layer4`-frozen schedule, which we have not yet run. On *per-corruption* CIFAR-10-C and within a single online stream (both helpful-dominated) it again *ties* the better fixed policy. The common rule across all of these: KGA approximately matches the better trivial policy, paying at most a small abstention cost when it declines a helpful but uncertifiable update. It *strictly* wins only where catastrophic, detectable harm is

actually present—and which method collapses depends on the benchmark. A tighter online (streaming) certificate is future work. (2) **Real-shift scope: no single-dataset natural beats-both.** Under a valid out-of-fold conformal radius, no natural shift is a beats-both; the four one-sided locked tracks are *no-harm* and the remaining three (PACS, ImageNet-R, CIFAR-10.1) are nulls or declared failures that lose to always-adapt — "no-harm" does not extend to them. Two fully instrumented pre-registered attempts (`NATURAL_WIN_PROTOCOL_v1`) fail their frozen bars with the false-adapt guarantee intact, so single-dataset beats-both wins remain confined to the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR). On *pooled* heterogeneous natural deployments, however, the pre-registered universal-gate arms are CI-robust beats-both wins (§X-E). Office-Home (Protocol M v2) and iWildCam (Protocol H v2) beat always-adapt with CIs excluding zero but *tie* always-freeze (Tables XXI, XXII); their earlier in-sample-radius beats-both was a calibration bug, corrected here. Camelyon17 Protocol G is likewise *not* a win: its pooled beats-both included in-distribution `id_val` cells, and on genuine OOD test KGA ties always-adapt (Table XX, no-harm). Earlier Camelyon diagnostics (§X-D) and multicandidate routes are not competing headline claims. **ImageNet-C** harmful SAR win is synthetic corruption. **CIFAR-10.1** is a within-seed low-margin probe; cross-seed transfer fails (§XI). **ImageNet-R** is a completed weak-evidence diagnostic: its estimated margin does not clear the empirical radius, so no candidate supports a promoted confidence-robust beats-both claim. This null does not establish structural non-identifiability; weak benefit estimation or calibration transfer could produce the same observation. The deployed **Office-Home** adapter audit (Table XIX) is an invalid-transfer regime, distinct from Protocol M v2. **PACS** is a completed three-seed domain-generalization breadth check and remains a null diagnostic; no leave-one-domain-out target supports a promoted multiseed win (Table XXVIII). We caution that the conformal radius can still fail to transfer under *severe* geographic or sensor shift, where in-distribution calibration need not hold out-of-distribution; characterizing that failure mode is future work. (3) The conformal radius assumes task exchangeability. (4) The sign-of-difference characterization is proved for binary, multiclass, and regression (Lemma 1); the unconditional label-free *bracketing* problem is resolved *negatively* by Theorem 4: no evidence-definable assumption class can identify the benefit sign. On the margin-monotone relative-calibration class the one-bit supplement certifies the sign *unconditionally*, and that class is *a* weakest such falsifiable class under a general-position assumption (Theorem 16); the *unconditional* weakest classes are characterized as an explicit finite family of dominance polytopes (Theorem 17), with no unique weakest class and General Position recovered as the collapsing face. (5) KGA's value scales with how often catastrophic, detectable harm occurs—precisely the quantity the theory characterizes. On helpful-dominated shifts KGA *ties* always-adapt; this is the certificate declining to commit a harmful update, not a failure—it approximately matches the better policy (at most a small abstention cost), so a near-tie where always-adapt is already oracle-optimal is the intended outcome, and the value proposition is conditional (insurance against detectable harm). (6) Abstention lowers *decision* coverage by construction but never withholds a prediction: an ABSTAIN routes to the safe default (keep f_0, the source model), so every input still receives an output, and the coverage/false-adapt trade-off is explicit and tunable via the α safety/coverage sweep reported above. (7) **These limitations are the frontier, not flaws.** Points (1) and (4)–(6) are not independent weaknesses but one phenomenon—the benefit-sign frontier (Theorem 3) seen from the operating side. Tying always-adapt on a helpful-dominated shift is the no-harm guarantee *holding* (one cannot beat an oracle that is already always-adapt); ImageNet-R in the *unknowable* regime is the impossibility direction firing where the observable margin falls below the drift budget and *no* label-free rule can decide the sign (Theorem 4); and the coverage reduction is the *mandatory* abstention of Corollary 1, not a tuning loss—committing there would be a false-adapt. None can be removed without violating the impossibility result: a method that "won everywhere" would be claiming to decide the sign of an unidentifiable quantity. The single axis that improves all three is a *richer label-free evidence map*—raising the observable margin $|M|$ so more shifts clear the budget (more knowable shifts, fewer abstentions, higher coverage)—together with a *tighter radius* (more calibration data or a separately validated finite-sample method) that reclaims knowable cells abstained only out of conservatism. Both are bounded by the frontier they cannot cross.

(8) **Replication and maturity hierarchy.** All promoted empirical artifacts were produced internally; the repository freezes an external procedure in `INDEPENDENT_REPLICATION_PROTOCOL.md`, but this release does not claim independent replication. The primary contribution is the binary split-observable frontier, its finite-sample certificate, and the registered empirical panel. Multicandidate, anytime, multiclass-capacity, regression, and auditable-budget results are companion extensions with explicit conditions and open formal clauses, not additional headline claims. PACS and ImageNet-R remain completed null/weak-evidence diagnostics. CIFAR-10-C SAR is quarantined and supplies no evidence unless the reinstatement gates in `CIFAR10C_SAR_QUARANTINE.md` are met.

(9) **The escape of §VIII is marginal, not conditional.** The episode budget controls a rate *across* deployments drawn from the episode law; it does not protect the deployment in front of you. Proposition 4(a) is explicit: conditional on the episode being one of the adversarial ones in its own fibre, the frontier commits and is wrong, and Lemma 3 says no label-free observable distinguishes that episode from a benign one. A deployer

told "the budget was estimated" has bought a long-run frequency, and Remark 12 states the same distinction that governs the certificate's own false-adapt guarantee. We state this in three places by design.

(10) **The budget route is worth taking only because it amortises, and that rationale is conditional on E2.** Proposition 6(b): at a *single* deployment a labeled probe of the same size dominates the episode budget, because the probe measures γ where it is needed and the budget measures it elsewhere. The episode route wins only across many deployments, so when E2 fails it loses its statistical guarantee and its economic rationale at the same time—and E2 is measured to fail exactly on the axis that defines a new deployment (§VIII-J).

(11) **E2 is not verifiable, only falsifiable, and its observable part is a one-sided screen.** Corollary 6 shows no label-free statistic certifies the conditional component of E2. The marginal component is testable: a cross-fitted history-versus-deployment separability AUC on the evidence alone reads 0.497 ± 0.033 on the exchangeable control, as it must, and correlates with realised coverage at Pearson $r = -0.35$ ($p = 1.2 \times 10^{-14}$, $n = 470$ splits). Used as a screen at $\Delta_{\text{sep}} < 0.70$ it gives mean coverage 0.905 against a nominal 0.900, with 7.7% of splits below 0.85; above the threshold, mean coverage 0.782 with 43.3% below 0.85. A low reading is informative; a high reading is a warning and nothing more. We report it alongside any episode budget and do not treat it as a certificate.

A. When to deploy K-Bound

The theory and experiments converge on a clear deployment rule: K-Bound's value is governed by how often *catastrophic, detectable* harm occurs—safety insurance in benign regimes, real wins in collapse-prone synthetic grids (Table XIV). The trichotomy reads off accordingly. **(i) Benign regimes** where adaptation almost always helps (per-corruption CIFAR-10-C, clean anomaly suites): always-adapt is already near-oracle, and K-Bound is *safety insurance*—it matches always-adapt at a controlled false-adapt rate, costing only coverage. **(ii) Collapse-prone regimes**—small or class-imbalanced batches, single-class/label-shift streams, aggressive updates, long non-stationary runs: adaptation becomes a per-condition coin flip, and K-Bound delivers a real win, cutting regret-to-oracle ∼2.5–5.4× versus always-adapt at 0% false-adapt on the CIFAR-10-C stress grid (Table XIV). **(iii) Self-protecting candidates** (e.g. SAR's reset): the headroom shrinks, and K-Bound *ties* rather than beats—approximately matching the better policy, at most a small abstention cost. The practical takeaway: deploy K-Bound as a default guard around any TTA candidate; its overhead is one forward pass, and its benefit scales with exactly the risk it is designed to detect.

XIII. Conclusion

A label-free adapt/freeze decision is possible only relative to a declared bound on calibration drift, so the question that decides the method is where that bound comes from. It cannot come from the deployment, and that is a theorem with an exact value rather than a limitation we regret. For every declared class of target laws the least label-free-certifiable budget is the fibre radius Γ_z, attained by a constant audit that reads none of its data (Theorem 6); on the paper's own deployment class that radius is β exactly (Corollary 4), so the best possible label-free audit of the budget returns the budget it was given, and the number it returns is simultaneously the radius of the band on which the frontier must abstain. Over the unrestricted class the audited rule's decision yield is at most the false-commitment budget δ, at every batch size and for every evidence map, and a fully labeled calibration sample from any other domain moves the minimax budget by exactly zero (Theorem 7). One lemma—that the label kernel on the disagreement region is unconstrained by every label-free observable (Lemma 3)—carries the matched-evidence impossibility, the abstention band and the budget impossibility as three readings of the same one-parameter family. Our own β-sweep is the predicted instance: on 6,885 real evaluation cells, declaring β the way this paper's method section prescribes under-covers by the size the theorem permits. Nor is the impossibility's configuration merely constructed: in logged deployments there are pairs whose label-free evidence cannot be told apart while their benefit has resolvably opposite sign, and about 30% of evaluable CIFAR-10-C cells sit in fibres whose sign genuinely goes both ways — though such pairs are 39× *rarer* than chance, so the evidence is informative and merely insufficient (§VI-F).

A budget can come from a *population* of deployments, and we price it: between 3 and 9 logged deployments at $\alpha = 0.10$ under retrospective outcome logging and episode exchangeability, first feasible on real data at exactly $K = 9$. The escape is localised to one logical move, averaging over the episode law instead of taking a supremum over the fibre, paid for with historical labels, and neither ingredient works alone (Proposition 4). The assumption that buys it is the one a shifted deployment violates: nominal 0.900 coverage holds under an exchangeable control (0.905) and along nuisance axes and collapses to 0.626 when the corruption family changes and 0.454 when severity changes, and the weighted-conformal relaxation restores coverage to 1.000 only by driving decision yield to 0.000 — the vacuity of Theorem 18 reappearing inside the method meant to escape it.

What remains achievable is a finite-sample certificate that trades decision yield for a false-adapt guarantee, and we measure that frontier rather than assert it. On the CIFAR-10-C stress grid it commits on 66.8% of 6,480 cells at zero false adaptations, cuts regret 5.00× below the better fixed policy, places its commitments on cells whose true effect is 24.5× larger than the ones it declines, and still returns zero false adaptations in all ten leave-one-corruption-out runs. The binding resource is calibration structure, not sample size: on Camelyon17 real-shift certifiability is *bias-limited*, and more unlabeled data cannot shrink a radius whose error is calibration drift. On natural shifts the outcome is scoped, not uniform: KGA is no-harm on the four one-sided tracks with locked held-out artifacts (Camelyon17, iWildCam, Office-Home, RxRx1), and on PACS, ImageNet-R and CIFAR-10.1 it is a conservative null or a declared transfer failure and *loses* to always-adapt — 1.76× worse on ImageNet-R — which we report rather than exclude. Protocols H v2 and M v2 beat always-adapt with CIs excluding zero but *tie* always-freeze, so neither is a beats-both (Tables XXI, XXII); their earlier in-sample-radius wins were a calibration bug, corrected here, and a re-audit withdrew the earlier Camelyon17 Protocol-G beats-both as a domain-pooling artifact (Table XX). The only beats-both results are the synthetic stress grids (CIFAR-10-C, ImageNet-C SAR). K-Bound is a safety, validity and abstention layer for test-time adaptation; the drift budget its own theory requires is not obtainable at a deployment, and the certificate is what remains when that is taken seriously.

XIV. Reproducibility

Code. All code, configs, and result artifacts are released in the repository linked from the supplementary material (see its `LICENSE`); the supplement includes the reproduction package and result manifests. *Anonymity note.* The released tree is **not** anonymized: its `CITATION.cff` carries the author's name and email, and several run manifests record machine-local paths. Earlier drafts described it as "an anonymized repository", which was false. For a double-blind venue the link and the `CITATION.cff` block must be withheld and the tree redacted before it is attached.

One-command reproduction. The CPU/numpy stages—theorem validators plus the anomaly-routing, regression, witness, and ablation experiments—run from a single driver in the release package. The deep-TTA stress grid runs on a single commodity GPU (16 GB) using the provided launcher, which builds the environment, fetches the data, and writes the tables and figures. Natural-shift Protocols H v2 and M v2 (dev-lock held-out scoring) run via `kbtrain.sh protocol-h-v2` and `kbtrain.sh protocol-m-v2`.

Environment. Experiments use a `uv/venv` environment, Python 3.12, with the deep-TTA runs pinned to `torch 2.5.1/torchvision 0.20.1` (**PyTorch** [60]); the CPU-only stages need only `numpy/scipy/scikit-learn/matplotlib` (**scikit-learn** [61]), and `pdflatex` compiles the paper. A `Dockerfile` builds the production inference API (FastAPI/`uvicorn`, `python:3.11-slim` pinned by digest).

Data. CIFAR-10/100 [49], ImageNet [50], and CIFAR-10.1 [59] are pulled via `torchvision`; the official CIFAR-10-C corruptions are the Hendrycks–Dietterich set, Zenodo DOI `10.5281/zenodo.2535967` and are auto-downloaded and checksummed by the release scripts.

Determinism and honesty. Single-run stages use fixed seeds; the locked deep-TTA stress grid reports seeds 0–4 with an out-of-fold (leave-one-out) conformal radius at $\alpha = 0.1$. The deep-TTA grid is *pre-registered* (declared before any result is observed) and reported in full, with no condition dropped. Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass. Result manifests and locked analysis artifacts back every reported number. Tables with multi-seed aggregates are marked explicitly in-caption (e.g., Tables XIV, XVI, XXIV); single-seed/scoping tables are labeled as such (e.g., Tables XVII, XVIII).

Appendix

Two verified extensions of the batch certificate (Theorem 2) are referred to elsewhere in this paper. We state them here; their proofs, their machine validators (`theory_v2/val_sequential_anytime.py`, `val_multicandidate.py`) and the remaining extension stack—multiclass and regression reductions, router and knowability capacity, the γ-meter route, the minimax and tight-constant closures, and the Lean formalisation inventory—are in the companion technical report, whose sources are the `theory_v2/` directory of the released tree. Nothing in the impossibility spine (§V–§VIII) and no empirical claim in this paper depends on any of them; they are carried as extensions with their own stated assumptions.

Theorem 14 (Anytime-valid sequential adaptation certificate)**.** *Let windows $t = 1, 2, \dots$ on a filtered probability space produce bounded, label-free-computable benefit summaries $X_t \in [a, b]$ with $a < 0 < b$, and write $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ for the conditional benefit of window t; no independence, identical distribution or stationarity across windows is assumed. Fix $\alpha \in (0, 1)$, a constant $c \in (0, 1)$, and predictable bets $\lambda_t \in [0, c/|a|]$, $\lambda_t^- \in [0, c/b]$, and set $E_t^+ = \prod_{i \le t}(1 + \lambda_i X_i)$ and $E_t^- = \prod_{i \le t}(1 - \lambda_i^- X_i)$. Return* ADAPT *at the first t with $E_t^+ \ge 1/\alpha$,* FREEZE *at the first t with $E_t^- \ge 1/\alpha$, and* ABSTAIN *otherwise, committing absorbingly at the first crossing. Then* (i) *under $H_0 : \Delta_t \le 0$ for all t, $\mathbb{P}(\exists t \ge 1 : \text{DECISION}_t = \text{ADAPT}) \le \alpha$;* (ii) *symmetrically, under $H_0' : \Delta_t \ge 0$ for all t, $\mathbb{P}(\exists t \ge 1 : \text{DECISION}_t = \text{FREEZE}) \le \alpha$. Both bounds are time-uniform, so they hold at an arbitrary, data-dependent stopping time τ. Run on a single window with the bet matching the conformal radius, the rule specialises to the level-α decision of Theorem 2. The guarantee is about the sign of $\Delta_t = \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$, the population object the stream targets; identifiability of the* true *benefit sign from X_t remains governed by the frontier of Theorem 3, and time-uniformity does not remove the drift budget.* Proof: companion report (one-sided betting supermartingale and Ville's inequality).

Theorem 15 (Multicandidate certified routing: family–wise false-adapt control)**.** *Fix f_0 and candidate adapters*

$f_a^{(1)}, \ldots, f_a^{(K)}$ with benefits $\Delta_k = R_T(f_0) - R_T(f_a^{(k)})$, and suppose each admits a label-free lower-confidence bound $L_k(\delta) = \widehat{\Delta}_k - \varepsilon_k(\cdot;\delta)$ with $\mathbb{P}[\Delta_k < L_k(\delta)] \le \delta$ for every $\delta \in (0,1)$—the one-sided half of the split-conformal coverage of Theorem 2, instantiated per candidate on a shared exchangeable calibration sample. Set the corrected level $\delta_K = \alpha/K$, form the certified-helpful set $S = \{k : L_k(\delta_K) > 0\}$, commit an arbitrary $\sigma \in S$ when $S \neq \varnothing$ and abstain otherwise. Then the family-wise false-adapt event $\{\sigma \in S \wedge \Delta_\sigma \le 0\}$ has probability at most α, under arbitrary dependence among the K coverage events and for every *selector σ, including data-dependent and adversarial ones. The Šidák level $\delta_K = 1 - (1-\alpha)^{1/K}$ requires mutual independence of the K under-coverage events, which generally fails under a shared calibration sample and under a common deployment input; Bonferroni is the assumption-light default.* Proof: companion report.

a) Real-data anytime demonstration (pre-registered).: The streaming certificate has a first real-data application (`research_lock/WIN_HUNT_v3_PROTOCOL.yaml`, arm D; bar frozen before scoring). On a native-order iWildCam test stream (35,370 images, 45 windows of 50 batches), continual Tent *collapses*: 75.6% of windows harmful, mean window benefit −0.370, always-adapt value 0.329 versus always-freeze 0.700. Two one-sided truncated-aGRAPA betting e-processes on the per-window benefit (the construction of Theorem 14) run at α=0.10: the FREEZE process crosses $1/\alpha$ at **window 6**, the gate locks the safe action, anytime false-adapt is **0**, and the realized stream value 0.6995 ties the oracle policy to four decimals. Per the frozen bar this is a *demonstration*, not a beats-both (freezing is the oracle on this stream, so no policy can beat it): the time-uniform guarantee holding while a 37-point collapse is caught within six windows on real data. Verdict: `research_lock/WIN_HUNT_v3_ARM_D_result.json`.

This appendix collects the detailed per-dataset tables for the ImageNet-C scale grid and the RxRx1 safety case that are part of the seven-dataset core panel (Office-Home, Camelyon17, CIFAR-10-C, ImageNet-C, iWildCam, RxRx1, ImageNet-R), together with evidence *beyond* it: an additional breadth check (PACS) and honest nulls. None of it is required for the paper's headline claims; it is retained for completeness and reviewer scrutiny.

A. Controlled multimodal corruption (Protocol D33)

Pre-registered MNIST two-view fusion (130 conditions: 13 Gaussian-corruption severities ×10 batches) splits each digit into left- and right-half modalities and corrupts the right modality. KGA mean accuracy is 0.8568 versus 0.5832 for always-fuse, 0.8536 for always-single-A, and 0.8573 for the oracle. It makes 9/119/2 adapt/freeze/abstain decisions, with observed false-adapt 0/130; the saved paired-bootstrap superiority probabilities against both fixed policies are 1.0. This is a controlled construction confirming routing when fusion harm is detectable, not a natural multimodal benchmark or evidence of universal accuracy improvement (`CONTROLLED_MULTIMODAL_PROTOCOL_D33_v1`; `KB-CLAIM-027`).

TABLE XXIII
ImageNet-R across ten backbones (4 seeds). Mean regret-to-oracle (lower is better) for KGA, always-adapt, always-freeze, and pooled false-adapt (FA). No backbone yields a Holm-corrected beats-both result. FA is descriptive within each 48-record candidate panel.

Backbone	KGA	always-adapt	always-freeze	FA
ConvNeXt-B	0.000	0.000	0.069	0.00
ConvNeXt-T	0.021	0.001	0.022	0.00
EfficientNet-B0	0.000	0.050	0.000	0.00
EfficientNet-B3	0.019	0.000	0.050	0.00
ResNet-101	0.015	0.001	0.024	0.00
ResNet-152	0.009	0.000	0.042	0.00
ResNeXt-101	0.006	0.000	0.052	0.00
Swin-B	0.023	0.000	0.037	0.00
Swin-T	0.004	0.011	0.005	0.00
ViT-B/16	0.016	0.000	0.024	0.021

B. ImageNet-R: architecture robustness across ten backbones

Table XXIII evaluates the ImageNet-R diverse panel across ten modern backbones (4 seeds, 12 conditions per seed, Holm-corrected paired bootstrap CIs, n_{boot}=10^4; `imagenetr_protocol_d_multiseed_v1`). A certified *beats-both* occurs on 0/10 backbones. The behavior is candidate-dependent: for example, KGA ties always-adapt on ConvNeXt-B, ties always-freeze on EfficientNet-B0, and improves over always-adapt but not significantly over always-freeze on Swin-T. Observed false-adapt is 1/480 descriptively pooled across candidate-condition-seed records. This is a completed weak-evidence diagnostic, not a policy win or a claim that the pooled records are independent.

C. Detailed core-panel tables: RxRx1, ImageNet-C scale grid, and PACS domain generalization

*a) Fourth natural-shift track: RxRx1 — the harmful-and-*detectable *corner at* 1,139*-class scale.:* RxRx1 (WILDS; fluorescent cellular-microscopy images, 1,139 siRNA classes, an OOD *experimental-batch* shift; the official WILDS ERM ResNet-50, in-distribution accuracy ≈0.36, frozen accuracy 0.283 on the OOD-test eval, mean over the three seeds) is the sharpest harmful-regime real shift in our suite. Over the full 9+ protocol (composition×batch×aggressiveness over 10 condition seeds and all 3 official WILDS base-model seeds = 360 conditions, 2,160 records, a commodity GPU (16 GB), complete 360/360), adaptation is *harmful-dominated and label-free detectable*: harmful base rate 81.5% ($B < 0$), mean benefit $\bar{B} = -0.063$, and—unlike ImageNet-R—the harm *is* detectable (best evidence AUC 0.78–0.84 across the three seeds, the BN-affine update-norm the top signal). Always-adapt is actively destructive: even the best candidate (Tent-episodic, 0.276) trails the frozen model

RxRx1 9+ grid (360 conditions, 3 model seeds)	value
harmful base rate ($B < 0$)	81.5%
mean true benefit $\bar{B}$	−0.063
harm detectability (best AUC)	0.78–0.84
frozen f_0 accuracy	0.283
best-adapter accuracy	0.276
SAR-online always-adapt (collapse)	0.024
per-condition oracle accuracy	0.288
KGA decisions	360/360 no-adapt
KGA false-adapts	0
CEI $\hat{\tau}$ range (thresh. $\tau^\star$)	1.06–2.71 (0.52)
bounded-drift fallback	289 abstain / 71 freeze

TABLE XXIV

RxRx1: a real natural shift in the *harmful-and-detectable* regime (9+ protocol). At 1,139-class scale adaptation is harmful-dominated (81.5% of records $B < 0$, $\bar{B}$=−0.063) and the harm is label-free detectable (AUC 0.78–0.84). Always-adapt is catastrophic—SAR-online collapses to 0.024 and even the best candidate (0.276) trails the frozen model (0.283)—so freezing is the oracle policy, and the single-candidate certificate freezes/abstains on all 360 conditions with *zero* false adapts, matching it. The multi-candidate agreement route abstains everywhere ($\hat{\tau} \gg \tau^\star$ on all 360): the six candidates share one backbone, the CEI-violating setting the diagnostic catches. Verdict replicates across all 3 official WILDS base-model seeds; complete 360/360, commodity GPU (16 GB).

(0.283), and SAR-online *collapses* to 0.024 ($\approx$ chance over 1,139 classes) on the single-class/tiny/aggressive cells. Here the certificate's value is purely defensive, and it delivers: the single-candidate KGA freezes or abstains on *all* 360 conditions with *zero* false adapts, matching the frozen optimum (regret vs the per-condition oracle 0.003–0.007) and avoiding the collapse outright—on SAR-online it pays regret 0.000 where always-adapt pays 0.259. It does *not* "beat both" because freezing already *is* the oracle policy (no helpful candidate exists to capture)—the honest does-no-harm outcome on the harmful side, the mirror image of Camelyon17's benign-side tie (Table XVI). As on ImageNet-R, the multi-candidate agreement route abstains on all 360 conditions ($\hat{\tau} \in [1.06, 2.71]$, all $\gg \tau^\star$=0.52): the six candidates share one backbone, the conditional-error-independence violation the CEI diagnostic exists to catch. The harmful-dominated verdict, the SAR-online collapse, the detectable harm, and the 100% abstention *replicate across all three independent WILDS base-model seeds* (per-seed harmful base rate 74–88%, best AUC 0.78–0.84). RxRx1 thus closes the real-data map of the harmful corner—*detectable* harm at 1,139-class scale where "adapt" is catastrophic and the zero-false-adapt guarantee carries the load (Table XXIV). These results are from the three archived official-seed manifests.

Fold-in note (2026-07-09). The full-scale ImageNet-C SAR run (27 cells; 3 noise corruptions × severities $\{1, 3, 5\}$ × regimes, seed 0) is now the headline configuration: KGA 0.0108 vs. always-adapt 0.0625 / always-freeze 0.0319, *CI-confirmed* beats-both (gap to freeze $[-0.032, -0.011]$, Holm p=0.0003; $\mathrm{FA_u} = 0$; harmful 14.8%; 12/27 abstain). The 36-cell tables below are the earlier locked configurations, retained for provenance; their point verdicts stand, and a decision-baseline re-derivation on the full-scale grid is a mechanical follow-up.

b) Decision baselines, executed.: Theorem 3(iv) shows prior decision styles are degenerate ($\varepsilon \equiv 0$) certificates; Table XXVI *runs* them, head-to-head with KGA, on the logged conditions of the mechanism-faithful ImageNet-C SAR grid (Table XXVII; per-condition evidence Z, a_0, a_a, identical metrics code). Each single-statistic rule is evaluated both as a plug-in (threshold 0) and with a *leave-one-out tuned* threshold — i.e. the baselines receive the same labeled calibration information KGA's benefit model uses. True AETTA (dropout passes) and agreement-on-the-line (a model family) are not computable from logged statistics and are represented by their decision-style surrogates; we say so rather than approximate silently. Findings: on the harmful candidate (SAR, 44% harmful cells under reference parity) *every realizable single-statistic rule loses* — the best (LOO-tuned EATA-filter, regret 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, while the plug-in ATC and entropy-progress rules *over-adapt* into SAR's collapse (regret 0.067/0.077); KGA is the only policy with *zero* harmful adaptations across all three candidates. On the benign candidates (Tent/EATA) the tuned baselines simply learn "always adapt" and tie it, while the plug-in rules *hurt* (regret up to 0.033). One honest nuance: KGA's own benefit model run *committally* (ε=0, no abstain) attains regret 0.001 on SAR — the multivariate evidence model carries the decision power — but it has no validity control and adapts into harmful cells uncontrolled; the conformal band's $\approx$ 0.02 regret on SAR is the measured price of the false-adapt guarantee.

c) ImageNet-C, mechanism-faithful SAR re-run: the harmful win is not an implementation artifact.: The SAR row of Table XXV was flagged for re-checking (§XII): its collapse concentrated atypically at the *larger* batch, raising the worry that the harmful regime—and thus KGA's only ImageNet-scale "beats both"—was an artifact of our first SAR port. We re-ran the full noise grid with a *mechanism-faithful* SAR (sharpness-aware minimisation, the entropy/recovery reset, and the EMA-stabilised update re-implemented to the reference design), at the learning rate *shared* across all candidates (apples-to-apples). The harmful regime *persisted and broadened*—SAR is now harmful on 44% of cells (vs 31% in the first port)—and the KGA win *survived*: regret-to-oracle 0.0229 versus always-adapt 0.1124 and always-freeze 0.0267, *beating both* trivial policies (Pareto crossover $p^\star$=0.2; Table XXVII). Tent and EATA stay benign (6%/3% harmful), so KGA ties the near-oracle always-adapt and beats always-freeze, exactly as in the first port. The implementation-artifact concern is thus removed: under a reference-faithful SAR the harmful-candidate win is real. *Residual check, stated honestly.* This holds at the matched/shared learning rate (mechanism-faithful, apples-to-apples); SAR's *official* gentler schedule

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.453**	0.0003		
	always-freeze	0.410	0.0440		
	K-Bound	0.449	0.0042	8%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0001		
	always-freeze	0.410	0.0349		
	K-Bound	0.444	0.0003	3%	no (ties adapt)
SAR	always-adapt	0.377	0.0606		
	always-freeze	0.410	0.0277		
	K-Bound	**0.429**	**0.0086**	31%	**yes**

TABLE XXV

ImageNet scale (ImageNet-C, ResNet-50), complete: 36 **cells,** 4000 **images/cell.** Full noise grid (gaussian/shot/impulse × severities $\{1,3,5\}$ × batch/aggressiveness regimes), same adapt/freeze/abstain protocol and metrics as Table XIV. The outcome splits by how robust each method is on this benchmark. *Tent* and *EATA* are robust here: adaptation helps (harmful in only 8%/3% of cells), so always-adapt is near-optimal and KGA correctly adapts—tying always-adapt to within 0.004 and beating always-freeze, the honest "does-no-harm" outcome. *SAR* collapses on 31% of cells (mean benefit −0.033): always-adapt falls to 0.377 while KGA certifies-or-abstains to 0.429, *beating both* trivial policies and approaching the oracle (0.437). Across all three methods KGA approximately matches the better trivial policy (at most a small abstention cost), and strictly wins on the one method that is harmful. This first ResNet-50 SAR port is corroborated by a mechanism-faithful re-run (Table XXVII).

decision rule	regret-to-oracle↓ Tent	EATA	SAR	SAR (harmful regime) false-adapt	harmful adapted	coverage
always-adapt	0.0007	0.0000	0.1124	0.44	1.00	1.00
always-freeze	0.0494	0.0386	0.0267	—	0.00	1.00
ATC-style confidence (plug-in)	0.0114	0.0328	0.0673	1.00	0.25	1.00
ATC-style confidence (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
entropy-progress (LOO-tuned)	0.0007	0.0000	0.0375	1.00	0.06	1.00
EATA-filter $\mathrm{frac}_{\mathrm{hi}}$ (LOO-tuned)	0.0007	0.0000	0.0369	1.00	0.06	1.00
drift-gate marginal-KL (LOO-tuned)	0.0028	0.0023	0.0382	1.00	0.12	1.00
benefit model, committal (ε=0)	0.0000	0.0000	**0.0010**	0.05	0.06	1.00
best single statistic (hindsight)	0.0007	0.0000	0.0181	0.17	0.06	1.00
K-Bound (KGA)	0.0060	0.0047	0.0229	**0.00**	**0.00**	0.39

TABLE XXVI

Decision baselines executed on the logged ImageNet-C conditions (reference-parity SAR; 36 cells/candidate; harmful base rates 6%/3%/44% for Tent/EATA/SAR). "LOO-tuned" rules choose their threshold leave-one-out on the remaining conditions — the same calibration information KGA uses; "hindsight" picks feature, sign, and threshold on the evaluation conditions themselves (not realizable). KGA is the only rule with zero false adapts on all three candidates; on the harmful candidate *every realizable single-statistic rule loses* — the best (EATA-filter, 0.0369) pays 1.6× KGA's 0.0229 and is still worse than plain always-freeze, because no single statistic cleanly separates SAR's harmful cells. The committal benefit model (row 8) shows the multivariate evidence model carries the decision power (0.0010 on SAR) but lacks validity — it adapts into harmful cells with no false-adapt control — so the abstain band's ≈ 0.02 regret on SAR is the measured price of the guarantee, matching Thm. 3(iv). *Reproducibility note:* the canonical KGA pipeline and this post-hoc re-derivation agree exactly on SAR; the leave-one-out benefit model is mildly sensitive to condition order, so the Tent/EATA KGA regrets differ from Table XXV by ≤ 0.004.

(lr 2.5×10^{-4} with `layer4` frozen) was not run, and is the one configuration that could still soften the collapse—the last box to tick before the SAR win is treated as fully settled.

Every one of the 65 cells behind the summary in Table XIII (KGA wraps Tent; EATA/SAR are the standalone candidates; ResNet-18, N=800 balanced held-out evaluation images per cell, online TTA) is released as `experiments/kbound/results/cifar10c_65cells.csv`, with columns `corruption, severity, frozen, tent, eata, sar, kga, oracle`. Its per-method means reproduce Table XIII exactly. We do not typeset the 65 rows here: no claim in this paper is made about an individual cell, and the summary row is the evidence.

D. Experiment and GPU-run inventory

Every track promoted in this paper has a results table, named in §X and in the guarantee accounting; the machine-readable index is `paper/generated/kbound_result_manifest.json`. Two pre-considered tracks were **not run** and contribute no evidence: the ImageNet-C blur and weather corruption families (the noise grid was judged sufficient for the harmful-candidate claim, so the ImageNet-C evidence covers three corruptions and not fifteen), and the official SAR schedule of Protocol E (a residual check on the SAR port, superseded by the mechanism-faithful re-run of Table XXVII). All runs were executed one job at a time on a single Apple MPS device or on CPU; per-track

candidate	policy	mean acc	regret↓	harmful cells	beats both?
Tent	always-adapt	**0.454**	0.0007		
	always-freeze	0.405	0.0494		
	K-Bound	0.446	0.0082	6%	no (ties adapt)
EATA	always-adapt	**0.444**	0.0000		
	always-freeze	0.405	0.0386		
	K-Bound	0.440	0.0038	3%	no (ties adapt)
SAR	always-adapt	0.319	0.1124		
	always-freeze	0.405	0.0267		
	K-Bound	**0.409**	**0.0229**	44%	**yes**

TABLE XXVII

ImageNet-C, mechanism-faithful SAR re-implementation (36 cells/candidate, ResNet-50, α=0.1). Same noise grid, protocol, and metrics as Table XXV; SAR re-implemented to the reference mechanism (SAM, recovery reset, EMA) at the learning rate shared across candidates. The harmful regime survives the faithful re-implementation (SAR harmful on 44% of cells, mean benefit −0.086): always-adapt collapses to 0.319 while KGA certifies-or-abstains to 0.409, *beating both* trivial policies (regret 0.0229 vs 0.1124/0.0267); Tent/EATA stay benign and KGA ties always-adapt. Oracle accuracy 0.455/0.444/0.432. *Caveat:* matched/shared LR; SAR's official lr 2.5×10^{-4}, layer4-frozen schedule not yet run.

TABLE XXVIII

PACS completed three-seed diagnostic (DomainBed leave-one-domain-out, 18 cells/domain/seed, α=0.10). Entries are means across seeds; regret-to-oracle is lower better. No domain earns a multiseed beats-both claim.

domain	adapt	freeze	**KGA**	mean $\mathrm{FA_u}$
art_painting	0.0241	0.0367	0.0309	0.037
cartoon	0.0250	0.0147	0.0147	0.000
photo	0.0106	0.0494	0.0494	0.000
sketch	0.0109	0.0776	0.0776	0.000
mean	0.0176	0.0446	0.0431	0.0093*

*Reported mean of per-seed/domain rates; the raw pooled false-adapt numerator was not retained, so no count or Wilson interval is reconstructed.

TABLE XXIX

Camelyon17 conformal-radius bias–variance decomposition (Protocol B diagnostic). Residual dominated by $\widehat{B}(Z)$ model error, not binomial measurement noise.

Quantity	n_{eval}=256	n_{eval}=1024
observed ε	0.112	0.029–0.038
measurement floor $\varepsilon_{\mathrm{meas}}$	0.044	0.022
obs./floor ratio	2.55×	≈1.0×
frac. variance = measurement	≈16%	≈16%
Protocol B $1/\sqrt{n}$ prediction	—	**fails**

hardware, wall-clock and the controller microbenchmark are in §XIV.

Theorem 1 is proved with a constructed witness and strengthened to a quantitative Le Cam minimax floor (part (ii)); Lemma 2 is a plug-in regret decomposition (exact identity) with a minimax floor; Theorem 2 is proved *conditional* on a valid label-free interval estimator and complemented by an anytime-valid e-process (manuscript, App. B); Corollary 2 proves the covariate case and Lemma 1 the binary, multiclass, and regression sign-of-difference results. The multiclass/regression label-free *bracketing* is closed under a bounded calibration-transfer assumption (manuscript, App. C); the *unconditional* bracketing (Conjecture 1) is resolved as a testability dichotomy (Theorem 4: no evidence-definable assumption identifies the sign; the minimal untestable supplement is one bit, which suffices under falsifiable structure). Every theorem ships a numerical sanity-check in the release package; all were re-run from a clean environment and pass: the Le Cam floor tracks the closed form, the regret identity holds to $\sim10^{-17}$, the multiclass identity to 10^{-16} with 100% sign agreement over 4000 trials, calibration-transfer sign-recovery is 100% when the confidence gap exceeds 2η, the router generalization gap decays as $\sim n^{-0.66}$, and the drift-corrected anytime false-adapt rate stays $\leq \alpha$. Where multi-seed runs are available, experiments report means ± std with paired t-tests; single-seed/scoping tracks are labeled explicitly in captions and discussed as regime-classification evidence. We also report evidence/α/batch ablations, a regression covariate-shift track, and a clean non-identifiability witness.

The benefit-sign frontier (Thm. 3) predicts that on the low-margin band any *committal* label-free rule must commit sign errors, whereas a valid certificate must *abstain* there. This appendix demonstrates that prediction on the two *synthetic-corruption* streams whose operating point the certificate is calibrated on (CIFAR-10-C, ImageNet-C); it makes *no* external-validation claim. The honest natural-shift (Camelyon17) finding—a limitation, not a win—closes the appendix (App. G).

E. CIFAR-10-C: K-Bound beats both trivial policies at zero false-adapt

On the pre-registered CIFAR-10-C stress grid (432 conditions/candidate), KGA adapts the helpful conditions and freezes the harmful ones with a *zero* false-adapt rate, so it strictly beats both always-adapt and always-freeze for the two benign candidates (Tent, EATA) and ties the collapse-resistant SAR (Table XXX).

candidate (harmful base)	regret-to-oracle ↓			
	KGA	always-adapt	always-freeze	false-a
Tent (34.5%)	**0.0016**	0.0086	0.1232	0.0
EATA (27.1%)	**0.0015**	0.0037	0.1311	0.0
SAR (15.7%)	0.0018	0.0018	0.1372	0.0

TABLE XXX

CIFAR-10-C stress grid (432 conditions/candidate, α=0.1). KGA *beats both* for Tent and EATA at a 0% false-adapt rate, and ties always-adapt for SAR.

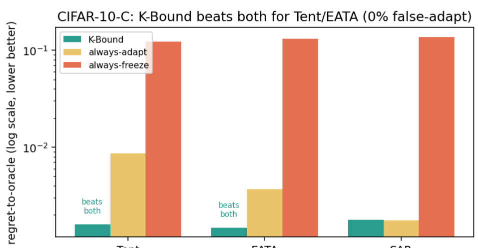

Fig. 8. **CIFAR-10-C frontier (regret-to-oracle, log scale).** KGA is below *both* always-adapt and always-freeze for Tent and EATA at 0% false-adapt, and ties always-adapt for SAR (Table XXX).

decision rule (SAR, harmful regime)	false-adapt	harmful-ad.	regret ↓	cov.
always-adapt	0.44	1.00	0.1124	1.00
always-freeze	—	0.00	0.0267	1.00
ATC-style confidence (plug-in)	1.00	0.25	0.0673	1.00
ATC-style confidence (LOO)	1.00	0.06	0.0375	1.00
K-Bound (KGA)	**0.00**	**0.00**	0.0229	0.39

TABLE XXXI

Committal label-free heuristics false-adapt exactly where KGA abstains (ImageNet-C, reference-parity SAR, 36 cells, α=0.1). ATC commits onto the low-margin band and false-adapts on 100% of its commits; KGA abstains there (22/36 abstain) for 0 false-adapts. On the benign candidates KGA also keeps 0 false-adapts at higher coverage (Tent: regret 0.0060, coverage 0.69; EATA: regret 0.0047, coverage 0.75).

F. ImageNet-C/SAR: the frontier on real corruptions—committal rules fail, K-Bound does not

On the logged ImageNet-C conditions (36 cells/candidate; SAR is harmful on 44.4% of cells), every *committal* label-free heuristic must place its commits somewhere on the low-margin band and therefore false-adapts: a plug-in **ATC** [13] confidence rule adapts into harmful cells on 100% of its commits (false-adapt 1.00, harmful-cells-adapted 0.25), and even its leave-one-out-tuned variant keeps false-adapt = 1.00. Among the rules compared, KGA reaches zero false-adapt and zero harmful-cells-adapted by *abstaining* on the band (coverage 0.39), consistent with the frontier prediction (Table XXXI).

G. Historical note (Camelyon17, frozen synthetic threshold)

Under a **frozen, Camelyon-free** multicandidate threshold $\tau^\star$=0.52 (calibrated only on synthetic grids), the certificate does *not* beat both trivial policies on an early

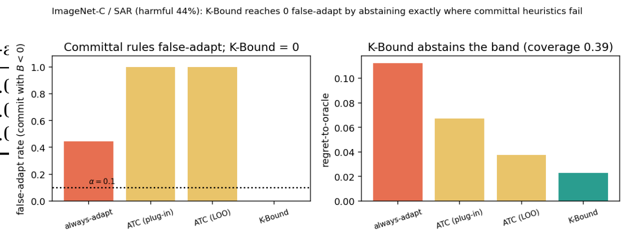

Fig. 9. **ImageNet-C/SAR frontier.** Committal label-free rules false-adapt on the harmful regime; KGA reaches zero false-adapt by abstaining the low-margin band (coverage 0.39; Table XXXI).

Camelyon17 composition grid (regret KGA 0.076 vs always-freeze 0.071). That failure motivated in-domain calibration and rich evidence; the headline result is Protocol G (main text, Table XX), which uses dev-hospital calibration and does not transplant $\tau^\star$ from synthetic runs.

The weakest one-bit class (resolving the open piece of Conjecture 1). Theorem 4 fixes the supplement at *one bit*; what remained open is the *weakest falsifiable class* on which that bit suffices. We settle it conditionally. On D write $a(x) := 2\eta_a(x) - 1$ with $\eta_a(x) = \mathbb{P}_T(f_a{=}Y \mid x)$, so $\Delta = \int_D a\, d\mu$ (Lemma 1). Let $m(x) := 2s(x) - 1$ be the observable relative (calibrated-score) margin, so $M = \frac{1}{2}\mathbb{E}_{\mu|D}[m]$, and let the observable *relative-calibration field* be $c(x) := w(x)\,(m(x) - m^\star)$ with observable nonnegative weight $w(x)$ and observable neutral anchor $m^\star$ (the margin at which $s = \frac{1}{2}$). Put $T_E := \int_D c\, d\mu$ (observable), and let E denote the law of all label-free observables; two targets are *evidence-identical* when they share E (equivalently (μ, m, s), since predictions and s are fixed maps of x).

Definition 13 (Margin-monotone relative-calibration class)**.** $\mathcal{C}_{\mathrm{mono}}$ is the class of targets for which $a(x) = \sigma\, c(x)$ μ-a.e. on D for a single global orientation $\sigma \in \{\pm 1\}$; equivalently the benefit is single-crossing in the observable margin at $m^\star$ ($\operatorname{sign} a(x) = \sigma \operatorname{sign}(m(x) - m^\star)$) with calibrated magnitude $|a(x)| = |c(x)|$. The class is *falsifiable but not verifiable*: a finite labelled probe can reject "$a = \pm c$", yet σ is never label-free identified.

Assumption 4 (General position)**.** Within an evidence fibre the regions of D whose benefit sign is free carry comparable, not-fully-E-pinned benefit mass: $T_E \neq 0$ and no proper subcollection of free regions has an E-certifiable dominant total $\int |a|\, d\mu$.

Lemma 4 (Structural sign-freedom = bit complexity)**.** *Fix E and a falsifiable class $\mathcal{C}$, and let $K(\mathcal{C}, E)$ be the number of disjoint μ-positive regions of D whose benefit signs are jointly free in $\mathcal{C}$ on the fibre E. Under Assumption 4, a falsifiable structural supplement of b bits pins* sign Δ *across the fibre* iff $b \geq K(\mathcal{C}, E)$.

Proof. If $b \geq K$, declaring the orientation of each free region fixes the sign field; the rest of a being E-pinned, $\Delta = \int_D a\, d\mu$ then has E-computable sign. If $b < K$, some free region R is undeclared; the two targets agreeing off R and differing in sign a on R are evidence-identical and share all b declared bits, while by comparability of $\int_R |a|$ with the complement (Assumption 4) their benefits have opposite sign—no decoder of the b declared bits separates them. □

Theorem 16 (Conditional resolution of Conjecture 1: a weakest one-bit class)**.** (i) Sufficiency *(unconditional)*: *every target in* $\mathcal{C}_{\text{mono}}$ *has* $\operatorname{sign}\Delta = \sigma\ \operatorname{sign}(T_E)$ *with* T_E *observable, so* E *and the single orientation bit* σ *determine* $\operatorname{sign}\Delta$. (ii) Necessity *(unconditional)*: σ *and* $-\sigma$ *are evidence-identical (*$\operatorname{Law}(E \mid \sigma) = \operatorname{Law}(E \mid -\sigma)$*) yet give* Δ *and* $-\Delta$*, so by Le Cam at least one bit is required. Thus* exactly one bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$, with no further assumption. (iii) Minimality *(under Assumption 4)*: *every falsifiable* $\mathcal{C} \supsetneq \mathcal{C}_{\text{mono}}$ *contains a non-margin-monotone target with* $K(\mathcal{C}, E) \geq 2$*, so by Lemma 4 at least two bits are required. Hence* $\mathcal{C}_{\text{mono}}$ *is a weakest falsifiable one-bit class—minimal in the single-crossing, calibrated-magnitude direction; it is not unique (Remark 19).*

Proof. (i) $\Delta = \int_D a\, d\mu = \sigma \int_D c\, d\mu = \sigma T_E$. (ii) Sending $\sigma \mapsto -\sigma$ is the D-local label complement of Theorem 4: it fixes (μ, m, s), hence E, while negating a and thus Δ; the two-point bound of Theorem 1 gives minimax committal error $\frac{1}{2}$. (iii) A target outside $\mathcal{C}_{\text{mono}}$ violates $\operatorname{sign} a = \sigma\ \operatorname{sign}(m - m^\star)$ on a μ-positive region, so its sign field is not one global orientation; in the swap-closed (falsifiable) fibre at least two regions are independently signable, i.e. $K \geq 2$, and Lemma 4 applies. □

Proposition 7 (Explicit non-identifiability witness)**.** *For each candidate single structural bit there are two targets with identical evidence law (*TV $= 0$*) sharing that bit's value but with opposite* $\operatorname{sign}\Delta$*. Take* $D = R_1 \cup R_2$, $\mu(R_1) = \mu(R_2) = \frac{1}{2}$*, with* R_1 *calibrated (*$c_1{=}1$*) and* R_2 *at the anchor (*$c_2{=}0$*, no observable benefit signal). For* $\beta = \operatorname{sign} a_1$*:* $a = (+1, -2)$ *vs.* $(+1, +2)$ *give* $\Delta = -\frac{1}{2}$ *vs.* $+\frac{3}{2}$*. For* $\beta = \operatorname{sign} a_2$*:* $a = (+1, +\frac{1}{5})$ *vs.* $(-1, +\frac{1}{5})$ *give* $\Delta = +\frac{3}{5}$ *vs.* $-\frac{2}{5}$*. No single bit certifies* $\operatorname{sign}\Delta$*, so two bits are necessary off* $\mathcal{C}_{\text{mono}}$*.*

Remark 19 (What is unconditional, and what rests on general position)**.** Parts (i)–(ii)—that one declared bit certifies $\operatorname{sign}\Delta$ on $\mathcal{C}_{\text{mono}}$—hold *unconditionally.* Only minimality (iii) uses Assumption 4: if it fails (one region's benefit mass is E-certifiably dominant) that region's orientation alone certifies $\operatorname{sign}\Delta$ even outside $\mathcal{C}_{\text{mono}}$, moving the boundary. Equivalently, Assumption 4 is the requirement that the certificate be *domination-free* (minimax-sound over the unobserved magnitudes); so the General-Position statement is only *conditional*; the unconditional weakest classes are characterized in Theorem 17 as an explicit finite family of dominance polytopes. Even under Assumption 4, $\mathcal{C}_{\text{mono}}$ is only *a* weakest one-bit class: any calibrated sign-aligned field $\{a = \sigma h : h \text{ observable}\}$ is equally one-bit and is incomparable to $\mathcal{C}_{\text{mono}}$. All six checks are machine-checked by a deterministic release validator: the unconditional core (sufficiency/necessity 20000/20000, $|\Delta - \sigma T_E| = 0$ exactly), the explicit minimality witnesses of (iii), the general-position ablation, and the non-uniqueness check.

Why general position cannot simply be dropped (the precise obstruction, and exactly what is open). The minimal counterexample is two equal-mass regions $D = R_0 \cup R_1$, $\mu(R_0) = \mu(R_1) = \frac{1}{2}$, with the observable field $c = +1$ on the *calibrated* region R_0 and $c = 0$ on the *anchor* region R_1 (no observable benefit signal there, so R_1 is sign-free in any evidence fibre). Enlarge $\mathcal{C}_{\text{mono}}$ to the falsifiable class $\mathcal{C}_{\text{dom}} = \{a_{R_0} = \sigma,\ |a_{R_1}| \leq \rho\}$ for a declared constant $\rho < 1$. Since $\Delta = \frac{1}{2}(a_{R_0} + a_{R_1})$ and $|a_{R_1}| \leq \rho < 1 = |a_{R_0}|$, the calibrated region dominates and $\operatorname{sign}\Delta = \sigma$ for *every* member, so one declared bit certifies $\operatorname{sign}\Delta$ on all of $\mathcal{C}_{\text{dom}}$. Yet $\mathcal{C}_{\text{dom}} \supsetneq \mathcal{C}_{\text{mono}}$ (it contains non-margin-monotone targets with $a_{R_1} \neq 0$, e.g. $a = (+1, \pm\frac{1}{2})$ giving $\Delta = \frac{3}{4}, \frac{1}{4}$, both positive) and is falsifiable (a labelled probe rejects targets that are uncalibrated on R_0 or violate $|a_{R_1}| \leq \rho$). This *contradicts* minimality once an E-certifiable dominant region is permitted: dropping $a = (+1, \pm 2) \Rightarrow \Delta = +\frac{3}{2}, -\frac{1}{2}$ shows that the moment $\rho \geq 1$ the anchor region can flip $\operatorname{sign}\Delta$ and two bits become necessary ($K{=}2$). The obstruction to an *unconditional* weakest class is therefore exactly this: characterising minimality across the *entire* lattice of dominant-region thresholds ρ (where the weakest one-bit class is not $\mathcal{C}_{\text{mono}}$ but a ρ-dependent family), which Assumption 4 sidesteps by fiat. Theorem 17 (below) resolves it: the ρ-lattice is the slice $\{r_0{=}1,\ r_1{=}\rho\}$ of the single dominance polytope $W^\star{=}\{r_0 \geq r_1\}$, with $\rho{=}1$ its boundary. The construction is machine-checked (`val_conj1_genpos.py`, seeds fixed): $\mathcal{C}_{\text{dom}}$ strictly contains $\mathcal{C}_{\text{mono}}$, is falsifiable (reject rate 1.0), is one-bit-certified (error rate 0 over 2×10^5 targets), and loses one-bit certifiability without the domination bound (error rate $\approx \frac{1}{4}$).

Unconditional resolution (General Position removed). Dropping Assumption 4 reveals, for each orientation pattern, a single maximal one-bit class—a *dominance polytope*—of which the ρ-family of Remark 19 is one slice.

Definition 14 (Orientation pattern and canonical class)**.** On a fibre E an *orientation pattern* is a tied set $P \subseteq \{i : c_i \neq 0\}$ with a tied sign-pattern $s \in \{\pm\}^P$; the free set is $Z = D \setminus P$. For an observable magnitude region $W \subseteq [0,1]^{|D|}$ the *canonical class* is $\mathcal{C}(P, s, W) = \{a : \operatorname{sign} a_i = \sigma\, s_i \operatorname{sign} c_i\ (i \in P),\ (|a_j|)_j \in W\}$ with one declared bit σ. Write $T(r) = \sum_{i \in P} s_i \operatorname{sign}(c_i) \mu_i r_i$ and $G(r) = \sum_{i \in Z} \mu_i r_i$.

Lemma 5 (Canonical form of a falsifiable class)**.** *A class is falsifiable* iff *it equals some* $\mathcal{C}(P, s, W)$*; anchors (*$c_i = 0$*) lie necessarily in* Z*.* (Swap-closure, Thm. 4(iii), forces the constraint on Z to depend only on $|a|$; forcing an anchor sign is not evidence-definable. Both aligned ($s_i{=}+$) and anti-aligned ($s_i{=}-$) ties are falsifiable since σ is unidentified.)

Theorem 17 (Unconditional one-bit criterion; weakest classes)**.** *Let $\mathcal{C}(P,s,W)$ be falsifiable.* (i) (Dominance margin.) *One declared bit certifies* $\operatorname{sign}\Delta$ *on $\mathcal{C}(P,s,W)$ iff* $\inf_{r\in W}\big(T(r)-G(r)\big)\ge 0$ *(with some member $\Delta>0$) or* $\sup_{r\in W}\big(T(r)+G(r)\big)\le 0$. (ii) (Weakest class.) *For a fixed pattern (P,s) the unique maximal one-bit class is the dominance polytope $\mathcal{C}^\star=\mathcal{C}(P,s,W^\star)$, $W^\star=\{r\in[0,1]^{|D|}: T(r)\ge G(r)\}$; $\mathcal{C}_{\mathrm{mono}}$, every $\mathcal{C}_{\mathrm{dom}}(\rho\le 1)$, and every box one-bit class are proper subclasses of one $\mathcal{C}^\star$.* (iii) (Finite family.) *The set of weakest falsifiable one-bit classes is the explicit finite family $\{\mathcal{C}^\star\}$ indexed by $(P\subseteq\{c\ne 0\}, s\in\{\pm\}^P)$ modulo the global flip $\sigma\mapsto-\sigma$ (at most $3^{|\{c\ne0\}|}$ patterns); distinct patterns give pairwise incomparable maxima, so there is no* unique *weakest class. $\mathcal{C}_{\mathrm{mono}}$ is the all-tied all-aligned member, and Assumption 4 is exactly the face on which the family collapses to $\mathcal{C}_{\mathrm{mono}}$, recovering Thm. 16(iii).*

Proof. Δ is linear, so over $\mathcal{C}(P,s,W)$ at bit $\sigma=+1$ it ranges over $[\inf_W(T-G),\sup_W(T+G)]$ (free signs independent by swap-closure; the $\sigma=-1$ fibre is its negation); a single bit certifies iff each fibre's strict sign set is a singleton, giving (i). For $W^\star$, $\inf_{W^\star}(T-G)\ge 0$ by construction, and any r_0 with $T(r_0)<G(r_0)$ admits, with opposing free signs, a member with $\Delta<0$ while $W^\star$ admits $\Delta>0$, so $W^\star$ is maximal; $\mathcal{C}_{\mathrm{dom}}(\rho)$ sits in $W^\star$ iff $\rho\le 1$, with $\rho=1$ the boundary, giving (ii). Incomparability (iii) is witnessed on $c=(1,1,0)$ by $a=(0,\frac13,0)$ and $a=(1,-\frac13,\frac13)$. On the general-position face the free block can overpower the tied block unless Z-magnitudes vanish, leaving only $\mathcal{C}_{\mathrm{mono}}$. ☐

Verification. The criterion (i) was checked against exact brute-force vertex truth on 2.8×10^5 box fibres and 3.2×10^3 non-box integer polytopes (exact-LP inf / sup) with 0 mismatches; sufficiency on $W^\star$ over 1.2×10^5 members (0 errors), necessity just outside $W^\star$ (error rate $\approx\frac12$), and the $\mathcal{C}_{\mathrm{dom}}(\rho)$ collapse ($\rho\le1$ one-bit, $\rho>1$ lost) are machine-checked (`val_unconditional_weakest.py`, seeds fixed; fails loudly). The single non-tautological input is Lemma 5, which rests on the swap involution (Thm. 4(iii)).

The main text treats the drift budget β as *declared*. This appendix settles when β can instead be *computed*: a vacuity theorem (label-free audits certify nothing), three purchasable audits (labels; smoothness×transport; rotation-uniform under unknown-direction index drift), a witness-class sufficient condition for useful root-n audits (Theorem 23), the composition into the decision guarantee, a fully empirical rule, and a domain-level verifiability protocol with an exact feasibility floor. Notation as in Assumption 1: $u=\eta_a-s\in[-1,1]$ on D, $\gamma(P)=\mathbb{E}_{\mu_P}[u\mid D]$; representations $\psi(x)$ with $\|\psi\|\le B$; all statements conditional on D.

a) Status and credit.: Proofs are complete below; the concentration ingredient of Theorem 22 is *known* in sharper form (Bartl–Mendelson; Boedihardjo; Nietert et al.) [77], [78], [79], [80] and is used as a lemma; the hierarchical-conformal backbone of Theorem 24 follows Dunn–Wasserman–Ramdas and Lee–Barber–Willett [81], [82]. The contribution is the composition into the adapt/freeze/abstain audit, the witness-class sufficient condition (Theorem 23), and the floor analysis; synthetic validation (coverage, non-vacuity, decision safety, Aud-F net panel) is in the repository artifact `audit_validation/results_audited_drift.json`.

Theorem 18 (Aud-A: label-free budget audits are vacuous)**.** *Fix the observables (μ_T,f_0,f_a,s), hence M, and let $\mathcal{P}(\mu_T)$ be all target laws with these observables. For every $\bar a\in[0,1]$ there is a law in $\mathcal{P}(\mu_T)$ with $\gamma=\bar a-\frac12-M$ and the same joint law of (Z,M); the achievable drift set is $[-(M+\frac12),\frac12-M]$ at matched evidence. Consequently any audit $\widehat\beta=g(Z)$ with $\mathbb{P}(|\gamma|\le\widehat\beta)\ge 1-\delta$ uniformly over $\mathcal{P}(\mu_T)$ satisfies, with probability $\ge 1-\delta$, $\widehat\beta\ge\frac12+|M|>|M|$: the audited frontier abstains with probability $\ge 1-\delta$ in every world.*

Proof. Extend the kernel construction in the proof of Theorem 1: for $\bar a\in[0,1]$ set $K^{\bar a}(x,f_a(x))=\bar a$, $K^{\bar a}(x,f_0(x))=1-\bar a$ on D (valid Bernoulli since binary labels give $\{f_0(x),f_a(x)\}=\{0,1\}$ on D), keeping the off-D kernel and all observables fixed. Then $\mathrm{Law}(Z,M)$ is identical across $\bar a$ while $\gamma=\bar a-\frac12-M$ sweeps the stated interval. Validity in the extreme world ($\bar a\in\{0,1\}$ as $\operatorname{sign}M$ dictates, where $|\gamma|=\frac12+|M|$) forces the $(1-\delta)$-quantile of $g(Z)$ to be at least $\frac12+|M|$; since $g(Z)$ has the same distribution in every world of $\mathcal{P}(\mu_T)$, the bound transfers, and on that event neither strict branch of $|M|>\widehat\beta$ can fire. ☐

Theorem 19 (Aud-B: label-budget audit)**.** *Let $X_1,\dots,X_n\sim\mu_T(\cdot\mid D)$ with labels revealed, and $\widehat u_i=\mathbf{1}\{f_a(X_i)=Y_i\}-s(X_i)\in[-1,1]$, $\widehat\gamma=\frac1n\sum\widehat u_i$, $t(n,\delta)=\sqrt{2\ln(2/\delta)/n}$. Then $\widehat\beta_{\mathrm{lab}}=|\widehat\gamma|+t(n,\delta)$ satisfies $\mathbb{P}(|\gamma|\le\widehat\beta_{\mathrm{lab}})\ge1-\delta$, and the audited frontier certifies a strict action whenever $|M|>|\gamma|+2t(n,\delta)$; in particular $n\ge 8\ln(2/\delta)/(|M|-|\gamma|)^2$ audit labels suffice when $|M|>|\gamma|$.*

Proof. $\mathbb{E}[\widehat u_i]=\gamma$; Hoeffding on range-2 variables gives $\mathbb{P}(|\widehat\gamma-\gamma|>t)\le 2e^{-nt^2/2}=\delta$; the triangle inequality gives both claims. (This quantifies, in the form the frontier consumes, the audit rate discussed in App. G.) ☐

Theorem 20 (Aud-C: transport-audited budget, finite-sample)**.** *Assume* (Lip)*: u is L-Lipschitz on D w.r.t. a bounded 1-d representation ($|\psi|\le B$). With n_{lab} labeled calibration points, $m_{\mathrm{cal}},m_{\mathrm{dep}}$ unlabeled points and empirical measures $\widehat\mu$, let $\varepsilon_{\mathrm{DKW}}(m,\delta)=\sqrt{\ln(2/\delta)/(2m)}$ and define*

$$\begin{aligned}\widehat\beta_W=|\widehat\gamma_{\mathrm{cal}}|&+t(n_{\mathrm{lab}},\delta/2)\\&+L\widehat W_1(\widehat\mu_T,\widehat\mu_{\mathrm{cal}})\\&+2BL\varepsilon_{\mathrm{DKW}}(m_{\mathrm{cal}},\delta/4)\\&+2BL\varepsilon_{\mathrm{DKW}}(m_{\mathrm{dep}},\delta/4).\end{aligned}$$

we have $\mathbb{P}(|\gamma(P_T)|\le\widehat\beta_W)\ge1-\delta$.

Proof. Population step: u/L is 1-Lipschitz, so Kantorovich–Rubinstein gives $|\gamma(P_T)-\gamma(P_{\mathrm{cal}})| \le L\, W_1(\mu_T,\mu_{\mathrm{cal}})$. Finite-sample: on a 1-d bounded representation, $W_1(\widehat{\mu},\mu) = \int|\widehat{F}-F| \le 2B \sup_t |\widehat{F}-F|$, bounded by DKW at level $\delta/4$ per measure; Hoeffding at $\delta/2$ for $\widehat{\gamma}_{\mathrm{cal}}$; union bound and the W_1 triangle inequality assemble $\widehat{\beta}_W$. (Lip) is declared, not verified—Theorem 18 guarantees some such purchase is unavoidable; multi-domain Lipschitz-ratio checks falsify but cannot verify it. ☐

Theorem 21 (Aud-D/E: composition and the fully empirical rule)**.** *Let $\widehat{\beta}$ be any audit with $\mathbb{P}(|\gamma| \le \widehat{\beta}) \ge 1-\delta$.* (D) *The audited population rule (adapt iff $M > \widehat{\beta}$; freeze iff $M < -\widehat{\beta}$; else abstain) has $\mathbb{P}$(false strict commitment) $\le \delta$.* (E) *With $\widehat{M} = \frac{1}{m}\sum s(X_i) - \frac{1}{2}$ from m unlabeled deployment points and $t_M(m,\delta') = \sqrt{\ln(2/\delta')/(2m)}$, the* empirical *audited rule (adapt iff $\widehat{M}-t_M > \widehat{\beta}$; freeze iff $\widehat{M}+t_M < -\widehat{\beta}$) has $\mathbb{P}$(false strict commitment) $\le \delta+\delta'$, and every quantity in the rule is computable. Gated by the finite-sample certificate of Theorem 2, each false-commit event is bounded by $\alpha + \delta(+\delta')$.*

Proof. (D): on $\{|\gamma| \le \widehat{\beta}\}$, $M > \widehat{\beta} \ge |\gamma|$ implies $M+\gamma > 0$, so $\operatorname{sign}\Delta = +1$ by Lemma 1; symmetric for freeze; a false commitment requires the audit event to fail. (E): Hoeffding for the $[0,1]$-valued $s(X_i)$ gives $\mathbb{P}(|\widehat{M}-M| > t_M) \le \delta'$; on the intersection of the two events the adapt branch gives $M \ge \widehat{M} - t_M > \widehat{\beta} \ge |\gamma|$; union bound. Validation: on the synthetic sweep the empirical rule makes 0/12 false strict commits (population-M rule 0/13; β=0 plug-in 3/20), with certification opening one grid step later than the population rule—the price of t_M (`audit_validation/results_audited_drift.json`). ☐

b) Rotation-closed classes and useful audits.: The maximality questions below need two notions that are used throughout the rest of the appendix.

Definition 15 (Rotation-closed drift class)**.** Fix a bounded feature map $\|\psi(x)\| \le B$ on D. A class $\mathcal{U}$ of drift functions $u : D \to [-1,1]$ is *rotation-closed* if $u \in \mathcal{U}$ implies $u \circ R_\psi \in \mathcal{U}$ for every orthogonal transformation R acting on the feature space (equivalently: if a single-index ridge $u(x) = g(\theta^\top\psi(x))$ is admitted for some $\theta \in S^{d-1}$, then the same ridge is admitted for every $\theta \in S^{d-1}$).

Definition 16 (Useful audit at rate r_n)**.** An audit $\widehat{\beta}$ is *valid* at level δ over a class $\mathcal{P}$ if $\mathbb{P}(|\gamma(P)| \le \widehat{\beta}) \ge 1-\delta$ for every $P \in \mathcal{P}$. It is *useful at rate* r_n if it is valid and, whenever the true drift vanishes ($\gamma \equiv 0$ and $\mu_T = \mu_{\mathrm{cal}}$), one has $\widehat{\beta} = O_P(r_n)$. Usefulness rules out the vacuous label-free bound $\widehat{\beta} \ge \frac{1}{2} + |M|$ of Theorem 18.

Theorem 22 (Aud-F: rotation-uniform audit under unknown-direction index drift)**.** *Assume $\|\psi(x)\| \le B$ on D and* (Idx)*: $u(x) = g(\theta_*^\top\psi(x))$ with g L-Lipschitz and $\theta_* \in S^{d-1}$* unknown. *Let $\mathrm{MS}_{\mathrm{net}}$ be the maximum of the empirical sliced distances $W_1(\theta\#\widehat{\mu}_{\mathrm{cal}}, \theta\#\widehat{\mu}_T)$ over an ϵ-net of S^{d-1}, and where $\varepsilon_{\mathrm{VC}}(n,\delta) = \sqrt{2((d+1)\ln(2n)+\ln(8/\delta))/n}$, define*

$$\begin{aligned}\widehat{\beta}_F = {} & |\widehat{\gamma}_{\mathrm{cal}}| + t(n_{\mathrm{lab}},\delta/4) \\ & + L(\mathrm{MS}_{\mathrm{net}} + 2B\epsilon) \\ & + 2BL\varepsilon_{\mathrm{VC}}(n,\delta/4) + 2BL\varepsilon_{\mathrm{VC}}(m,\delta/4).\end{aligned}$$

Then $\mathbb{P}(|\gamma(P_T)| \le \widehat{\beta}_F) \ge 1-\delta$. The same statement holds for k-index drift $u = \sum_{j\le k} g_j(\theta_j^\top\psi)$ with $\sum_j \mathrm{Lip}(g_j) \le L$.

Proof. (i) *Index transfer:* g/L is 1-Lipschitz on the projected line, so 1-d Kantorovich–Rubinstein gives $|\gamma(P_T) - \gamma(P_{\mathrm{cal}})| \le L\, W_1(\theta_*\#\mu_{\mathrm{cal}}, \theta_*\#\mu_T) \le L \sup_\theta W_1(\theta\#\mu_{\mathrm{cal}}, \theta\#\mu_T) = L \cdot \mathrm{MS}$; the supremum covers the unknown θ_*. (ii) *Uniform deviation:* for every θ simultaneously, deviations of the empirical sliced distances are controlled by $W_1(\widehat{\nu},\nu) = \int_{[-B,B]}|\widehat{F}-F| \le 2B\sup_t|\widehat{F}-F|$ (the 1-d identity, valid with atoms), and $\sup_{\theta,t}|\widehat{F}_\theta(t) - F_\theta(t)|$ is the uniform deviation over the halfspace class $\{x : \psi(x)\cdot\theta \le t\}$, of VC dimension $\le d+1$; the VC inequality at level $\delta/4$ per measure gives the two $\varepsilon_{\mathrm{VC}}$ terms via $|\sup_\theta a - \sup_\theta b| \le \sup_\theta |a-b|$ and the per-θ W_1 triangle inequality. (iii) *Net:* $|W_1(\theta\#\cdot) - W_1(\theta'\#\cdot)| \le 2B\|\theta-\theta'\|$ (identity coupling and Cauchy–Schwarz, both measures), so $\mathrm{MS} \le \mathrm{MS}_{\mathrm{net}} + 2B\epsilon$; the net-inflated value enters $\widehat{\beta}_F$. (iv) *Budget:* three random events (Hoeffding; VC for calibration; VC for deployment) at $\delta/4$ each, total failure $\le 3\delta/4 < \delta$. For k-index, bound each term by its own slice and use $\sum_j \mathrm{Lip}(g_j) \le L$. Sharper constants (and removal of the $\ln(2n)$ factor) follow by replacing the VC step with the max-sliced concentration of [77], [79]; the $O((3/\epsilon)^d)$ net is computational only—the statistical bound is net-free. ☐

Remark 20 (Aud-F audits adaptive max-sliced W_1)**.** The population quantity controlled by Theorem 22 is the *max-sliced* (projection-robust) distance $\Delta_1(\mu_{\mathrm{cal}},\mu_T) = \sup_{\theta\in S^{d-1}} W_1(\theta\#\mu_{\mathrm{cal}}, \theta\#\mu_T)$, not a predeclared finite set of directions. The unknown ridge θ_* is therefore covered adaptively; the ϵ-net enters only to produce a *computable* upper certificate $\mathrm{MS}_{\mathrm{net}} + 2B\epsilon$ on $\Delta_1(\widehat{\mu}_{\mathrm{cal}},\widehat{\mu}_T)$. Sharp $n^{-1/2}$ MSW_1 bounds under bounded support are available from [79], [77]; Aud-F's contribution is the composition into a useful drift-budget audit for the adapt/freeze/abstain frontier. Numerical coverage of the net certificate on unknown-direction worlds is recorded in `audit_validation/results_audited_drift.json` (panel `F_index_msw`).

Theorem 23 (Aud-H: useful audits for low-complexity witness classes)**.** *Let $\mathcal{F}$ be a symmetric class of measurable functions $D \to [-L,L]$ that contains the unknown drift u and has Rademacher complexity satisfying $\mathfrak{R}_n(\mathcal{F}) \le C\sqrt{(v+\ln(2/\delta))/n}$ for constants C, v (alternatively: VC-subgraph dimension at most v, which yields the same root-n scale). Define the integral probability metric $\Delta_{\mathcal{F}}(P,Q) =$*

$\sup_{f\in\mathcal{F}} |Pf - Qf|$. *With* $n_{\rm lab}$ *labeled calibration points,* n *unlabeled calibration points, and* m *unlabeled deployment points, set*

$$\widehat{\beta}_{\mathcal{F}} = |\widehat{\gamma}_{\rm cal}| + t(n_{\rm lab}, \delta/3) + \Delta_{\mathcal{F}}(\widehat{\mu}_{\rm cal}, \widehat{\mu}_T) + 2\mathfrak{R}_n(\mathcal{F}) + 2\mathfrak{R}_m(\mathcal{F}) + L\sqrt{\tfrac{\ln(6/\delta)}{n}} + L\sqrt{\tfrac{\ln(6/\delta)}{m}}.$$

Then $\mathbb{P}(|\gamma(P_T)| \le \widehat{\beta}_{\mathcal{F}}) \ge 1-\delta$. *In particular the audit is useful at rate* $O(\sqrt{(v+\ln(1/\delta))/n})$ *whenever* $\mu_T = \mu_{\rm cal}$ *and* $\gamma = 0$. *Specializations:* (i) *projected* 1*-Lipschitz / halfspace-CDF witnesses recover the adaptive* MSW_1 *budget of Theorem 22 with* $v \asymp d$; (ii) s*-sparse ridge witnesses replace* d *by* $s\log(ed/s)$; (iii) *the full* 1*-Lipschitz class on a* d*-dimensional ball has metric entropy forcing the* $n^{-1/d}$ *empirical-*W_1 *rate [83], [85] and is therefore* not *useful at* $O(\sqrt{d/n})$.

Proof. (i) *Population transfer.* Since $u \in \mathcal{F}$ and $\mathcal{F} = -\mathcal{F}$,

$$|\gamma(P_T) - \gamma(P_{\rm cal})| = \left|\mathbb{E}_{\mu_T}[u] - \mathbb{E}_{\mu_{\rm cal}}[u]\right| \le \Delta_{\mathcal{F}}(\mu_T, \mu_{\rm cal}).$$

(ii) *Empirical IPM.* The triangle inequality gives $\Delta_{\mathcal{F}}(\mu_T,\mu_{\rm cal}) \le \Delta_{\mathcal{F}}(\widehat{\mu}_T,\widehat{\mu}_{\rm cal}) + \Delta_{\mathcal{F}}(\widehat{\mu}_T,\mu_T) + \Delta_{\mathcal{F}}(\widehat{\mu}_{\rm cal},\mu_{\rm cal})$. Standard symmetrization yields $\mathbb{E}\Delta_{\mathcal{F}}(\widehat{\mu},\mu) \le 2\mathfrak{R}_n(\mathcal{F})$; a bounded-difference / McDiarmid upgrade (range $2L/n$ per sample) adds the two $L\sqrt{\ln(6/\delta)/(2n)}$ concentration terms at total failure $\delta/3$. (iii) *Labeled calibration.* Hoeffding as in Theorem 19 at level $\delta/3$ controls $|\widehat{\gamma}_{\rm cal} - \gamma(P_{\rm cal})| \le t(n_{\rm lab}, \delta/3)$. (iv) *Union.* On the intersection of the three events, $|\gamma(P_T)| \le |\widehat{\gamma}_{\rm cal}| + t + \Delta_{\mathcal{F}}(\widehat{\mu}_{\rm cal},\widehat{\mu}_T) + 2\mathfrak{R}_n + 2\mathfrak{R}_m +$concentration, which is $\widehat{\beta}_{\mathcal{F}}$. Specialization (i) follows because $W_1(\theta\#P, \theta\#Q) = \sup\{|Pf - Qf| : f(x) = h(\theta^\top\psi(x)), \mathrm{Lip}(h) \le 1\}$ and the halfspace thresholds that realize the CDF form of 1-d W_1 have VC dimension $\le d+1$, recovering Theorem 22's $\varepsilon_{\rm VC}$ scale. Specialization (iii) is the classical empirical-W_1 rate on bounded domains. ☐

Theorem 24 (Aud-G: domain-level verifiable alignment with an exact floor)**.** *Assume* (G1) *the* K *calibration domains and the deployment domain are exchangeable tuples* (P_k, Z_k, γ_k)*;* (G2) *each calibration domain carries a labeled sample of* common *size* $n_{\rm lab}$*, giving* $\widehat{\gamma}_k$ *with common Hoeffding radius* e *at level* δ*;* (G3) *the label-free drift predictor* $\gamma_{\rm pred}(\cdot)$ *is fixed and independent of all* $K{+}1$ *domains. Let* $s_k = |\widehat{\gamma}_k - \gamma_{\rm pred}(Z_k)|$ *and* $j = \lceil(1-\alpha+\delta)(K+1)\rceil$*, feasible iff* $\alpha \ge \frac{1}{K+1} + \delta$*. Then*

$$\mathbb{P}\Big(\,|\gamma_{K+1}| \le |\gamma_{\rm pred}(Z_{K+1})| + s_{(j)} + e\Big) \;\ge\; 1-\alpha,$$

with every right-hand quantity computable without deployment labels.

Proof. The computable scores $s_1, \dots, s_{K+1}$ are exchangeable (common $n_{\rm lab}$ and estimator under (G1)–(G3); if per-domain labeled sizes differ, exchangeability of the scores fails and validity is lost). Split conformal at level $\alpha' = \alpha - \delta$ gives $\mathbb{P}(s_{K+1} \le s_{(j)}) \ge 1 - \alpha'$. One deployment-side Hoeffding event bounds the ideal score $S_{K+1} = |\gamma_{K+1} - \gamma_{\rm pred}(Z_{K+1})| \le s_{K+1} + e$ (the counterfactual size-$n_{\rm lab}$ sample interpretation of e); the deterministic triangle $|\gamma_{K+1}| \le |\gamma_{\rm pred}(Z_{K+1})| + S_{K+1}$ and a union bound finish. *Floor:* the conformal index requires $j \le K$, i.e. $\alpha \ge \frac{1}{K+1} + \delta$ exactly; with $K = 5$ domains nothing below $\alpha \approx 0.17{+}\delta$ is certifiable, with $K = 8$ nothing below $\approx 0.11{+}\delta$. Per-deployment verification remains impossible (Theorem 18); verifiability is *priced in exchangeable domains.* (G3) is essential: a predictor fit on the calibration domains breaks exchangeability. ☐

Conjecture 3 (Maximal auditable class)**.** *Characterize the largest rotation-closed class* $\mathcal{U}$ *(Definition 15) that admits a* useful *(Definition 16) label-free-at-deployment audit at rate* $O(\sqrt{d/n})$.

c) Proved bracket.: Useful audits at that rate exist whenever the dual witness class has VC/Rademacher complexity $O(\sqrt{d/n})$ *(Theorem 23), including the rotation-closed unknown-direction / bounded-*k*-index ridge classes of Theorem 22. Full Lipschitz / unrestricted* W_1 *witness classes are excluded from useful* $O(\sqrt{d/n})$ *audits by the* $n^{-1/d}$ *empirical-*W_1 *rates on bounded domains [83], [85].*

d) Why estimation lower bounds do not finish the question.: One-sided upper certificates (audits) and two-sided estimation are not equivalent: for Wasserstein geometry on the cube, identity testing *can require roughly the square root of the samples needed for estimation [86]. Thus* W_1 *estimation lower bounds alone cannot prove that no intermediate rotation-closed class admits a useful root-n audit.*

e) Open.: Whether there exists a rotation-closed class strictly larger than finite-index / adaptive MSW_1*-ridge (e.g. slowly growing* $k(n)$*, constrained low-rank potentials) that still admits a useful root-n audit; and whether one-sided audits can beat the spiked-transport estimation boundary [84]. A resolution plausibly requires a new lower-bound construction over direction mixtures that targets useful UCBs rather than two-sided estimators.*

Conjecture 4 (Certified supremum, polynomial time)**.** *Is there a* $\mathrm{poly}(d,n)$*-time algorithm that, from unlabeled samples, returns a number* U *satisfying both* (i) soundness: $\sup_\theta W_1(\theta\#\widehat{\mu}_{\rm cal}, \theta\#\widehat{\mu}_T) \le U$ *with probability* $\ge 2/3$*, and* (ii) null-completeness: $U = O(\sqrt{d/n})$ *with probability* $\ge 2/3$ *whenever* $\mu_{\rm cal} = \mu_T$ *(bounded support)?*

f) Status.: Open for linear *max-sliced* W_1*. The strongest nearby evidence is the statistical-query gap for spiked transport [84]; exact NP-hardness is known for a* kernel *max-sliced* W_2 *variant [87] and does* not *transfer to the linear* MSW_1 *certificate required by Aud-F. Stationarity guarantees for 1-d max-slicing exist [80] but are not global certificates.*

g) Operational rule.: Until settled, gradient-ascent surrogates for $\mathrm{MS}_{\rm net}$ *are heuristics; only the net value (or an equally certified relaxation) may enter* $\widehat{\beta}_F$ *in Theorem 22.*

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